HIGH-PRESSURE PUMP JETS 3.8/60-COMPACT

MANUAL FOR OPERATION AND MAINTENANCE
SAFETY RULES



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1. INTRODUCTION

1.1 TECHNICAL PARAMETERS

Technical parameters with use of multiplier 60K:

Maximum surrounding temperature:	+35℃
Minimum surrounding temperature:	+5℃ (+15℃ after putting the pump into operation)
Maximum surrounding humidity:	85%
Maximum depth:	1350 mm
Maximum width:	1690 mm
Maximum height:	1550 mm
Total dry weight:	1800 kg
Maximum pressure of output water:	4100 bar
Maximum quantity of water for cutting:	3.8 l/min
Number of multipliers:	1 or 2 - type 60K
Maximum total consumption of water:	3.8 l/min
Minimum pressure of input water:	3.5 bar
Maximum pressure of input water:	6 bar
Input water pH value:	6 - 8
Degree of cutting water filtration:	1 μm and 0.5 μm cascade
Maximum total input:	39.3 kW
Electromotor:	37 kW; 3x400/230V; 50Hz; 1500revs/minute
Cooler motor:	1.5 kW; 3x400/230V; 50Hz; 900revs/minute
Filling of hydraulic oil:	160 l (170 l for change of oil)
Type of oil:	SHELL TELLUS 46 or OHHM 46
Oil pump:	PARKER PV063 - 63,0 cm^3/rev
Maximum working temperature of oil:	+52 ℃
Maximum working pressure of oil:	215 bar
Maximum working temp. of bleed down valve:	+50 ℃
Water input:	G 3/4“ external thread (hose ¾“)
Water output:	G 3/4“ external thread (hose ¾“)
High-pressure water output:	UNF 3/4“-16 internal thread

Technical parameters with use of multiplier 40K:

Maximum surrounding temperature:	+35℃
Minimum surrounding temperature:	+5℃ (+15℃ after putting the pump to operation)
Maximum surrounding humidity:	85%
Maximum depth:	1350 mm
Maximum width:	1690 mm
Maximum height:	1550 mm
Total dry weight:	1800 kg
Maximum input water pressure:	4100 bar
Maximum quantity of water for cutting:	6.2 l/min
Number of multipliers:	1 or 2 - type 40K
Maximum total consumption of water:	6.2 l/min
Minimum pressure of input water:	3.5 bar
Maximum pressure of input water:	6 bar
Input water pH value:	6 - 8
Degree of cutting water filtration:	1 μm and 0.5 μm cascade
Maximum total input:	39.3 kW
Electromotor:	37 kW; 3x400/230V; 50Hz; 1500revs/minute
Cooler motor:	1.5 kW; 3x400/230V; 50Hz; 900revs/minute
Filling of hydraulic oil:	160 l (170 l for change of oil)
Type of oil:	SHELL TELLUS 46 or OHHM 46
Oil pump:	PARKER PV063 – 63.0 cm^3/rev
Maximum working temperature of oil:	+52 ℃
Maximum working pressure of oil:	215 bar
Maximum working temp. of bleed down valve:	+50 ℃
Water input:	G 3/4“ external thread (hose ¾“)
Water output:	G 3/4“ external thread (hose ¾“)
High pressure output:	UNF 3/4“-16 internal thread

1.2 CONDITIONS FOR INSTALLATION AND OPERATION AND OPERATION FILLINGS

1.2.1 Location.

Pump JETS 3.8/60-COMPACT is determined for indoor and ventilated or compulsorily ventilated environment with permanent temperature from **+10℃ to +30℃** and humidity **50%** at the temperature of **+30℃** and altitudinal height up to **1000 above sea level.** For air renewal for cooling of oil and the electromotor it is necessary to observe the minimum distance from the left part of the pump from any obstacle of **550 mm**. This distance may be reduced to **400mm** at the rare side of the pump. The necessary distance from the right side and from the front side of the pump is **1100mm** for operation and control of the distributor. To prevent leakage of oil to the indoor environment it is necessary to provide the floor and walls of the machine room with antiskid coating which is resistant to oil substances up to the minimum height of **250 mm** above the floor. Necessary floor capacity for the installation is **9000 kg/m².** The surface, on which the high-pressure pump shall be installed, must not show the roughness higher than **10mm/m² (±5mm/m²)**. Producer strictly recommends locating the high-pressure pump JETS 3.8/60-COMPACT to a separate machine room to prevent excessive noisiness in the workplace, cleanness at the maintenance and service and safety at work. The ideal location of the machine is drawn on the picture of sample layout of the pump (serial number: PTV-011-V-0190). The pump is equipped with an air-cooler of the hydraulic oil, which is integrated in the machine. The machine must not be exposed to the microwave, ultraviolet, laser or roentgen rays. The machine shall be separated from the floor by means of rubber silentblocks; vibrations of other machines transmitted by the floor on which the machine is placed shall have negative influence on the lifetime of individual elements of the machine. Storage of the machine is possible in a dry environment with temperature ranging between +5℃ and +35℃ and in a suitable packing. Manipulation with the machine may be carried out only by means of a lift-truck; two openings are in the bottom part of the frame. These are designed for lifting forks of the truck. Any other method of manipulation is not allowed.

1.2.2 Power supplies.

According to the drawing no. PTV-011-V-0190.

Water supply for cutting (Water supply „A“): it is necessary to use only **clean filtered water** or **potable water** from the water feeder. The maximum consumption of water is **4.0 l/min**. The maximum pressure of input water is **6bar,** the minimum pressure is **3.5bar** and value of **pH is 6-8**. The connecting point has to be installed 1m from the floor level and ended with closing valve G3/4“ with internal thread. This water will be used to supply the high-pressure pump JETS 3.8/60-COMPACT. The ideal place of installation of the supply is marked on the picture „Sample layout of the machine room“ (see drawing no. PTV-011-V-0190).

Water outlet from the pump: this waste water must be drained to a tank, because the water may be mixed here with the hydraulic oil. After this tank is filled, it is necessary to liquidate it ecologically. The connection of the drain has to be installed from 150 to 500mm from the floor level and ended by external thread G3/4“. The ideal place of installation of this supply is marked on the picture of sample layout of the machine room (drawing number: PTV-011-V-0190).

Connection of power supply („A“): the rated input of the pump is **37 kW**, the supply voltage **3x400/230V; 50Hz**, the electric input of the cooler is **1.5 kW,** knife fuses **80A** are installed in the main distributor. The distributor has to be placed on the wall of the room according valid regulations for electric devices of this type. Electric distributors of the machine are located in the side part of the machine in a metal box.

Connection of the pump to the electric network may be made by the cable **CYKY 4x25mm² or 5x25mm² in the length of the cable up to 15 meters and if longer, the cable CYKY 4x35mm² or 5x35mm² shall be used**. Supply protection must be made by the circuit breaker or fuses with the rated value of **100A at the execution of the supply with cable 4x25mm²(5x25mm²) or 125A at the execution of the supply with cable 4x35mm²(5x35mm²).** The electric power supply must be made in a distance of 2500mm from the machine room floor at least. The ideal place of installation of this connection is marked on the picture of the sample layout of the machine room (serial number: PTV-011-V-0190) and it should be always in the immediate proximity of the distributor. Operator of the machine is **obliged** to carry the yellow-green wire **CYA 16mm²** to the main distributor for execution of the central connection of the electric system.

Power supply („B“): in this place it is necessary to attach the socket 3x400/230V; 50Hz; 25A on the wall of the machine room and a socket 230V; 50Hz; 25A for necessary assembly, service, maintenance and auxiliary works..

Outlet of the high-pressure water from the pump: any heat influence caused by any external source may result in the change of material structure of the high-pressure water part and its devaluation this way – avoid working with heat sources (welding machine, oxyacetylene burner, etc.) in the near proximity of these parts. This output and distribution of the high pressure water has to be regularly checked for leakages of water.

Any interferences, maintenance, and service of the electric connection and the supply must be carried out only by a person with electrical qualification (according the Czech public notice no. 50/78 Col., article 6). Service and repairs of the machine are permitted only with the consent of the producer. In case of an unauthorized intervention to the connection and setting of the electric distributions the users shall loose the guarantee provided by the producer!

1.2.3 Operation fillings.

Operation filling of the pump is the hydraulic oil **OHHM 46**, **SHELL TELLUS 46** or Chevron hydraulic AW-46, Exxon Nuto H-46, Texaco Rando HD-46. The net volume of the tank is **210 l**. For change of oil it is necessary to prepare approximately **170 l** of oil. When filling the tank it is necessary to use the filling filtering mean with the final minimum filtration **5 μm**. Change of the hydraulic oil must be carried out precisely according the operation instructions and pay attention to the maximum cleanness. At the change of oil it is necessary to dispose off the used oil filling ecologically. The maximum operating pressure of the hydraulic oil is **215 bar**.

All necessary data and proposal for location of the machine is shown on the picture of the sample layout of the pump (drawing no. PTV-011-V-0190). Installation, assembly, putting the machine to operation and training of operators shall be provided exclusively by the producer or the supplier of the machine!

1.3 MACHINE CAPACITY AND REQUIREMENTS FOR OPERATION

The stable high-pressure pump **JETS - 3.8/60-COMPACT** with the fitted multiplier of type **„60K“ 3.8 l/min** delivers the water at the maximum pressure **4100 bar**. This device serves for cutting of material by means of a water jet. For operation and maintenance of this machine it is necessary to train one operator (training of the operator must be carried out by the producer of the machine), who will learn the safety instructions related to this machine as well. These instructions are included in this manual. This operator must keep the machine in **clean and functional condition** and **check the observance of safety and operation instructions** by other workers who move around the machine as well.

1.4 MACHINE DESCRIPTION

1.4.1 Hydraulic part.

Scheme of this part creates the appendix of this manual – see drawing no. PTV-011-D-0173.

-Electromotor (note1 - M1) powers the hydraulic pump

-Suction filter (note 23 - SF) serves for filtration of oil sucked by the hydraulic pump

-Hydraulic pump (note 4 - P) supplies necessary pressure and quantity of hydraulic oil for the multiplier.

-Hydraulic distribution box (notes 5 and 42 – HK1 and 2) serves as a holder and distribution for other hydraulic elements.

-Safety valve (note 9 - PV) protects the pump and the multiplier against possible overpressure in the distribution.

-Proportional valve (note 46 - PPV) controlled by the electronic unit; ensures smooth functioning of the pump and correct output parameters of the hydraulic oil; it is integrated directly on the hydraulic pump.

-Closing electromagnetic valve (note 45 - EV) ensures closing of delivery of the hydraulic pump.

-Electromagnetic distributor (note 6 - DEV) ensures operation of bleed down valve (note 27 – BD) at the reduction of output water pressure.

-Panel of choke valves with check valves (note 8 - DSV) reduces the speed of oil flow from and to the bleed down valve.

-Panel of check valve (note 7 – DZV) ensures correct direction of hydraulic oil flow for control of the bleed down valve.

-Manometer (note 44 – MP2) serves for measuring of pressure on the output of the hydraulic pump and setting of its working level.

-Choke valve (note 43 - SV) limits vibration of manometer hand at the operation of the machine.

-Pressure sensor (note 10 – P1) reads the pressure of hydraulic oil on the output of the pump and delivers this information to the automatic control.

-Bleed down valve (note 27 - BD) ensures reduction of pressure from the high pressure pump conduct at the reduction of the pressure and the machine standstill.

-Multiplier (note 25 - VTM) by means of hydraulic oil pressure (max. 215 bar) increases the water pressure to the maximum pressure of 4100 bar (with the multiplier type 60K). Number of presets of multiplier at the operation of the machine is namely 80/min (max. 110/min).

-Hydraulic oil cooler (note 24 - CH) by means of ventilator, own circulatory pump and electromotor (M2) cools the hydraulic oil to the appropriate operation temperature (approximately 50℃). Hydraulic oil cooler is integrated in the machine.

-Reverse filter (note 15 - ZF) ensures fine filtration (10μm) of hydraulic oil before it enters to the tank.

-Filler (note13 - NZ) ensures safe filling of the tank with the hydraulic oil.

-Tank (note 19 - N) ensures sufficient operation stock of hydraulic oil (approximately 160 l).

-Openings (note 18 - PN) allow access to inside of the tank.

-Ball valves (note 21 – KV2) ensure safe draining of the oil filling from the tank and closing of suction conduction.

-Oil level sensor (note 17 – L1) allows automatic putting the machine out of operation at the decrease of oil level under the critical level.

-Local thermometer and level indicator (note 12 – T2+L2) allow visual checking of temperature and level of oil in the tank.

-Oil temperature sensor (note 16 – T1) reads the temperature of the hydraulic oil and controls the operation of the pump according the measured value.
-Pressure switch (note 14 – PF) sends a signal to the automatic control in case the cartridge of the check valve is fouled up.
-Plug (note 11 – Z) closes connection for service measuring of pressure at the output from the pump.
-Coupling (note 3 – S) transmits the torsion moment between the electromotor and the pump.
-Holder (note 2 – D) ensures firm connection of electromotor boxes and the pump.
-Slide valve (note 22 – SO1) ensures closing of suction of the pump when servicing the hydraulic circuit.
-Temperature sensor (note 26 – T3+T4) reads the temperature of check valves and indicates the level of their wear.
-Oil temperature sensor (note 28 – T5) reads the temperature of the bleed down valve and indicates this way the level of its wear.
-High pressure sensor (note 29 – P4) reads the pressure value at the output of water from the pump – shall be delivered on request.

1.4.2 Part of water supply to multipliers – filtering unit and increase of pressure of input water.

Scheme of this part creates the annex of this manual – see drawing number PTV-011-D-0173.
Filtering unit (note 30 – FJ):
-Manometers at the input and output (note 32 – MP1) measure the input and output pressure of the filtering unit and indicate this way the rate of fouling up of the filters.
-Ball valves (note 31 - V) allow closing of water supply to the multiplier and serve as the prevention against water leakage at the change of filters and draining of the water from the unit.
-Filter (note 33 – FV1) filters water to the mechanical cleanness < 1μm
-Filter (note 34 – FV0.5) filters the water to the mechanical cleanness < 0.5μm
-Pressure sensor (note 35 – P2) indicates level of pressure of the input water; in case of pressure decrease under the critical value the machine stops.
-Increase of input water pressure (note 36 – TVV) – shall be delivered on request:
-Pump (note 37 – PV) increases input water pressure.
-Electromotor (note 38 – M3) powers the pump.
-Pressure switch (note 39 – P3) switches the electromotor with the pump on and off based on reached water pressure.
-Pressure tank with bag (note 40 – TN) ensures sufficient stock of water at the required level of pressure.
-Little pressure valve (note 41 – TV) for service purposes – supplements the air pressure working against the bag.
-Electromagnetic valve (note 47 – EVV) closes the entry to the multiplier when the machine is put out of operation.

1.4.3 High-pressure part.

Scheme of this part creates the annex of this manual – see drawing number: PTV-011-D-0173.
-Multiplier (note 25 - VTM) increases pressure of water to the required level – the machine is standardly fitted with one multiplier. The maximum pressure of input water from the multiplier is 4100 bar (60KPSI) and the number of circles (presets) is namely 80/min according to the used water nozzle, maximum number of cycles is 110/min.
-Bleed down valve (note 27 - BD) at the requirement for low input pressure of water ensures decrease of pressure in the high-pressure conduit and when the machine is out of operation it releases all the pressure from the high-pressure distributions.
-Temperature sensor (note 28 – T5) indicates the temperature of the bleed down valve on the control panel.
-Temperature sensor (note 26 – T3, T4) indicates temperature of check valves on the control panel.

1.4.4 Electric part.

Scheme of this part creates the Annex of this manual, see drawing number: PTV-011-D-0192.
-Electromotor (note 1 – M1) powers the hydraulic pump (note 4 – P).
-Electromotor (note 24 - M2) powers the circulatory pump and ventilator of the cooler of the hydraulic oil (note 24 - CH).
-Electromotor (note 38 – M3) powers the water pump increasing pressure of input water (note 37 – PV); shall be delivered on request.
-Main distributor (see scheme no. PTV-011-D-0192) ensures safe power supply for electromotor and control distributor of the machine. The main distributor is placed on the right part inside the machine.
Q1- main switch
P1- knife fuses
K1, K2, K3, K4- contactors
M1, M2, M3- electromotor
F1- current protection
F2, F3, F4, F5, F6- circuit breakers
R1, R2- relay
Z- socket 230V
F7- time relay (contactor)
TR1- isolating transformer
X0, X1, X2, X3- terminal
-Control distributor of the machine (see scheme number: PTV-011-D-0192) ensures appropriate and safe operation of all control and operation elements of the machine and the automatic control is fitted in it. Control distributor is placed on the right part inside the machine.
TR2- feeding transformer
F10- fuse
U1- automatic control PLC
U3- controlling electronics of the proportional valve
S0, S1, S5, S6, S7, S10, S11, S12, S21, S22, S23, S24- sensors
O1, O2- relay solid state
V1, V2, V3, V4, V5, V6- controlled elements
R3- relay
TB1, TB2, TB3, TB4, TB5- terminal of the automatic control
X4, X5, X6- terminal
-Control panel (see scheme number: PTV-011-D-0192) delivers information on current state of the machine to the operator and allows its control. The control panel is located on the right part of the front end of the machine.
U4- control panel with display
H1- green pilot light on the control panel
S3- start button
S2- stop button
S1- mushroom total stop button

2. MACHINE FUNCTION DESCRIPTION

2.1 Hydraulic part.

Scheme of this part creates the Annex of this manual – see drawing number: PTV-011-D-0173.
Electromotor (note 1 – M1) is connected by means of a fitting (note 3 – S) and a holder (note 2 – D) with the hydraulic pump (note 3 - P) and it powers it. Suction of the hydraulic pump is realized through the suction filter (note 23 – SF) in the tank (note 19 - N) and it is possible to close the suction by means of the slide valve (note 22 – SO1). A proportional valve (note 46 – PPV is fitted on the pump (note 4 – P) which controls the pressure level on the output from the pump based on a signal from the electronic module and the automatic control. The output from the pump is fitted with hydraulic distribution boxes (notes 5 and 42 – HK1 and HK2). The safety valve (note 9 – PV) is installed in HK1. It ensures the delivery of the pump in case the maximum preset pressure is exceeded and it serves as the safety element as well. Waste from the safety valve is drained directly to the tank by means of T-piece through the ball valve (note 21 – KV2). The HK2 is fitted with electromagnetic valve (note 45 – EV), this valve ensures closing of the pump delivery and it separates it this way from the multiplier. This closing is important at the pressure level 0 bar, because the pump works with the minimum pressure – approximately 30 bar. The HK2 is further fitted with control elements of the bleed down valve – panel of the check valve (note 7 – DZC), panel of the butterfly valves (note 8 – DSV) and electromagnetic distributor (note 9 – DEV). All these elements ensure smooth operation of the bleed down valve at the reduction of pressure level and putting the pump out of operation. Waste from these elements is drained to the waste line of the pump (leakage) by means of T-pieces. The pressure sensor (note 10 – P1) is connected to the HK2. It reads the real pressure of the hydraulic oil. The signal is being transmitted to the automatic control and based on this information the device controls the proportional valve. The local manometer (note 44 – MP2) is connected by means of the T-piece to the HK2. It serves for the informational measuring of the real pressure of the hydraulic oil. The butterfly valve (note 43 – SV) is placed in front of the manometer. It softens pressure jumps and prevents excessive vibration of the manometer hand. The main pressure line leads from HK2. The pump delivers necessary quantity and the oil pressure for the multiplier (note 25 – VTM). The waste hose (so called leakage) leads from the pump as well. It leads directly to the tank of the hydraulic oil and it is connected to the T-piece inside the emptying ball valve (note 21 – KV2) and then to the tank. The main pressure line brings the necessary pressure and quantity of hydraulic oil to the multiplier and the multiplier subsequently increases water pressure to the required level. The waste oil from the multiplier is drained to the tank through the reverse filter (note 15 – ZF), which ensures the fine filtration of the oil and protects this way the system against dirt and excessive wear caused by the dirt. The reverse filter is fitted with the pressure switch (note 15 – P3) which indicates pollution of the filter and reports the need to change the cartridge to the control panel. Cooling of the hydraulic oil proceeds in the closed circuit by means of air cooler (note 24 – CH) which ensures the right operation temperature of the oil. Electromotor of the cooler (M2) is being switched on and of from the automatic control which ensures current temperature by means of temperature sensor (note 16 – T1) fitted on the tank, the automatic control evaluates and switches on or off the cooler electromotor. The cooler has its own circulatory pump and it is possible to close the cooling circle by means of ball valves (notes 20 and 21 – KV1 and KV2) to separate tanks this way. This shall be used at the servicing of the cooling circle. The tank of the hydraulic oil is further fitted with the surface switch (note 17 – L1), which protects the hydraulic circle against the damage by the oil leakage. Further there is the filler (note 13 – NZ) which ensures functioning of the filtered deaeration and possibility of safe filling of the tank with the oil. Another elements of the tank are openings (note 18 – PN) which serve for cleaning of the tank at the change of oil and the local level indicator and the thermometer (note 12 – L2, T2) – information meters for operation of the machine.

2.2 Part of water supply to multipliers – filtering unit and increase of input water pressure.

Scheme of this part creates the Annex of this manual – see drawing number PTV-011-D-0173.
Filtering unit (note 30 – FJ):
Water pressure at the input and output from the filtering unit is being measured by manometers (note 32 – MP1), which measure pollution of filters in the unit by optic difference of pressures. Supply of water to multipliers is lead through the ball valve (note 31 – KV3) which serves as a prevention against leakage of water when changing filters and then to own filters (note 33 – FV1 and 34 – FV0.5) in which the final filtration of water up to the level < 0.5µm is carried out. Before the water enters the multiplier the sensor of input water pressure (note 35 – P2) is fitted in the line which in case of pressure decreases under the critical level stops the machine and reports this action by lighting up the pilot light on the control panel.
Increasing of pressure of input water (note 36 – TVV) – shall be delivered on request:
This compact water feeder is fitted with own pump (note 37 – PV) which is powered by electromotor (note 38 – M3).
The electromotor is being put into operation by means of pressure switch (note 39 – P3) at the decrease of water pressure under the minimum level. This switch may be mounted as a part of the filtering unit as well. The water is pressurized to the tank with bag (note 40 – TN), in which a sufficient stock of water with required pressure will be maintained. The pressure is maintained in the tank by means of air pressure from the second part of the bag. This air pressure may be checked and replenished by means of little pressure valve (note 41 – TV). To prevent leakage of water when the machine was stopped the electromagnetic stop valve (note 47 – EVV) is mounted in the filtering unit. It closes the supply to multiplier by pressing the stop button.

2.3 High-pressure part.

The water from the filtering unit is carried by hoses to the multiplier (note 12 - VTM) where the pressure is being increased up to the maximum pressure of 4100 bar (60K). The HP water is carried to the countervailing cylinder (note 48 – VTV) which softens hydraulic impacts in the pipe. Behind the cylinder there is the output of the water to the supply pipe of the cutting place. The bleed down valve (note 27 – BD) is parallelly connected to this output. The bleed down valve functions as the depressurizing element for reduction of pressure and putting the machine out of operation. The electric temperature sensor (note 28 – T5) is connected to the waste water line leading from this valve. The sensor warningly indicates overheating of the bleed down valve on the control panel which prevents its damage. The pressure sensor (note 29 – P4) may be installed (on request) to the line of HP water output. The sensor measures the real pressure of water at the output.

2.4 Electric part.

Power supply and power control is secured by the main distributor on which the main switch is mounted with the control light which indicates whether the machine is alive or not. The scheme of the main distributor creates the annex of this manual – see drawing number: PTV-011-D-0192 pages 2 and 3. Description of parts of the distributor and their function:

Q1- main switch allows switching of power supply to the main distributor.

P1- knife fuses in the holder protect the power supply to the distributor against the short circuit and excess current.

K1, K2, K3- power contactors ensure running-up of the main motor M1 to YD (star/triangle)

F1- current protection of the main motor M1

M1- the main electromotor driving the hydraulic pump P

X0- terminal for connection of supply conductors

X1- terminal for connection of conductors of the main motor

X2- terminal for connection of other conductors of motors and control

X3- terminal for connection of control conductors

F2- circuit breaker of the backup 3f power supply

F3- circuit breaker of hydraulic oil cooler motor power supply M2

K4- conductor switching the hydraulic oil motor cooler M2

M2- hydraulic oil cooler electromotor CH

F4- circuit breaker for control distributor power supply

F5- circuit breaker for power supply of socket 230V

Z- socket 230V

F6- circuit breaker of power supply for the motor of the unit which increases the input water pressure M3 – installed on request

R2- relay for switching the motor of the unit which increases the input water pressure M3 – installed on request

M3- motor of the unit which increases the input water pressure TVV – installed on request

F7- circuit breaker of the power supply for the primary of the isolating transformer TR1

TR1- transformer ensures galvanic separation of power and control elements of the main distributor

F8- circuit breaker of the secondary of the isolating transformer TR1 and power supply control elements of the main distributor

K7- time relay switching Y and D

R1- relay which switches contactors K1-K3 for run-up YD

Other elements of the electric part of the machine are placed in the control distributor of the machine and on the control panel. Scheme of the connection of electric elements of the control distributor and the control panel create the annex of this manual – see drawing number: PTV-011-D-0192 list 4-10.

Description of elements of the control distributor and their function – see drawing number: PTV-011-D-0192 pages 4 up to 10:

TR2- power supply transformer for supply of the control panel and the control distributor

F10- fuse of the control secures the circuits and control against the short circuit and the excess current

U1- the automatic control controls and operates all active elements of the machine based on instructions of the operator, evaluates measured figures. It consists of several modules – basic PLC unit, module RS485, temperature modules, analogue modules, etc.

TB1, TB2, TB3, TB4, TB5- special terminal of the automatic control serve for connection of active regulation elements (meters, valves, contactors, etc.) to the automatic control

X4, X5, X6- standard terminal for connection of power supply automatic control and active regulatory elements and meters

S0, S1- sensors of level of hydraulic piston of multiplier – when using multiplier with electric shifting of the slide valve

S5- switch of the state of low level of hydraulic oil in the tank – L1

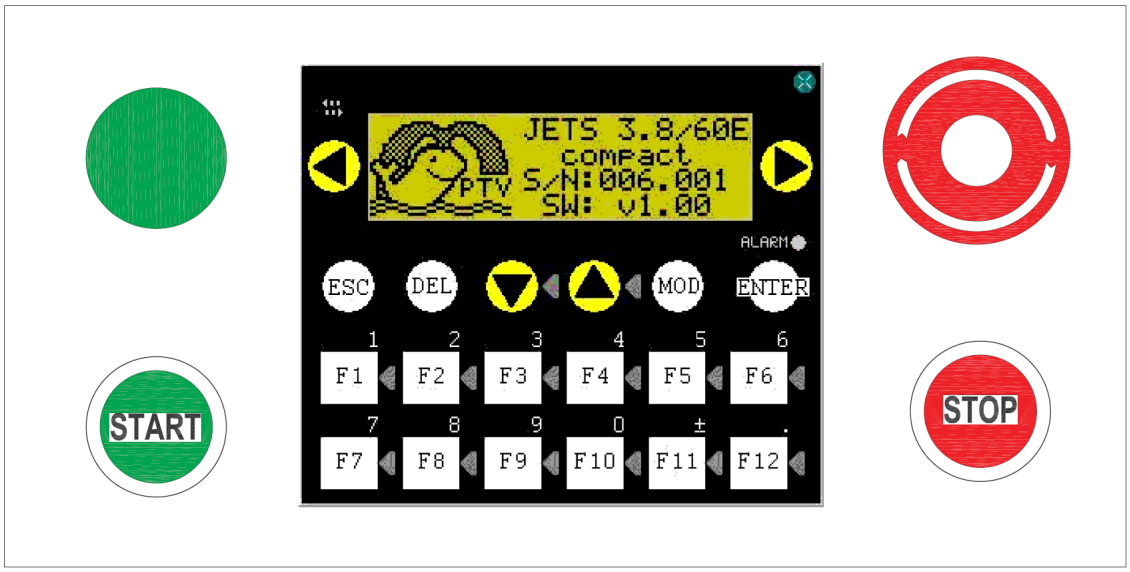
S6- switch of the state of polluted cartridge of reverse filter – P3

S7- sensor of level of the pressure comparator – prepared for additional assembly

O1, O2- relay for electric switching of multiplier – when using multiplier with electric shifting slide valve

V1, V2- coils of the electromagnetic slide valve of the multiplier switching – when using multiplier with electric shifting of the slide valve
V3- electromagnetic distributor - DEV control of BD valve – BD
V4- electromagnetic valve – EVV input water control
V5- electromagnetic valve – EV control of delivery of the hydraulic oil from the pump
V6- proportional valve - PPV
S10- sensor of input water pressure – P2
S11- sensor of hydraulic oil pressure – P1
S12- sensor of input water pressure – P4 – installed on request, prepared for additional assembly
S21, S22- sensors of check valves temperature – T3 a T4
S23- sensor of hydraulic oil temperature – T1
S24- sensor of bleed down valve temperature – T5
U3- electronic module for control of proportional valve
Description of elements of the control panel and their function–see drawing number:PTV-011-D-0192page 4:
U4- control panel with display serves for communication between the operator and the machine. It displays the current state of the machine, values of monitored figures and allows comfortable control.
H1- pilot lamp – green – indicates the state „machine in operation“ when lighting
S3- Start button; it starts the machine when pressing it.
S2- Stop button; it stops the machine when pressing it.
S1- The mushroom Total Stop button; for quick stopping the machine in case of breakdown of the machine or of the high-pressure distribution.
Description of buttons on the control panel:
a) display – gives information on current state of the machine, with colour backlight indicates the emergency (orange colour of the backlight) and breakdown (red colour of the backlight) states. By means of text it informs on breakdowns and problems
b) buttons „◄“ and „►“ allow transition between individual menu and selection of individual items of menu for editing
c) button „ESC“ when pressing once, you return to the previous menu or to the display and 2 quick presses mean return to the basic display with PTV logo.
d) button „DEL“ deletes the edited value by one character back – only in edit mode
e) buttons „▲“ and „▼“ allow browsing in the multi-row text up and down, when editing increase of value, if the button is usable, the LED diode is lighting on it.
f) buttons „MOD“ serves for selection of field for editing
g) button „ENTER“ serves for saving of edited value
h) pilot lamp „ALARM“ indicates excess of value of some of monitored figures – state of warning or breakdown
i) buttons „1“ up to „9“ (F1-F9) serves for selection of pressure level, button „0“ (F10) is the pressure of 0bar
j) button „±“ (F11) serves for quick switch over between preset values of low and high pressure, in edit more is the sigh ±
k) button „.“ (F12) serves for switching over for language versions, in the edit mode it is a decimal point
l) button „START“ serves for starting the machine
m) button „STOP“ serves for stopping the machine
n) button „TOTAL STOP“ serves for stopping the machine in case of emergency
o) the green pilot lamp is lighting when the machine is in operation

Detail of the control panel

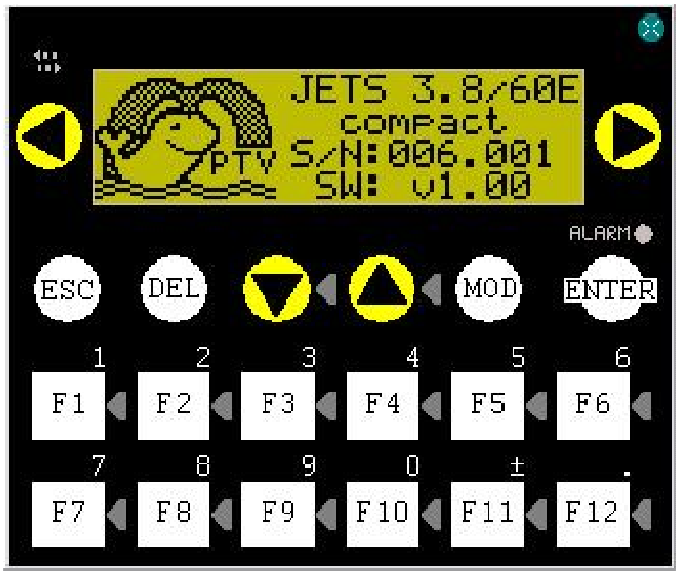


3. MACHINE OPERATION

3.1 Machine control

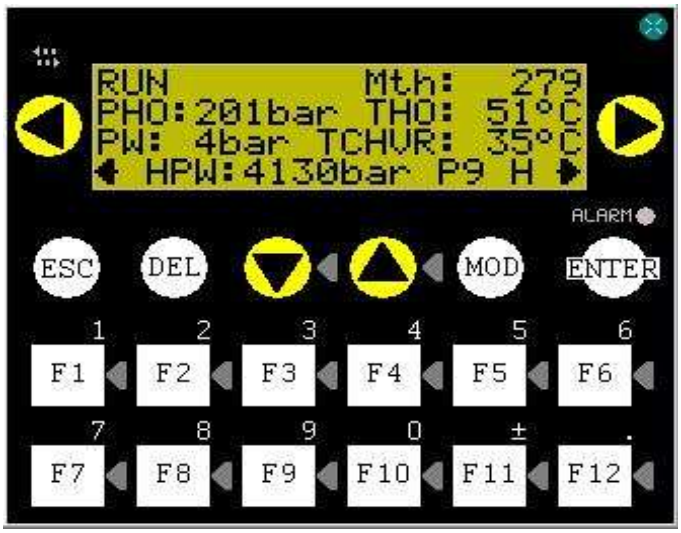
3.1.1 Manual control by means of control panel

The control is realized by means of control panel with display which is interconnected with the communication cable with automatic control in the control distributor. Function of individual buttons was described in the previous chapter.



The basic display

Appears immediately when switching the main switch of the machine and then it switches to the operation display. It provides the information on type and the serial number of the pump, producer, and software version.



Operation display

It appears immediately after the basic display switches off.

The machine shall be put into operation by the button Start and it starts the machine to the pressure level 0bar. Then, buttons 1-9 (F1-F9) and button 0 (F10) will be available:

When pressing one of buttons 1 up to 9 and 0, the arrow of the selected pressure level is flashing until the moment when the machine reaches this level. Then the arrow is constantly on and it indicates the current selected pressure level. Pressure level – see the table, at the bottom on the left.

Operating condition:
RUN – in operation, change of pressure level is permitted
FAULT – fault condition, start is not allowed
STOP – put out of operation, preparation for start
CHECKING – test of input water pressure
STARTING – putting the machine to operation (motor start)
CHANGING – change of pressure level is in progress
STOPPING – putting out of operation, depressurization of the high-pressure distribution
Mth – number of worked out machine hours
PHO – current pressure of hydraulic oil
THO – current temperature of hydraulic oil
PW – current pressure of input water
TCHVR or TCHVL – current temperature of Check Valves – right or left
TBD – current temperature of Bleed Down valve
HPW – current pressure of output high-pressure water (at the absence of high-pressure sensor P4 the pressure is recalculated from the current pressure of hydraulic oil).
P9 – indication of preset pressure level (1 up to 9 and 0)
H – indication of preset pressure level of quick switch, LO – low pressure or HI – high pressure or without indication – another pressure level than LO or HI.
◄ - switch over to menu of MONITOR
► - switch over to the MAIN MENU
▲▼ – browsing in menu up and down

Button	*Hydraulic oil*	*Input water pressure*
1 (F1)	*30 bar*	*600 bar*
2 (F2)	*51 bar*	*1000 bar*
3 (F3)	*77 bar*	*1500 bar*
4 (F4)	*103 bar*	*2000 bar*
5 (F5)	*120 bar*	*2350 bar*
6 (F6)	*138 bar*	*2700 bar (40KPSI)*
7 (F7)	*164 bar*	*3200 bar*
8 (F8)	*195 bar*	*3800 bar (55KPSI)*
9 (F9)	*212 bar*	*4130 bar (60KPSI)*
0 (F10)	*0 bar*	*0 bar*
± (F11)	*Low/high*	*Low/High*
. (F12)	*Selection of language of communication*	

Description of individual menu:

The system has some items which may be changed by means of a protective access password. Passwords entered to the system have three levels – A, B and C.

The access to the system without a password is determined for ordinary operation of the machine and allows setting of date and time and presetting of two pressure levels for quick remote switch, L – low pressure and H – high pressure. The access to the system using password of A level serves for ordinary service and allows zeroing of values of Mth in the item of service for interval of change of hydraulic oil and lapping of check valves. This has to be done at the execution of these operations so that the report on need of service do not appears repeatedly on the display. Other intervals of change or service start to run automatically after zeroing the counter.

Preset intervals:

Check valves – lapping always after 500Mth

Change of oil and cartridge of the filter – first change after 500Mth and each other after 2000Mth.

Further it is possible it is possible to change the time constant of delay of shift of the multiplier's slide valve under the password of A level (when using the slide valve with electric shift) in menu Settings – Switch delay. This option is naturally determined for experienced users, only at the special service. During the current operation it is not necessary to interfere in this setting.

The access to the system using the password of B level allows presetting of all items in menu Settings. This choice is determined for service engineers of the producer for detection of failures and operation states of the machine. This choice is not necessary for an ordinary user. The user of the machine may interfere to presetting of items in menu Settings only with an explicit consent of the producer and with his assistance.

The access to the system with use of the C level password allows making fundamental changes in parameters of the machine and interfering to the control program of the machine. This option is determined only for service engineers of the producer – producer of the control program.

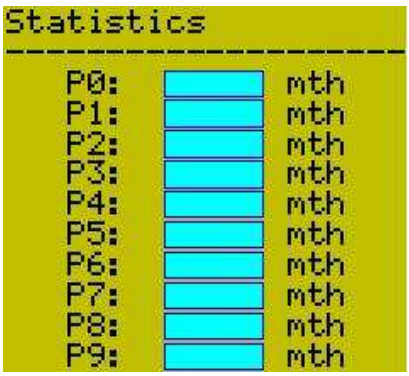


Main Menu

▲▼ – selection of item in the menu
◄► – confirmation of the selection
ALARM LIST – current state of the machine – indicates alarms
ALARM HISTORY – history of alarms
HI/LO PRESET – presetting of quick switch over to LO – low pressure or HI – high pressure
CLOCK – setting of data and time
STATISTICS – states of counters of Mth in individual levels of pressure
PASSWORD – access to protected items, setting of the password in levels A, B and C
SERVICING – allows zeroing of counters of Mth after completion of service of check valves and after change of hydraulic oil and cartridge of the reverse filter. Access under the password of A level
SETTING – setting of basic parameters of the machine, access under the password of following level:
A – only Switch delay (time constant of the delay of shift of the slide valve of the multiplier – at the electric transfer).
B – complete access to the entire menu and option to preset all basic parameters of the machine.
C – complete access to the whole menu, option to preset all basic parameters of the machine and option to interfere to the control program of the machine.



Preset of pressure levels for quick switch (HI/LO preset):
With button MOD you will switch to the required window.
With buttons 1 up to 9 or 0 or arrows ▲▼ you will select the required value in the window.
With the button ENTER you will confirm values entered in the window.
With the button MOD you will switch to another window and repeat the procedure.
When you enter both values to the windows and press the button MOD you will switch to the window x→ and pressing the button ENTER or ► you will confirm the command SAVE (saving valued to EPROM memory). These values remain permanently saved in the memory of the control system of the machine.



Statistics:
This display shows numbers of completed Mth (machine hours) in individual pressure levels, it is only informative.



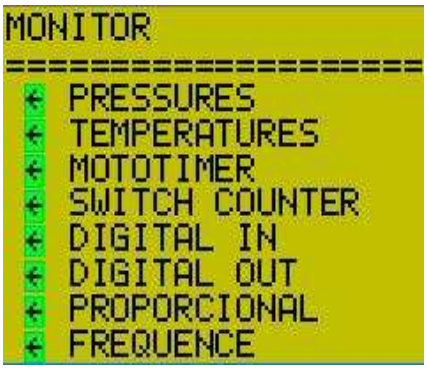
Password:
pressing the button MOD the window for entering the password appears; using buttons 1 up to 9 or 0 or using arrows ◄►▲▼ you will enter the password of the required level A, B or C and confirm it with key ENTER. If the password was entered correctly, the indication of the selected level (A, B or C) will appear at the bottom on the left. Access to protected items of the menu is permitted then only for 5 minutes. entered values shall be permanently saved in the EPROM memory of the control system of the machine.



Servicing:
of the control system is preset so that after the relevant number of completed Mth the system notifies the operator that it is necessary to carry our basic servicing, such as:
- lapping of check valves (after each 500 Mth)
- change of hydraulic oil and the cartridge of the reverse filter (first after 500 Mth and then after each 2000 Mth).
If any or both services were carried out, it is necessary to zero the relevant counter of Mth so that the control system may count further servicing intervals. Each such operation shall not be carried out if the system keeps to report the need of servicing.
Zeroing shall be done by means of buttons MOD, DEL and arrows ◄►▲▼.

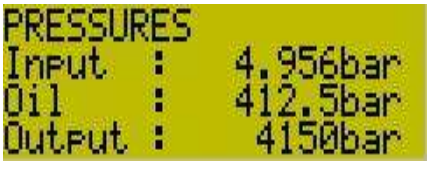


Settings:
Configuration – configuration of parameters and values of the system. Access by means of B level password.
Bleed down timer (BD timer) – setting of time for opening the BD valve at the decreasing the pressure from level 9 to the level 0. Access by means of B level password.
Timer of the multiplier (Switch delay) – setting of delay time at the switching of the electric slide valve of the multiplier. Only when using the multiplier with electric shift of the multiplier. Access by means of A level password.
Timer of the gradual change of levels of pressure (Change timers) – setting of individual times for changes of the pressure of the hydraulic oil to individual pressure levels. Access by means of B level password.
Pressure levels (Presets) – presetting of values of pressure in individual pressure levels. Access by means of B level password.
Linearization of the change of the pressure (Linearization) – presetting and configuration of linearization of the change of the hydraulic oil pressure to individual pressure levels. Access by means of B level password.
Constant (Constant mult.) – presetting of constant for recalculation of pressure between hydraulic oil pressure and output pressure of high-pressure water. Access by means of B level password.
Saving of values (Save) – confirmation of preset values and their saving to the system memory EPROM. Saved values remain valid even when the machine is switched off and switched on again.
Starting values (Default) – reading of original values preset by the producer of the machine based on practical tests. This selection allows restoration of all original functional presetting of all values in the control system and to remove all problems incurred when bad or senseless values were entered. Access by means of A level password.



Monitoring

Menu serves for information on current state of the machine and on all measured values. Move in menu by means of arrows ◄►▲▼.
PRESSURES – information on measured pressure
TEMPERATURES – information on measured temperatures
MOTOTIMERS – information on completed Mth
SWITCH COUNTER – number of settings of the multiplier – this function is accessible only for the multiplier with electric shift of the slide valve
DIGITAL IN – information on state of individual signal inputs to the control system of the machine
DIGITAL OUT – information on state of individual signal outputs from the control system of the machine
PROPORCIONAL – information on current setting-up of the proportional valve
FREQUENCE – information on number of shifts (cycles) of the multiplier per minute – this function is accessible only in case of multiplier with electric shift of the slide valve



Pressures:
Input – pressure of input water in bars
Oil – pressure o hydraulic oil in bars
Output – pressure of output high-pressure water for cutting. In case of absence of the high-pressure sensor P4 the pressure is being calculated by means of Constant mult. from the pressure of the hydraulic oil.



Temperatures:
Oil – temperature of hydraulic oil in ℃
Bleed down – temperature of the bleed down valve in ℃
CHV – temperature of check valves – left & right in ℃



Machine hours (Mototimer):
All – total number of completed machine hours
Last – number of machine hours completed from the last start-up of the machine



Pre-setting (cycles) of the multiplier (Switch Counter): only when multiplier with electric shifting of the slide valve
All – total number of pre-settings (of cycles)
Last – number of pre-settings (of cycles) from the last start-up of the machine



Digital inputs (Digital In):
this display is determined primarily for service engineers of the producer of the machine and provides the information on the current state on digital inputs to the machine control system. In windows of individual inputs 10 up to 19 values 1 or 0 may appear.



Digital outputs (Digital Out):
this display is determined particularly for service engineers of the producer of the machine and it provides the information on the current state on digital outputs from the machine control system. In windows of individual inputs I0 up to I9 values 1 and 0 may appear.



Proportional valve (Proportional):
Preset – preset current pressure level
Output – value of output signal for control of the proportional valve



Frequency:
F – number of pre-settings (cycles) of the multiplier per minute. Only in case of use of the multiplier with the electric shift of the slide valve

3.1.2 Control by means of cutting table control system

3.1.2.1 Manual control from the cutting table control panel



PTV produces cutting tables for areal cutting by the high-pressure jet including own control system. Buttons for manual control of the high-pressure pump from the panel of the table control system are on the left of the picture. Their number, design, and function may differ in the connection to the concrete type and version of the table control system. In the simplest version of the control (relay) it is possible to control the pump manually from the control panes of the table control system as follows:

Start HP pump – putting the pump into operation (pressure 0 bar)
Stop HP pump – putting the pump out of operation
Press + switching to the high pressure (according to the Hi preset)
Press - switching to the low pressure (according to the Lo preset)
Press 1, 2, 3 and 4 – these buttons are not used

In the higher (communication) variant of the pump control the buttons Press 1, 2, 3 and 4 are used for pressure levels selected by the user and this way they allow quick switching among them. Buttons Press + a Press – are used for increasing and decreasing the pressure by one level up and down. Communication shall proceed mutually between the table and the pump and the full takeover of the control panel of the HP pump to the display of the control system of the table is fully possible and this way the remote control becomes adequate.

The current state of the high pressure pump may be displayed on the display of the cutting table at both variants of the remote control - **HP pump status.** State in operation or stop and high or low pressure or in case of higher variant of control the current levels of pressure P1 up to P9 and P0 are displayed there.

3.1.2.2 Automatic control from the control system of the cutting table based on cutting NC program

It is possible to integrate commands for the high-pressure pump directly to the created cutting program is possible by means of the postprocessor (e.g. WRYKRYS).

In the simplest (relay) variant it is possible to control the pump from the program by means of these commands:

Start – putting the pump into operation (pressure 0 bar)
Stop – putting the pump out of operation
Low pressure – switching to the low-pressure mode (according to the Lo preset)

High pressure – switching to the high-pressure mode (according to the Hi preset)
In the higher (communication) variant is possible to make selection of the pressure level of the pump directly in the cutting program (P1 up to P9 and P0) for each element of the product which shall be cut and this way it is possible to optimize the cutting performance and markedly reduce operating expenses. This variant may be applied particularly for products with different required quality of the cut and in case of products and materials with special characters.

3.2 Putting the machine into operation.

NOTICE: the first initiation of the machine may be carried out only by a professional engineer of the producer of the device! First initiation of the machine and further starting after the change of oil of after longer shutdown of the machine it is necessary to carry out following procedures:

1. It is necessary to fill in the oil pump with the hydraulic oil to its delivery and to the filler of the waste line – leakage (when hoses are disconnected), otherwise it may result in a serious damage of this pump. Then the hoses may be connected again.
2. Fill in the tank with the hydraulic oil, approximately 5 mm under the upper level of the local level indicator.
3. Open the water input to the multiplier, disconnect the supply hoses to the multiplier, and let the water off for 3 minutes at least to wash up dirt from supply hoses.
4. Close the water supply and connect hoses to the multiplier.
5. Check visually the entire machine for any oil and water leakages – in case of need to remove defects by tightening of hoses..
6. Open the water supply to multipliers and make sure by means of manometer on the control panel that the water pressure is sufficient and that water filters are not polluted.
7. Close the HP valve on the HP water supply to the cutting workplace.

!!!ATTENTION!!! BEFORE EACH START OF THE MACHINE CHECK BALL VALVES AND THE SLIDE VALVE ON THE BUTTOM PART OF THE TANK – ALL MUST BE OPEN IN FULL BEFORE THE PUMP STARTS. OTHERWISE A SERIOUS DAMAGE OF THE HYDRALIC PUMP AND THE CIRCULATORY PUMP OF THE COOLER MAY HAPPEN !!!

8. Switch the main switch on the main distributor on.
9. Press the START button and the machine will starts operating. Press the STOP button after approximately 10 seconds. Repeat this procedure 4 times at least so that hydraulic circuits smoothly fill with the oil. Then start the machine with START button.
10. Wait until the operation of the machine becomes stable and concurrently make visual check of all its parts. Press button „1“ on the control panel. This way you will let the hydraulic oil flow to the multiplier at the approximate pressure of 30 bar. Wait until the operation of the machine is stable.
11. Open slowly the HP valve on the HP supply to the cutting workplace and let the water to flow to the HP distribution system.
12. Open the ON/OFF valve on the cutting table and leave the HP water to flow freely through the water jet nozzle to the tank of the table. Use only the water jet nozzle with the size up to the maximum capacity of the pump. In case of use of larger nozzle the pump may be damaged!
13. Pressing the button „2“ on the control panel you increase the HO pressure to approximately 51 bar.
14. Leave the machine in operation at this pressure for approximately 5 minutes. Concurrently carry out visual check of the whole machine.
15. Pressing the button „4“ on the control panel you increase the HO pressure to approximately 103 bar.
16. Leave the machine in operation for approximately 5 minutes. In case of any prove of possible failure, leakage of liquids or irregular operation, stop the machine immediately.
17. Pressing the button „6“ on the control panel you increase the HO pressure to approximately 138 bar.
18. Leave the machine in operation for approximately 5 minutes. In case of any prove of possible failure, leakage of liquids or irregular operation, stop the machine immediately.
19. Pressing the button „8“ on the control panel you increase the HO pressure to approximately 195 bar.
20. Leave the machine in operation for approximately 5 minutes. In case of any prove of possible failure, leakage of liquids or irregular operation, stop the machine immediately.
21. Pressing the button „9“ on the control panel you increase the HO pressure to approximately 212 bar.
22. Carry out the visual check of the whole machine and then close covers.
23. Now you can close the ON/OFF valve on the cutting table and the machine is prepared for the necessary work.

NOTE: In case vibrations in the hydraulic circle occur at the start-up of the machine and at the pressure increase, it is caused by aeration of regulatory elements of the hydraulic circle. Then it is necessary to stop the machine and start again several times until the circle deaerates by itself. If this failure repeats, please contact the machine producer.

NOTE: Operational pressure of the hydraulic oil when starting the cold machine is

approximately 215bar at the closed ON/OFF valve at the output from the machine. When the oil warms up the pressure decreases up to approximately 205-212bar. When the ON/OFF valve is opened, it is at the ordinary operation, the pressure then ranges between approximately 180-210bar.

NOTE: during the first 160 Mth of the operation of the machine it is necessary to pay particular attention to the checking of the machine, regularly check tightness of all connections and immediately remove all failures.

NOTE: the temperature sensor T1 mounted on the tank does not allow increase of the hydraulic oil pressure by means of the potentiometer if the oil temperature does not reach 10℃ at least. This situation may occur i f the temperature in the machine room is lower than 10ºC. Then it is necessary to let the machine idle until the oil warms up or establish heating of the machine room!

3.3 Ordinary putting the machine into operation.

Before putting the machine into operation it is necessary to carry out visual check of the machine for leakage of the hydraulic oil and water. If any leakage is detected, it is necessary to localize immediately the place of leakage and to remove the failure. Further, it is necessary to check the level of the hydraulic oil in the tank and a superficial examination of the machine for tightness of connections, hoses etc. The machine shall not allow its starting when the temperature of the machine room is lower than 10℃.

Starting procedure:

1. open the water supply to the filtering unit and make sure according manometers that the water pressure is sufficient (3-6bar)
2. check visually both water supplies to the multiplier for any possible leakage
3. close the HP valve on the HP supply to the cutting table .

!!! CHECK OPENING OF BALL VALVES AND THE SLIDE VALVE ON THE UNDERSIDE OF THE TANK – IT MUST BE FULLY OPEN !!!

4. switch the main switch on the main distributor
5. press the START button
6. wait until the operation of the machine becomes stable and then make one more visual examination of the whole machine
7. press the button „1“ on the control panel by means of which you set the pressure of oil to approximately 30bar
8. wait until the machine becomes stable and then examine the entire machine visually again
9. pressing buttons „0-9“ you will set the required pressure level
10. wait until the machine stops and then examine visually the entire machine once again
11. let the HP water out from the machine by slow opening of the HP valve on the output of the machine to the distribution system of the HP pipe
12. examine the entire machine and the high-pressure distribution visually and close all cover of the machine

! ATTENTION: THE MACHINE MUST NOT RUN WITHOUT NOZZLES MOUNTED ON THE CUTTING HEAD !

13. variation of the information manometer which is fitted on the HK2 (oil pressure) should be dependent on the operation of the multiplier, maximally between values 170-210 bar
14. during the standard operation of the machine the prescribed maximum number of cycles of the multiplier is approximately **110 cycles per minute**; in case of damage of nozzles the machine will be overstrained by the quantity of supplied water and it is necessary to stop the machine immediately. The normal number of cycles at the operation of the machine is 80-90/min.

! Never start the machine in case you detect any failures !

3.4 Machine standstill.

1. decrease the pressure of oil by button „0“ to the zero value
2. leave the machine run for approximately 1 minute at the zero pressure (to stabilize the temperature of the oil)
3. press the Stop button
4. switch off the main switch on the main distributor
5. now the machine is put out of operation
6. close the water supply to the filtering unit
7. open covers of the machine
8. examine the entire machine visually for any leakages of oil, water, closeness and tightness of connections etc.
9. close covers and the machine is prepared for further start

3.5 Emergency machine standstill.

In case of failure or defect caused outside the machine or on the machine:

1. press the button TOTAL STOP (button is adjusted in the pressed level)
2. switch off the main switch on the main distributor
3. now it is possible to detect the incurred defect and start the repair

4. remove the incurred defect and check the whole machine visually
5. the machine is now again prepared for start

3.6 Automatic emergency machine standstill.

The machine has two levels for reporting of failures which are being recorded to the memory in the main menu under the item - alarm history:

Warning – warns the operator of exceeding the critical level of any of measured values or necessity of service. The machine shall create the report on the display of the control panel and indicates it by orange flashing of the entire display.

Fault – stops the machine automatically at the exceeding of the critical level of any of measured values. The machine shall create the report on the display of the control panel and indicates it by red flashing of the entire display.

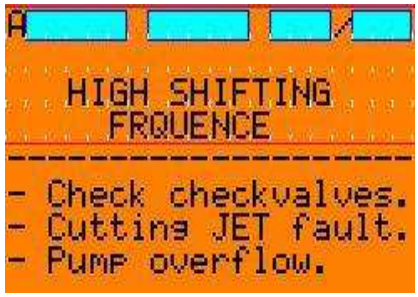
Warning reports:



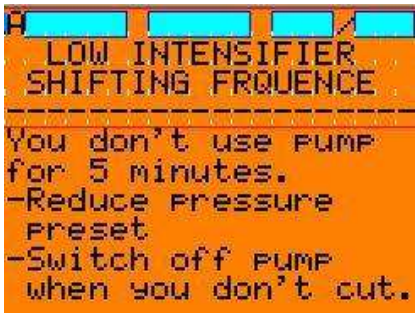
High temperature of oil – the causes may be as follows:
- too high temperature in the machine room – it is necessary to ventilate it
- choked cooling lamellas of the cooler – it is necessary to clean them
- failure of the cooler or its pump and the ventilator – it is necessary to repair it
- main hydraulic pump worn – it is necessary to change it
- wear or failure of the low-pressure part of the multiplier – it is necessary to repair it
- wear or failure of the hydraulic control – it is necessary to repair it



Polluted filter of oil – the cause is polluted cartridge of the reverse filter and therefore it is necessary to replace it.



High frequency of presetting of the multiplier – it is possible only when the multiplier is used with electric shifting of the slide valve. Causes may be as follows:
- a water jet nozzle was which exceeds possibilities of the quantitative delivery of the HP pump put in the cutting head (the maximum is 3,8 l/min at the pressure of 4130 bar) – it is necessary to replace it with a nozzle of the correct size
- water jet nozzle is damage and has to be replaced by the new one
- one or two check valves are damaged and let the water pass back – it is necessary to repair it or to change it
- the value for delay of the switching over of the electric slide valve of the multiplier in the control system of the HP pump is set in incorrectly (Main menu-Settings-Switch delay) – it is necessary to preset the correct value. This is valid only in case of use of multiplier with electric shifting of the slide valve.



Low frequency of presetting of the multiplier – this warning is here due to economy of the operation. If the pump is not used for longer time (5 and more minutes) it is better to switch to the 0 bar level and this way you can save electric energy. This is valid only in case of use of the multiplier with electric shifting of the slide valve.



High temperature of the bleed down valve – at the repeated decreasing of the pressure the bleed down valve is often used for draining of the waste water due to which it is warming up. It is better to optimize the cutting NC program so that the reduction of the pressure level occurs only in the most necessary cases – this will markedly lengthen the lifetime of active parts of the BD valve. The second cause may be leakage of BD valves caused by wear of sealing surfaces. In both cases it is good to examine the correct function of the BD valve and possibly to carry out its repair or change.





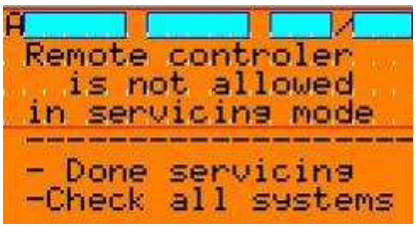
Temperature of the right/left (front/rear) check valves of the multiplier – indicates need of repair (lapping) or change of check valves. In case of large wear the leakage on the input and output of the water from the multiplier incurs which causes excessive warming and wear of check valves – it is necessary to repair it or replace it.



Servis of check valves – this report warns the operator of necessity to service check valves (lapping) after each completed 500 Mth. After servicing it is necessary to reset the counter of Mth in the Main menu-Servicing-Check valve. If the reset is not carried out, the system displays the warning permanently. After resetting the system starts to count another 500 Mth



Servis of oil and filter – after the first 500 Mth and each other 2000 Mth the system warns the operator of necessity to carry out the hydraulic oil service. After each change of hydraulic oit and the cartridge of the reverse filter it is necessary to reset the counter of Mth in the Main menu-Servicing-Oil&filt. If the reset is not carried out, the system displays the warning permanently. After resetting the system starts to count another 2000 Mth

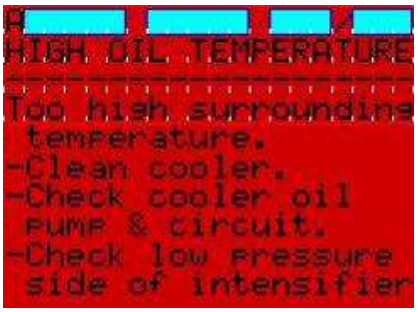


Servicing mode is used for local repair and service. When you lock Safety Key, the machine is switched to Servicing mode, and isn't possible to control it remotely.



First start on day must be locally from HP pump control panel

Error message:



High temperature of the hydraulic oil – causes are as follows:
- too high temperature in the machine room – it is necessary to ventilate it
- polluted cooling lamellas of the cooler – it is necessary to clean it
- failure of the cooler or its pump and ventilator – it is necessary to repair it
- wear of the main hydraulic pump – it is necessary to repair it
- wear or failure of the low-pressure part of multiplier – it is necessary to repair it
- wear or failure of the hydraulic control – it is necessary to repair it



Low temperature of hydraulic oil. The cause is the surrounding temperature (of the machine room) which is lower than 10℃. The control system shall not permit the start until the temperature rises above this level. Then the machine is protected against the excessive wear caused by too cold (and thick this way) oil.



Low level of hydraulic oil in the tank. The control system protects the hydraulic circle against the "dry operation" which could cause damage of active elements of the hydraulics. This situation may occur when some hose, screwed joint or hydraulic boxes in the hydraulic circuit are damaged.



Low pressure of the input water; the system protects the multiplier against "dry operation" without sufficient pressure of the input supply water. Without these the high-pressure parts in the multiplier could be damaged. Causes may be as follows:
- closed cock of the water supply – it is necessary to open it
- polluted cartridges of water filters in the filtering unit - it is necessary to replace it
- failure of the sensor of input water pressure – it is necessary to replace it
- failure of function of the electromagnetic valve (only at the installed increase of pressure of the input water – water feeder – assembled on request) – repair/replace
- leakage of water on the input – it is necessary to repair it



High frequency of setting of the multiplier – it is only when using the multiplier with the electric shifting of the slide valve. Causes may be as follows:
- a water jet which exceeds possibilities of the capacity of the HP pump (the maximum is 3,8 l/min at the pressure of 4130 bar) was put in the cutting head – it is necessary to replace it by the one with the correct size
- the water jet nozzle is damaged and it is necessary to replace it with the new one
- one of both check valves are damaged and lets water pass through back – it is necessary to repair it or to replace it
- the value for delay of switching of the electric slide valve of the multiplier is wrongly set in the control system of the HP pump (Main menu-Settings-Switch delay) – it is necessary to set the correct value. This is valid only in case of use of multiplier with electric shifting of the slide valve.



Low frequency of setting of the multiplier – this warning is here due to economy of the operation, if the pump is not used for a longer time (5 and more minutes) it is better to switch it to the 0 bar level and save the energy. This is valid only at the use of the multiplier with electric shifting of the slide valve.



Current congestion of the motor M1 (main motor) and Emergency Stop – causes are as follows:
- emergency stop button are ON
- failure of the electromotor – it is necessary to have it checked by an authorized service firm (Siemens) and have it repaired or changed.
- failure or bad presetting of current protection of the motor – it is necessary to set it or replace it
- failure of one or more phases of the energy supply – it is necessary to examine knife fuses P1 and check supply phases for any failure
- failure in the hydraulic circuit may be caused by failure of the hydraulic pump P or some of the active elements of the hydraulic circle – it is necessary to check and repair or replace according need

ATTENTION: ANY INTERFERENCE TO ELECTRIC CONNECTIONS MAY BE CARRIED OUT ONLY BY A PERSON WHO IS QUILIFIED IN THE ELECTROTECHNICAL FIELD!
ATTENTION: ANY INTERFERENCE THE SETTING OF THE MACHINE IS POSSIBLE ONLY WITH THE PRODUCER'S CONSENT!

The machine reports a possible problem or evolving errors when:
a) the temperature of the hydraulic oil raises above 55℃
b) number of shifts (cycles) of the multiplier increases above the critical value - 110/min for a time longer than 10 minutes. This is valid at the use of the multiplier with the electric shift of the slide valve.
c) the number of shifts (cycles) of the multiplier decreases under the level of 10/min for a period longer than 20 minutes. This is valid in case of use of the multiplier with the electric shifting of the slide valve.
d) the time for servicing the check valves comes – after each 500 Mth
e) the temperature of the bleed down valve reaches to 40℃
f) the cartridge of the reverse filter ZF is polluted
g) temperature of the left or the right check valve reaches to 40℃
h) the time for the regular change of the oil and the cartridge of the reverse filter ZP comes – first 500 and each other 2000 Mth

The machine stops automatically or does not permit the start when:
a) the level of the hydraulic oil falls under the critical level
b) the temperature of the hydraulic oil grows above the critical level - 60℃
c) the pressure of the input water fall under the critical level – 3.0 bar
d) the number of shifts (cycles) of the multiplier raises above the critical level - 110/min for a time longer than 10 minutes. This is valid at the use of the multiplier with the electric shifting of the slide valve.
e) the temperature protection of the electromotor is overstrained.
f) the number of shifts (circles) of the multiplier falls under the level of 10/min for a period longer than 20 minutes. This is valid at the use of the multiplier with the electric shifting of the slide valve.
g) the temperature of the hydraulic oil is lower than 10℃
In case of any such an emergency stop the failure is indicated by the flashing on the control panel by the red colour and the description on the display. If the machine stops automatically, it is necessary to carry out the same operations as in case of emergency stop.

3.7 Cooling of hydraulic oil.

The machine is equipped with the air cooler of the hydraulic oil with own circulatory pump, the cooler is placed in the machine. The temperature of the hydraulic oil must range at the operation between 45 and 50℃. This temperature may be red from the display of the control system and from the local thermometer placed on the tank (position no. 12 – T2 and L2). The thermometer is visible after you remove the upper folded cover placed on the front part of the machine. The temperature sensor (note 16 – T1) is located on the tank of the hydraulic oil. It switches the ventilator of the cooler on when the temperature reaches to 50℃ and switches it off at the temperature of 45℃.

! It is explicitly prohibited to interfere to the setting of the temperature sensor !

4. SERVICE AND MAINTENANCE

For reliable maintenance and service of the machine it is necessary to set up the service and operation logbook to which the operator of the machine shall record all service and maintenance work and possible failures of the machine, including records on number of worked off machine hours and dates of their realisation. This book shall be delivered by the producer together with the machine. Any interference to the presetting of hydraulic and electric elements so as to the hydraulic and electric connection of the machine is inadmissible without previous consultation with the producer.
The operator of the machine is obliged to keep the entire machine clean and examine regularly its condition at lease once a week. The operator is obliged to record any suspicion of possible origin of failure to the service and operation logbook and report this fact immediately to the producer of the machine. In case this principle is not observed the producer may not provide his guaranty for any faults.

4.1 Electric part.

The electromotor doesn't require any special maintenance. Maintenance and service of all electric parts, devices, and connections it is necessary to observe current valid standards; the basic standard is ČSN EN 60204-1. The correct presetting was carried out and tested by the producer. Any interference to preset values may be carried out only after consultation and agreement of the producer.

Any interference, maintenance, and services may be carried out on the electrical connection and supply only by a person with electro-technical qualification in accordance with the notice number: 50/78 Col., article 6. Services and repairs of the machine may be carried out only with the consent of the producer. With any unauthorized interference to the connection and setting-up of the electric distribution the user looses the guaranty provided by the producer!

4.2 Hydraulic oil part and filtering unit.

Change of hydraulic oil shall be carried out when the machine is not running. Loose the emptying plug on the emptying valve (note 21 – KV2) of the tank of the hydraulic oil (note 19 - N), screw on the ending and hose and drain out the oil by means of the filling filtering device with the pump to the prepared tank (barrel is the best). The change shall be carried out immediately after the work when the oil is still warm and flows well. For change of oil you should have approximately **170 l** of oil prepared. When filling the tank it is necessary to use the filling filtering device fitted with filters with final filtration **min. 5 μm** to prevent penetration of dirt to the tank through the filler. It is necessary to liquidate the old oil ecologically. Fill in the tank with the oil approximately **15mm** under the upper part of the local level indicator.
Change of cartridge of the hydraulic reverse filter (note 15 - ZF) shall be carried out after draining off the hydraulic oil. For this you have to loosen the upper cover of the filter and take off the cover. Then take out the cartridge from the body. After you put in the new filter cartridge, screw on the cover back on it. The reverse filter should be changed always at the change of the hydraulic oil.
Cleaning or change of the suction filter in the hydraulic tank (note 23 - SF) shall be carried out after the all hydraulic oil was drained out. First you have to take out the cover of the right opening (note 18 - PN) and in the bottom part inside the hydraulic tank you will find the filter. Unscrew the suction filter from the suction filler in the tank and wash it in the degreaser and then rinse it by hot pressure water against the direction of flow of oil and leave it to dry properly. Then attach the filter and the cove of the opening back in the tank. The producer recommends replacing the suction filter with the new one at each second change of the hydraulic oil.
It is unconditionally necessary to take off the cover of the opening on the tank before filling the new oil and clean the tank thoroughly from all sediments. Fill the tank with the new oil by means of the filling means with filtration (recommended filter 5 μm) approximately 15 mm under the upper edge of the local level indicator and leave it for a while. Then start the machine without increasing the pressure with open output of the HP water – INCREASE THE PRESSURE OF THE HYDRAULIC OIL BY PRESSING THE BUTTON „1" TO APPROXIMATELY 30 BAR AND LEAVE THE MACHINE RUNNING FOR 10 SECONDS AT LEAST. Proceed according the instruction for the first start of the machine which is stated above in this manual. IF ANY HOSES WERE DISCONNECTED IN THE HYDRAULIC CIRCLE IT IS NECESSARY TO POUR THE HYDRAULIC PUMP WITH OIL DIRECTLY TO ITS DELIVERY AND THE CASE (LEAKAGE) BEFORE STARTING THE MACHINE – THIS SHALL BE CARRIED OUT BY DISCONNECTION OF HOSES AT THE OUTPUT FROM THE PUMP ON THE SIDE FOR CONNECTION TO THE HYDRAULIC DISTRIBUTION BOX AND DISCONECTION OF WASTE HOSES FROM THE PUM AND POURING THE OIL DIRECTLY TO THESE HOSES.
NEVER START THE MACHINE WITHOUT FULL FILLING OF THE HYDRAULIC OIL !
The change of the hydraulic oil should be carried out:
- after the first 500 Mth – change the hydraulic oil and the reverse filter fitted on the oil tank and clean the tank.

- after another 2000 Mth – change the oil and the reverse filter and clean the tank.
- after each another 2000 Mth – change the oil and the reverse filter regularly.
Clean the suction filter after each change of oil. Change the suction filter regularly after each 4000 Mth.
- it is recommended to carry out (at the change of oil) washing of the entire hydraulic system by means of external filtering unit (this service should be carried out by a professional firm specialized in this field) and cleaning of the entire hydraulic tank from caught dirt.
In case of leakage of the oil through leaky connections you may only refill the oil. In case of suspicion of bad quality or pollution of the oil it is necessary to consult it immediately with the supplier or producer of the oil and send its sample to the producer or some other tribo-technical laboratory for analysis. This will not prevent possible failures or damage of the machine.
Maintenance of individual parts according the hydraulic scheme PTV-011-D-0173:
Suction filter (note 23-SF): at the change of oil it is always necessary to clean (wash) it against the direction of flow of the oil. Replace it in case of any damage. Replacements shall be carried our regularly after each 4000 Mth.
Slide valve (note 22-SO1): does not require any special maintenance.
Pump holder (note 2-D): does not require any special maintenance.
Coupling (note 3-S): in case of change of oil check visually the state of the plastic element. Replace it in case of considerable wear or damage. At this operation it is necessary to disconnect the hydraulic pump from the holder and connected hoses, take it out, replace the plastic element of the coupling and put the pump back and reconnect hoses. It is necessary to fill in the pump with the hydraulic oil to the box before the first start. Further you should proceed in accordance with point 3.1.
Hydraulic pump (note 4-P): does not require any special maintenance. **Any interference, repair, or setting-up of this valve may be carried out only by a professional engineer of the producer of the machine – report any failure to the producer.**
Proportional valve (note 46-PPV): do not require any special maintenance. **Any interference, repair, or setting-up of this valve may be carried out only by a professional engineer of the producer of the machine – report any failure to the producer.**
Electromagnetic valve (note 45-EV): does not require any special maintenance. **Any interference, repair, or setting-up of this valve may be carried out only by a professional engineer of the producer of the machine – report any failure to the producer.**
Hydraulic distribution boxes (notes 5 and 42-HK1 and HK2): do not require any special maintenance.
Panel of the reverse valve (note 7-DZV): do not require any special maintenance.
Safety valve (note 9-PV): does not require any special maintenance. **Any interference, repair, or setting-up of this valve may be carried out only by a professional engineer of the producer of the machine – report any failure to the producer.**
Electromagnetic distributor (note 6-DEV): does not require any special maintenance. **Any interference, repair, or setting-up of this valve may be carried out only by a professional engineer of the producer of the machine – report any failure to the producer.**
Panel of butterfly valves (note 8-DSV): does not require any special maintenance. **Any interference, repair, or setting-up of these valves may be carried out only by a professional engineer of the producer of the machine – report any failure to the producer.**
Butterfly valve (note 43-SV): does not require any special maintenance. Operator of the machine is allowed to set this valve so that the vibration of the manometer on the control panel is not too quick. Replace in case of malfunction.
Manometer (note 11-MP2): does not require any special maintenance. Replace in case of failure.
Bleed down valve (note 27-BD): maintenance and service are described in the separate (high-pressure) part of this chapter.
Multiplier (note 25-VTM): maintenance and service are described in the separate (high-pressure) part of this chapter.
Cooler of the hydraulic oil (note 24-CH): it is necessary to carry our visual examination of the cooler of any leakage of oil to the surrounding environment. In case of detected leakage or in case of failure stop the machine and inform the producer. After each 2000 Mth clean the body of the cooler from outside by the pressure air. Polluted lamellas of the body of the cooler may cause overheating of the hydraulic oil.

! ATTENTION: The cooler of the hydraulic oil is fitted with the ventilator which is powered by the motor with operating voltage 380V; any interference to the connection of electricity may be carried out only by producer of the machine or an electric professional trained according valid regulations and codes !

Reverser filter (note 15-ZF): change the cartridge of the filter at each change of hydraulic oil.
Filler (note 13-NZ): does not require any special maintenance. In case of change of oil check the sieve and in case of its damage replace it already before you pour the oil filling thought it. In case of excessive pollution, demount the sieve and clean it.
HO tank (note 19-N): at each change of oil you should clean the tank through the openings. In case the oil is leaking from the tank, stop the machine immediately, drain the oil out and contact the producer.
Openings (note 18-PN): do not require any special maintenance. Replace in change of damage.

Ball valves (notes 21 and 22-KV1 and KV2): do not require any special maintenance. Replace in case of damage.
Temperature sensor (note 16-T1): does not require any special maintenance. Replace in case of damage or malfunction.
Level switch (note 17-L1): does not require any special maintenance. Replace in case of damage or malfunction.
HP equalizing cylinder (note 48-VTV): it is maintenance-free, in case of any damage or leakage of HP water stop the machine immediately, and contact the producer. It is PROHIBITED to interfere to the cylinder in any way or to dismantle it without an explicit permission of the producer of the machine!
Filtering unit (note 30-FJ):
Manometers (note 32-MP1): do not require any special maintenance. Replace in case of malfunction.
Ball valves (note 31-KV3): does not require any special maintenance. Replace in case of malfunction.
Filters (note 33 and 34; FV1 and FV0,5): check regularly the difference of pressures in front of the filtration and behind it. It must not exceed 1.2 bar when the machine is running. Before you replace the filter you must remove the front cover of the machine and loosen covers of filters to the left. Before you loose covers of filters you have to close the water supply by means of the closing valve. Take old cartridges our and put new ones and tighten elements again. Replace both cartridges ones a year at least.
Pressure sensor (note 35-P2): does not require any special maintenance. Replace in case of malfunction. Any interference or setting-up may be carried out only by a professional engineer of the producer.
Increasing the input water pressure (note 39-TVV) – shall be delivered on request:
Water pump (note 37-PV): does not require any special maintenance – it must not run dry, without input water!
Electromotor (note 38-M3): does not require any special maintenance.
Pressure switch (note 39-P3): does not require any special maintenance. It is preset by the producer to the switch on pressure of approximately 3.7bar and the switch off pressure of approximately 4.5 bar. Do not change the setting-up!
Pressure tank with bag and the little pressure valve (notes 40 and 41-TN a TV): check the pressure in the tank once a year; it must be approximately 1.2bar. You may refill the air pressure by means of standard tyre inflator and check it by means of standard manometer.
Electromagnetic valve (note 47-EV): does not require any special maintenance. Replace in case of malfunction.

4.3 High-pressure part.

! ANY UNPROFESSIONAL INTERVENTIONS TO THE HIGH-PRESSURE ARE QUITE INADMISSIBLE SO AS UTILIZATION OF OTHER THAN ORIGINAL SPARE PARTS !

All faults and interventions to the HP water parts have to be consulted with the producer of the machine. Description of the maintenance of the multiplier, HP closing valves, and bleed down valve is stated below.
Compensatory HP cylinder is maintenance-free and in case of its failure it is necessary to stop the machine IMMEDIATELY and contact the producer.

! IT IS EXPLICITLY PROHIBITED TO DISMANTLE THIS CYLINDER AND TO INTERFERE IN IT IN ANY WAY !

Service and maintenance of the multiplier: described below.
Service and maintenance of the bleed down valve: described below.
Service and maintenance of HP valves in the HP distribution: when the water flows thought the head of the valve, stop the machine and release the pressure of the HP pipe, dismantle the head of the valve and change seals or the closing needle according need.
Service and maintenance of HP distribution: check (minimally once in three months) tightening of all joints of the HP pipes. In case of burst of HP pipe stop the machine immediately and change the relevant part.

Safety recommendations

Before the initiation of the system make sure that all persons are in the sufficient distance from the machine.

All safety protections or covers must be permanently on its place on the machine.

First aid means have to be always available in suitable places in the workplace. Their placement must be known to all persons in the workplace.

According the established practice we recommend using safe footwear, glasses, and anti-noise protection of ears.

We do not recommend wearing rings, watches, or chains while operating or maintaining the machine during its operation.

Mechanical systems

Any leakages of liquids from pipes, couplings, or screw joints must be immediately removed.

Do not carry out any maintenance or cleaning of the machine while it is running.

The system remains under pressure for longer time even after the source of the high-pressure water is switched off.

Before repair of any part of the system make sure that the pressure in the system was released.

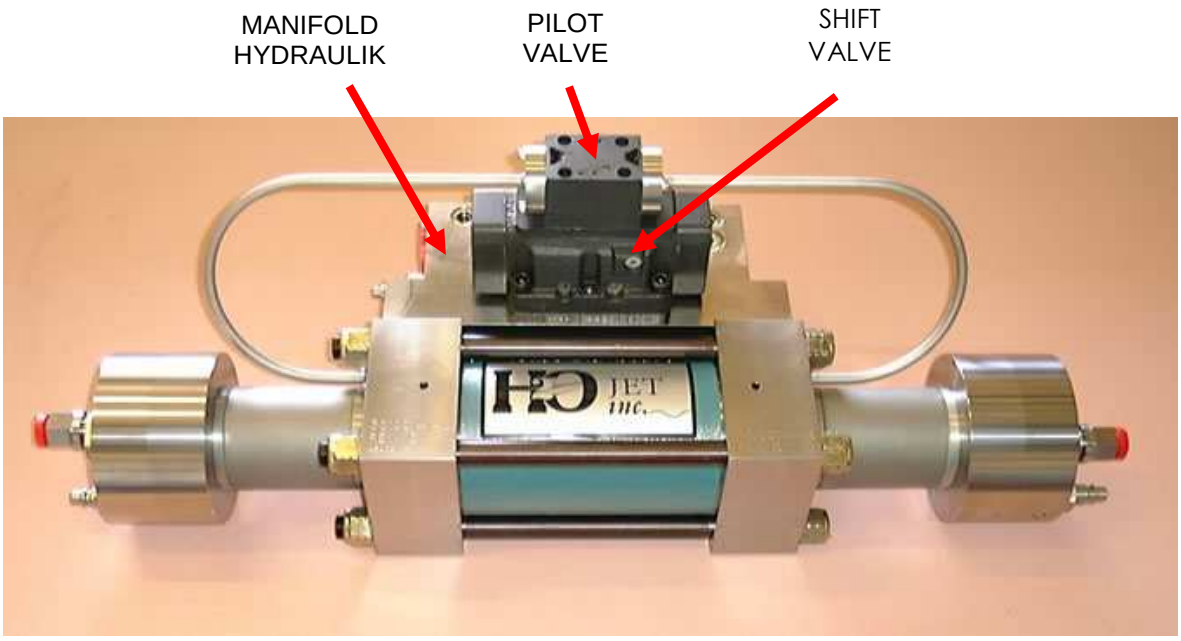
In case you are going to make change on the high-pressure water system or you add anything to it, use only high-pressure pipes, couplings or valves certified to 60 000 PSI (4130 bar).

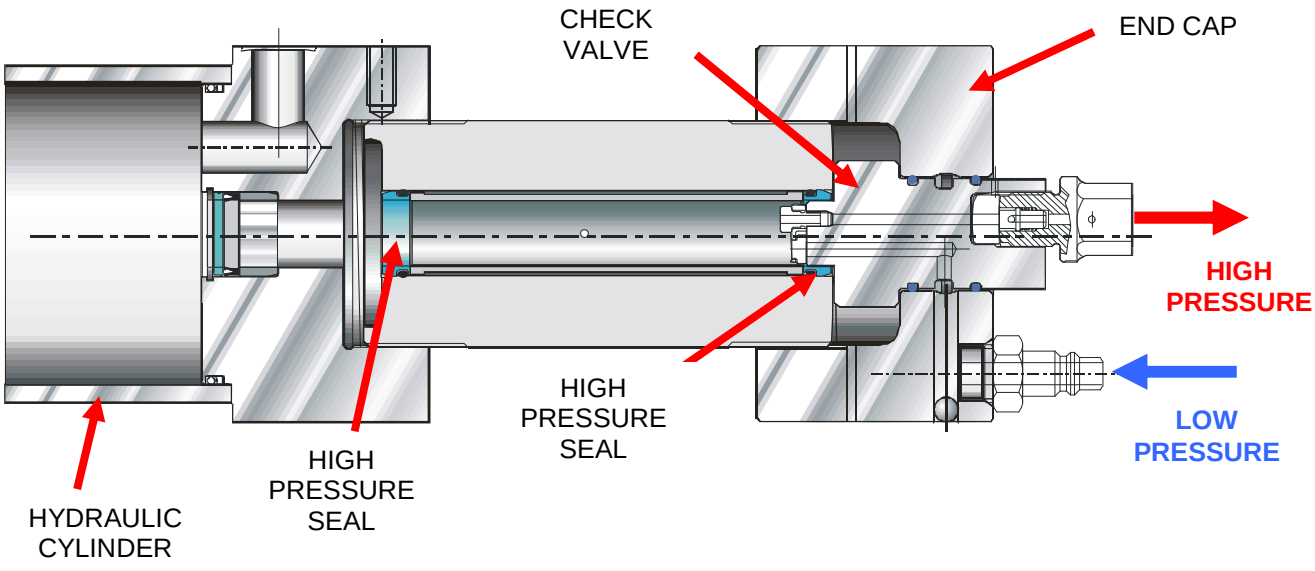
Do not exceed the determined operating pressure of the water system.

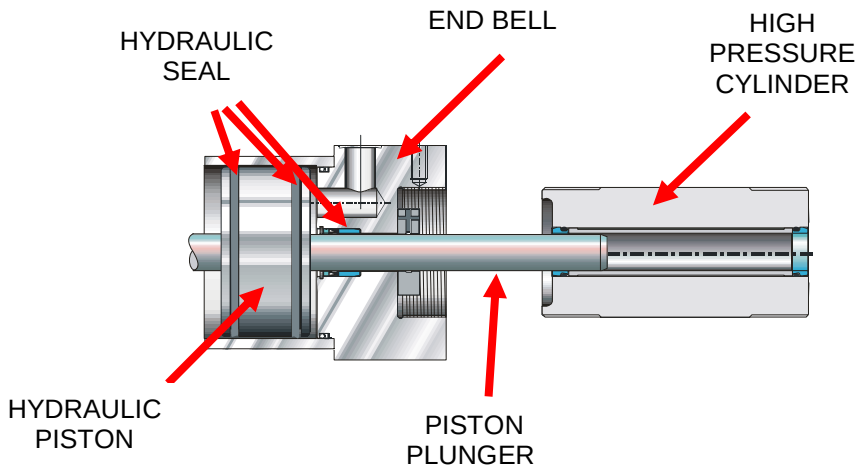
Observe recommendation of the producer for maintenance of the machine and in case of replacement of parts use always parts of the producer.

Description of multiplier

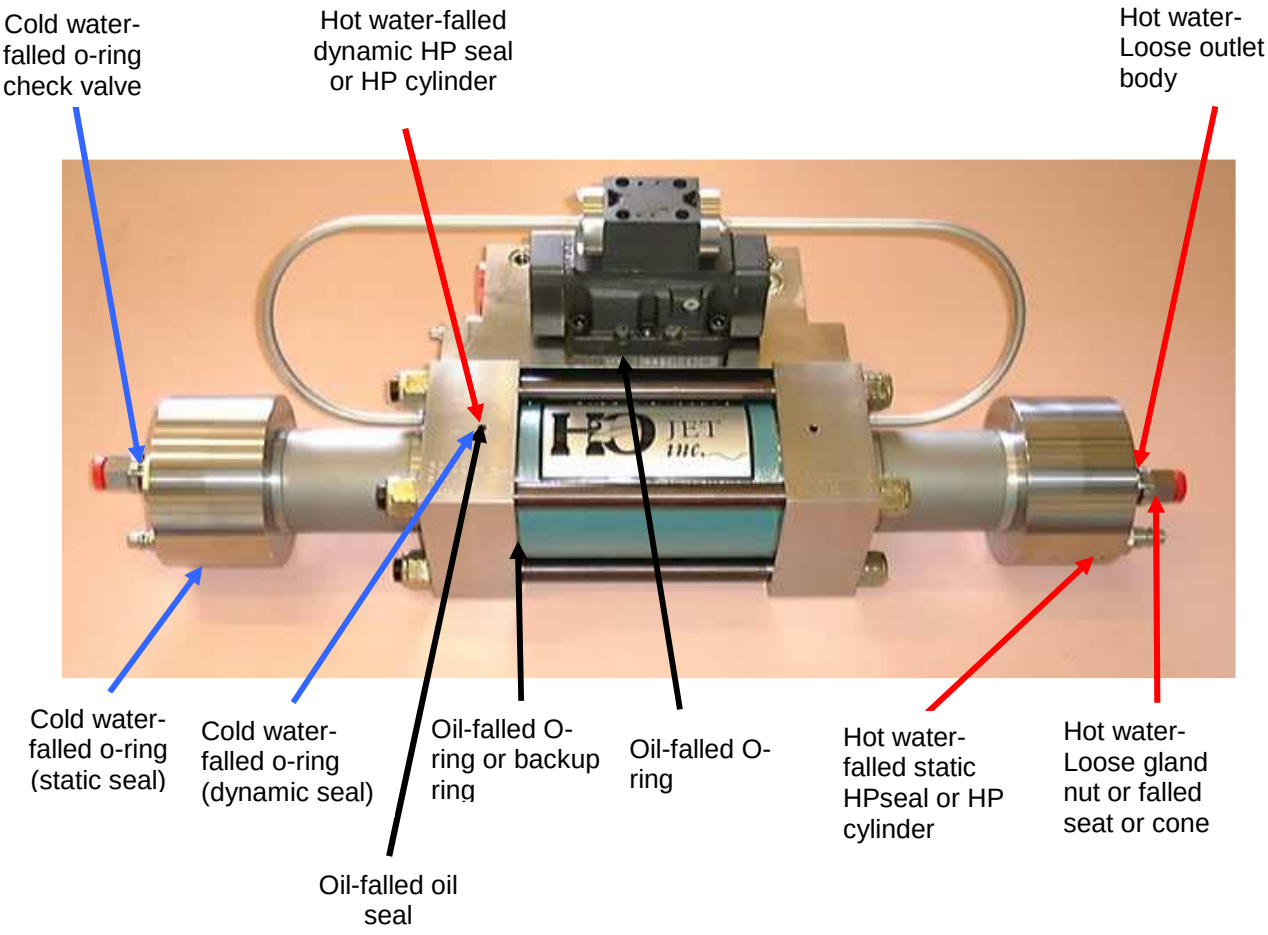
High-pressure intensifier consists of double-acting hydraulic cylinder, transfer valve assembly, operating valve block and two high-pressure cylinders of water which are equipped with input and output closing flaps.







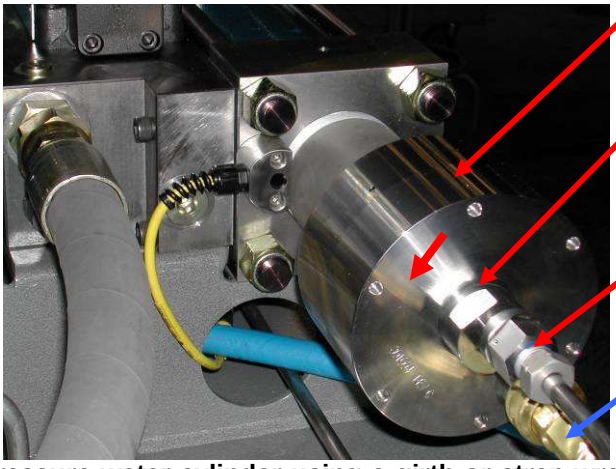
Intensifier troubleshooting guide



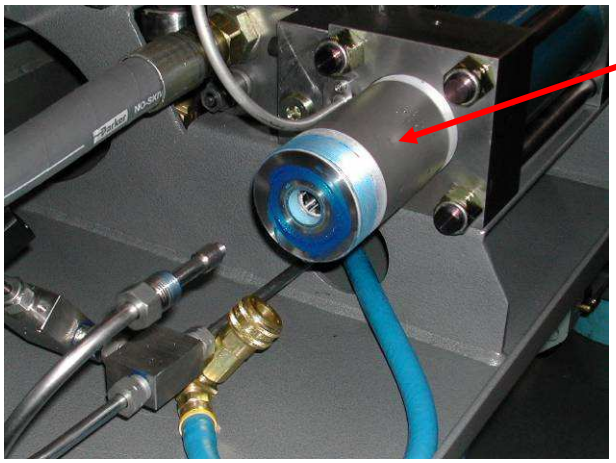
High-Pressure Water Cylinder and Seals Service

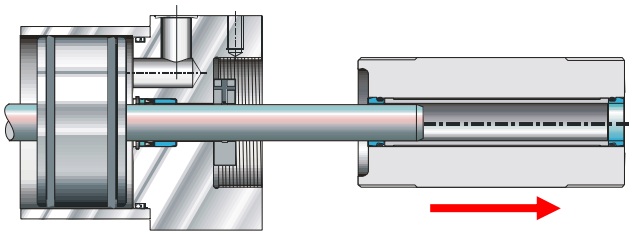
The intensifier itself requires no periodic maintenance. However, when the seals leak excessively or a stress fracture is evident, the intesifier must be repaired immediately. If only the high-pressure water cylinder and related components are being serviced, we recommed that the intesifier remain mounted in the pump frame. Be sure to keep the work area cleen and free of airborne fiber and dust.

1. Disconnect the supply water hose at the quick-disconnects and, using two wrenches, disconnect the high-pressure water line from the check valve outlet bodies. Loosen the other end of the tubing and move the tubing out of the way.Note the orientation of the main check valve body, then loosen the end cap by tapping the spanner wrench with a mallet. If required, remove the cable guides, then restrain the high-pressure water cylinder with a strap or girth wrench.Unscrew the end cap from the cylinder by hand, ensuring that the check valve assembly rotates with the end cap as an assembly



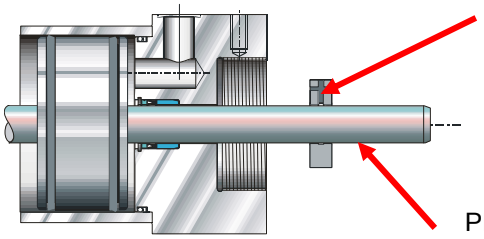
2. Loosen the high-pressure water cylinder using a girth or strap wrench. Remove the cylinder by hand, being especially careful when sliding the cylinder off the plunger to avoid scoring the plunger or the seal backup sleeve located inside the cylinder.



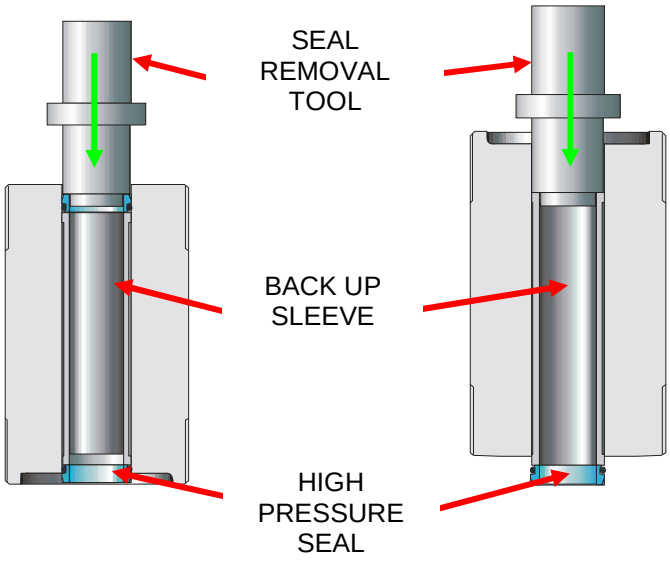


3. Using the seal removal tool, remove the high-pressure seals from the cylinders. Check the seals for flaws and replace as needed. Remove the opposite end cap and cylinder. Lightly tap the plunger with a mallet to expose the plunger. Remove the seal ring and discard. The backup ring is a wear item and is replaced at every seal change. Clean and inspect the plunger. The surface must have a mirror-like finish with no scratches, pits, nicks, corrosion, or other surface disruptions.

SEAL
BUTTRESS



4. **Place the cylinder on a clean, dry shop towel. Place the shouldered end of the seal installation tool into the end of the cylinder and tap with a mallet to remove the seal located in the opposite end. Repeat this step for the seal in the other end of the cylinder. Remove the seal backup sleeve. Clean the high-pressure water cylinder and backup sleeve. Be careful not to scratch, nick, or otherwise damage either part during the cleaning process. Inspect all surfaces of the cylinder for any signs of damage or fatigue. Inspect the inner diameter by looking through the cylinder angled toward a brightly lit white surface. If there is any damage to the interiér surface, the cylinder must by discarded and replaced. Inspect the sealing surfaces of the high-pressure cylinder for roughness or damage. Minor build-up or erosion can be polished out if done evenly. No ncks or scratches are permitted in these areas. Inspect the seal backup sleeve for any signs of damage. The sleeve is a nonstressed part.**

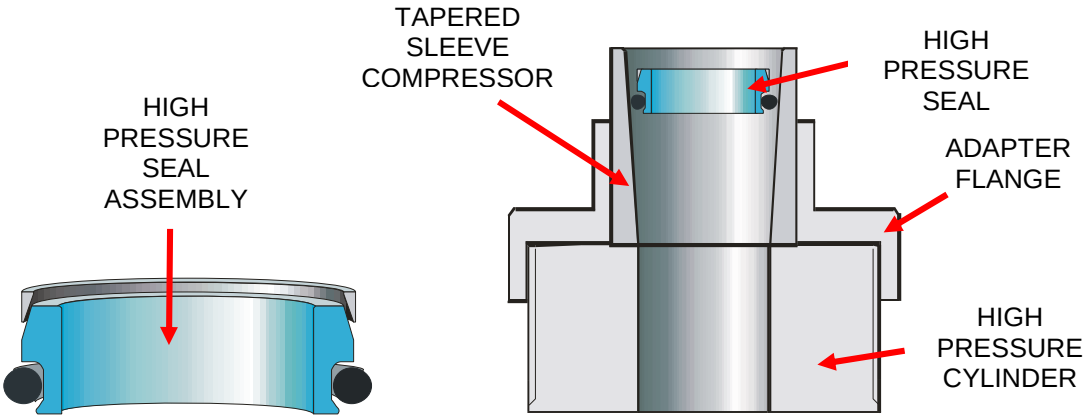


5. **Lubricate each end of the cylinder inner surface with White Food Grade O-Ring Lube. Apply Blue Lubricant evenly to the threads on the cylinder and thread the sleeve guide onto the nonrecessed end of the clinder.**

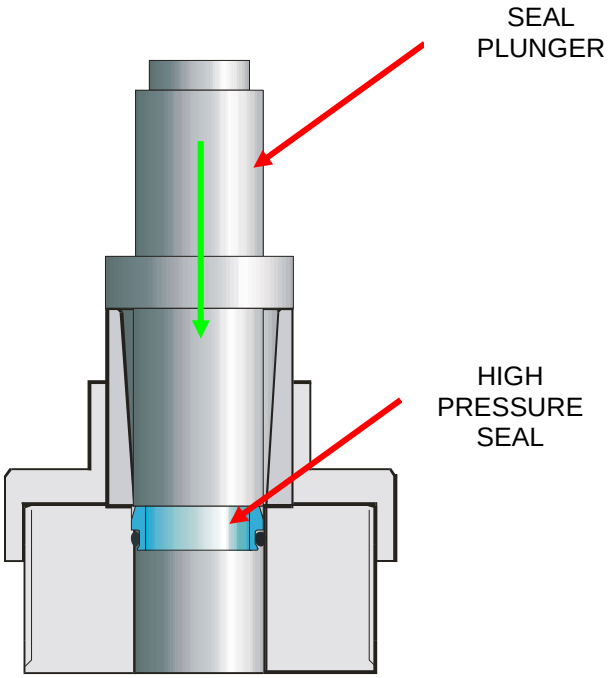
6. **The reassembly of the intensifier begins at this point. The area around the intensifier must be free or surface or airborne grit. Having clean hands, tools, shop coat, solvent and work surface all contribute to reduced downtime and lower maintenance costs. Remember that when instructed to clean parts, it is necessary to first wipe all components with a rag to remove most of the Blue Lubricant and debris. Next, clean all components in an approved solvent, and air dry.**



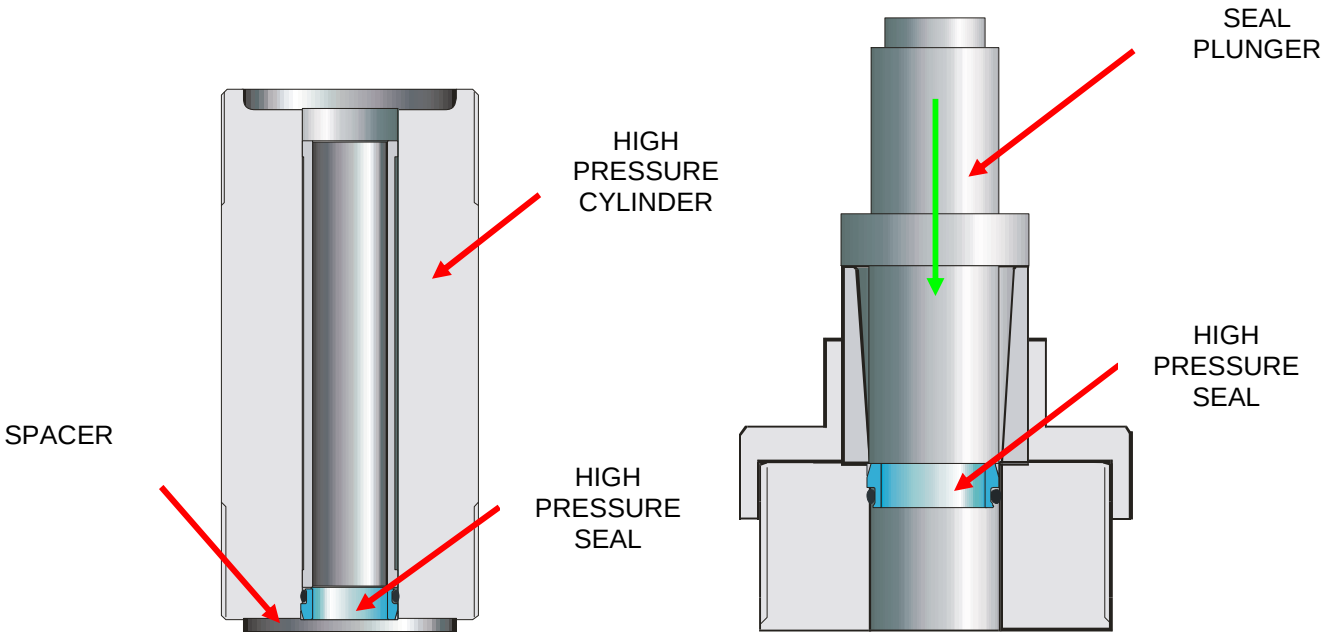
7. **Wipe the inner surface of the seal installation sleeve with White Food Grade O-Ring Lube and place it into the sleeve guide, small ID first. Lubricate the seal O-ring with White Food Grade O-Ring Lube and place it in the groove on the seal. Set the mental hoop aside as it is not installed through the sleeve. Lubricate the high-pressure seal and the O-Ring with White Food Grade O-Ring Lube and then place the seal (o-ring end first) into the sleeve.**



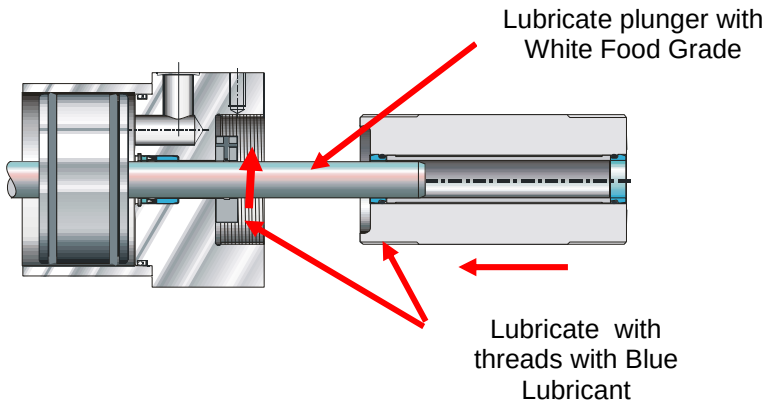
8. **Insert the non-shouldered end of the seal installation tool into the sleeve and press the seal into the cylinder. Remove the seal installation tool, seal installation sleeve and sleeve guide form the cylinder.**



9. **Lubricate the hoop, inside and out, with White Food Grade O-Ring Lube and press the hoop, sharp edge first, into the cylinder using thumbs. Additional lubricant may be used to hold the hoop in positionIf the hoop is too large to fit in by hand, reassemble the sleeve guide and sleeve onto the cylinder, place the hoop in the sleeve, and insert the hoop by tapping the seal installation tool lightly with a mallet. Invert the cylinder so that the flat end is on the aluminium disk. Lubricate each end of the backup sleeve with White Food Grade O-Ring Lube and insert the sleeve into the cylinder. Take care not to damage parts or dislodge the seal and hoop. Place the aluminium disk on the non-recessed end of the cylinder. Thread the sleeve guide over it onto the cylinder. The disk prevents the seals, hoops and sleeve from being dislodged as the cylinder is being installed.**



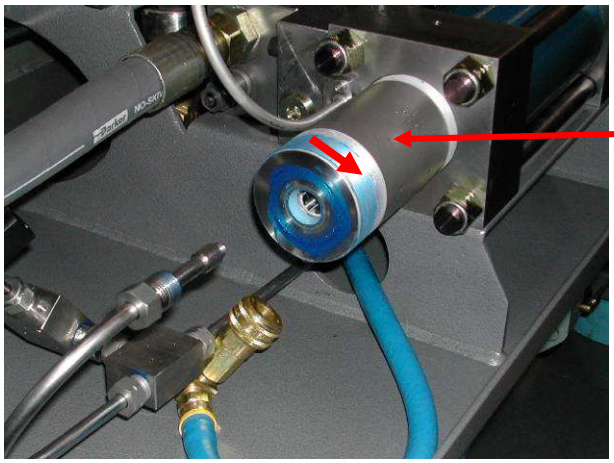
10. **Lubricate both sides of the new seal ring with Blue Lubricant. Slide the new seal ring onto the plunger. The seal ring has a tight fit but should slide along the plunger with only slight, intermittent resistance. If the seal ring is in satisfactory condition, leave it on the plunger. The seal ring is a wear item which must be replaced at every seal change.Coat the pluger evenly with White Food Grade O-Ring Lube and slide the seal ring onto the plunger.Coat the threads of the end bell with Blue Lubricant. Carefully slide the cylinder onto the plunger until it can be threaded into the end bell. If the plunger has been fully retracted, the plunger will not contact the seal until the cylinder has been partially threaded onto the end bell. This will make cylinder installation easier. Thread the cylinder on until hand tight. The cylinder will thread on easily and stop firmly. If the cylinder can be turned and additional one-eight revolution on more, the hoop may have become dislodged. Remove the cylinder and check for problems.**



11. Tighten the cylinder using a strap wrench. The strap wrench is torque limiting – it will begin to slip after reaching a certain torque.

THE STRAP WRENCH





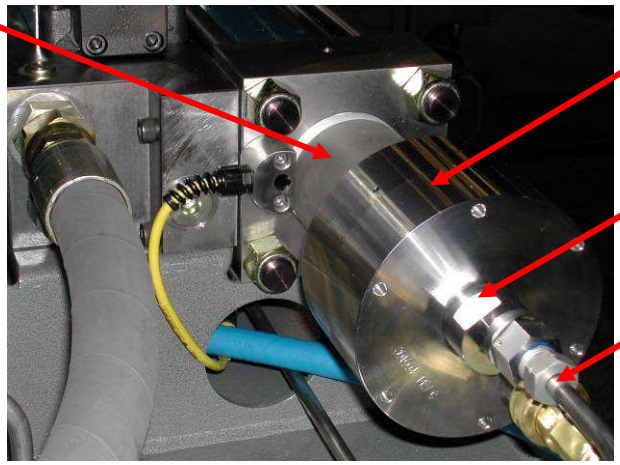
TIGHTEN THE CYLINDER USING A STRAP WRENCH

12. Remove the seal guide and aluminium disk from the cylinder and thread the end cap onto the cylinder. Ensure that the check valve assembly rotates with the end cap as an assembly. Check that the check valve fitting, which is drilled odd center, will be oriented as originally installed. Back off the end cap one-half turn if the check valve must be rotated to avoid damaging the seal and o-rings, then retighten. Installation of the heigh-pressure tubing is easier if the check valve is in the original position. Torque the end cap by tapping on the spanner wrench with a mallet or 8 oz. hammer. Using two wrenches, connect the high-pressure tubing at both ends and connect the supply water horse at the quick disconnect fitting. Remove tools, parts, and rags from around the pump. Make sure the covers are in place and closed. Turn the pressure control knob fully counter-clockwise to the lowest pressure setting and start the pump.

HIGH PRESSURE CYLINDER

SPANNER WRENCH INTENSIFIER END CAP





END CAP

CHECK VALVE

CONNECT THE HIGH-PRESSURE TUBING

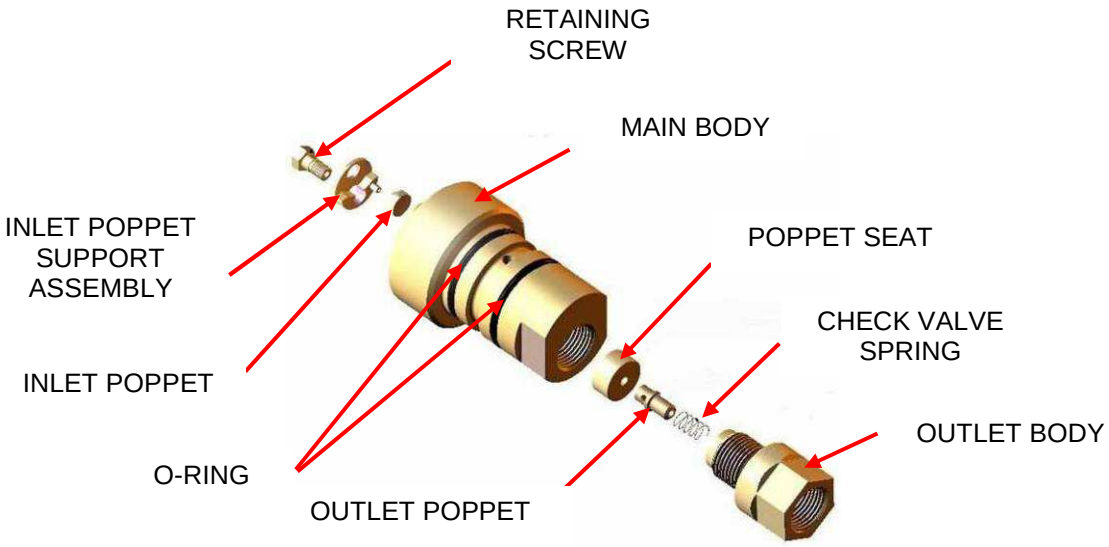
End Cap and Check Valve Service

The check valves are contained in a two-piece unit in the end cap. The components of the check valve are shown below. When the check valves require service, one or more of the following conditions are usually present.

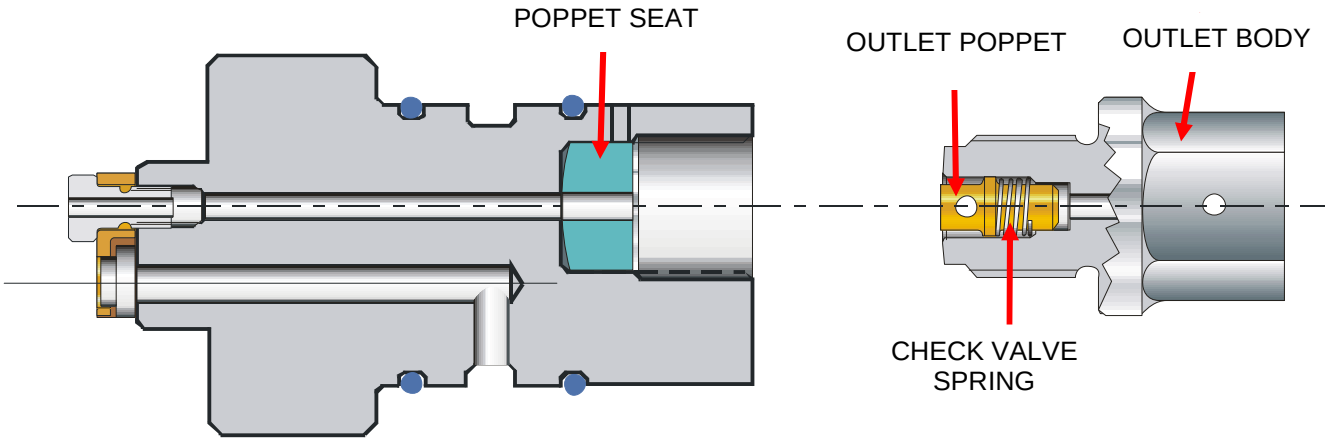
a) High-pressure water temperature at the outlet fitting exceeds 120℉ (49℃) indicating excessive back-flow through the outlet check valve.

b) Hydraulic piston slams to the end of travel indicating excessive back-flow through the inlet check valve.

c) Repetitive spiking of the high-pressure water pressure indicating that one or both of the valves may be leaking excessively.



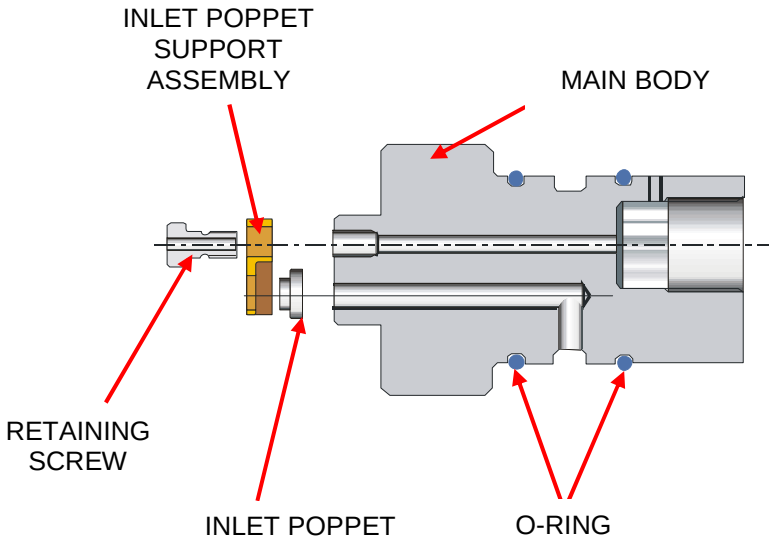
1. Shut down the systém. Disconnect the supply water hose at the quick-disconnect fitting and, using two wrenches, disconnect the high-pressure water line from the check valve outlet body. Loosen the other end of the tubing and move the tubing out of the way. Using two wrenches remove the outlet body. The outlet body contains the outlet poppet and a spring. Remove the insert from the main body. Note that the lapped surface is visible when the insert is in place.



2. Note the orientation of the main check valve body, then loosen the end cap by tapping the spanner wrench with a mallet. If required, remove the cable guides, then restrain the high-pressure water cylinder with a strap or girth wrench. Unscrew the end cap from the cylinder by hand, ensuring that the check valve assembly rotates with the end cap as an asembly. Failure to rotate the check valve assembly with the end cap may damage the o-rings located on the check valve body. Use a wrench if required to ensure that the check valve rotates with the end cap. Check the corners of the wrench flats on the main body. All burrs must be filed off to prevent scratching the end cap bore during removal. Using your thumbs, press the check valve main body out through the end cap. The only resistance you should feel is when the o-rings on the main body contact the inner bore of the end cap.

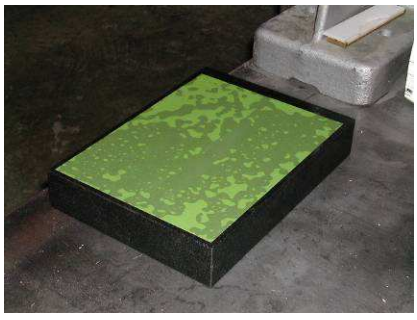


3. **Remove the retainer screw. Remove the support assembly and inlet poppet. Clean all parts with fresh, clean, non-residue type solvent and air dry. Clean each part separately to prevent damaging parts. Inspect the o-rings located on the main body and replace if required. If the o-rings show signs of damage, check tha passages in the main body for debris. Inspect the sealing surfaces on the main body for erosion, cracks, and any other damage. Replace the main body if required. Inspect the support assembly and inlet poppet. Do not reuse the support assembly if the poppet has caused extensive wear to the mating face of the support.**

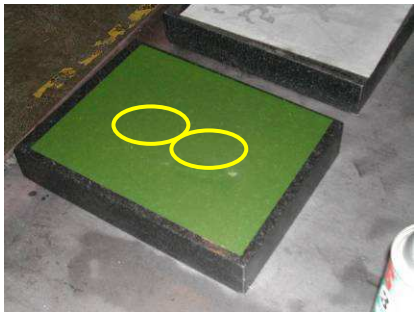


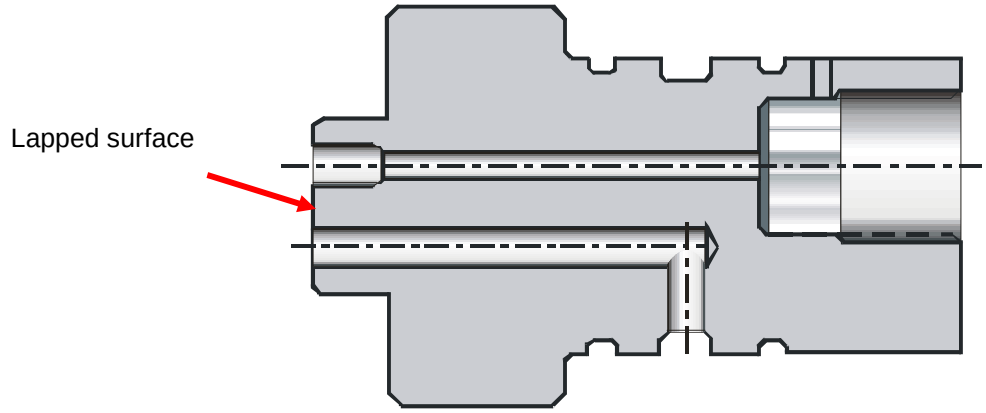
4. **Use the check valve lapping kit and the following procedure to lap the sealing surface:**

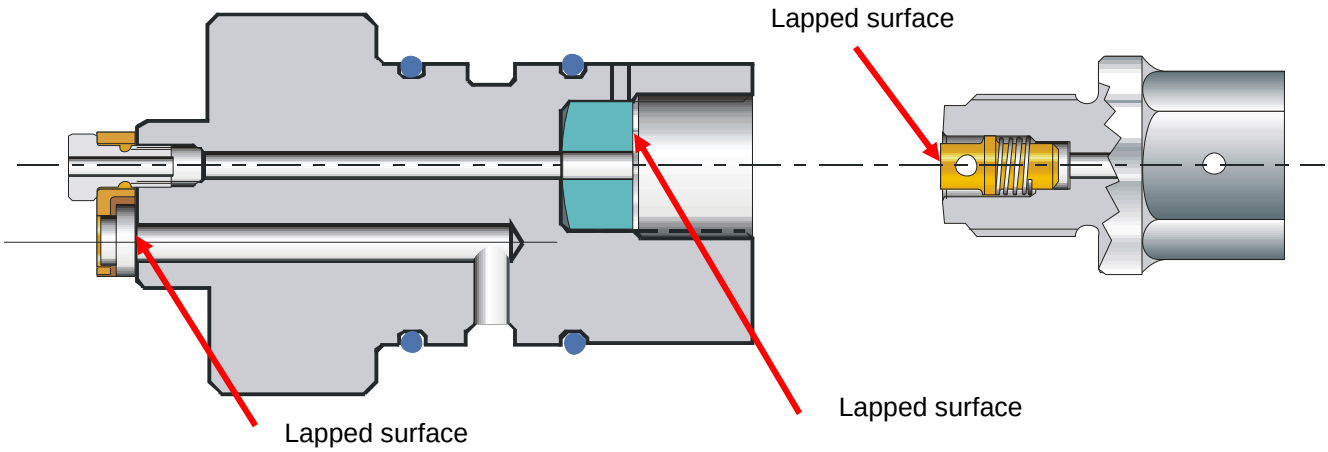
a) **Remove the glass and one piece each of the 320 and 600 grit adhesive back abrasive strip. Attach one of the abrasive strips to each side of the glass plate.**

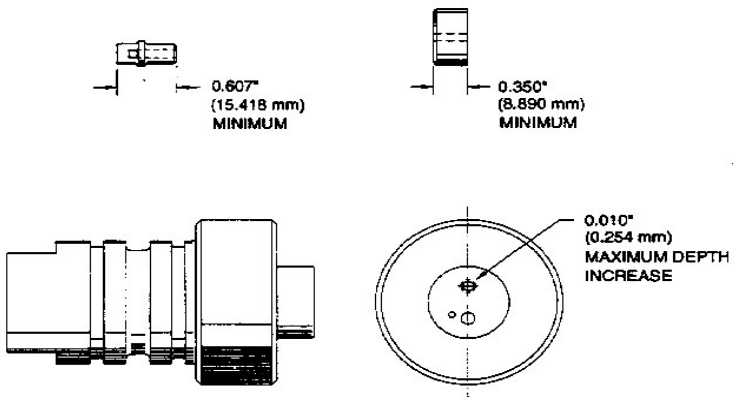


b) **Place a few drops of water on the 320 grit abrasive strip and lap the surface of the part using a figure-eight pattern. Turn the glass plate over to the 600 grit side. Place a few drops of water on the 600 grit abrasive strip and lap the surface of the part using a figure-eight pattern. Lap until an even surface is attained.**

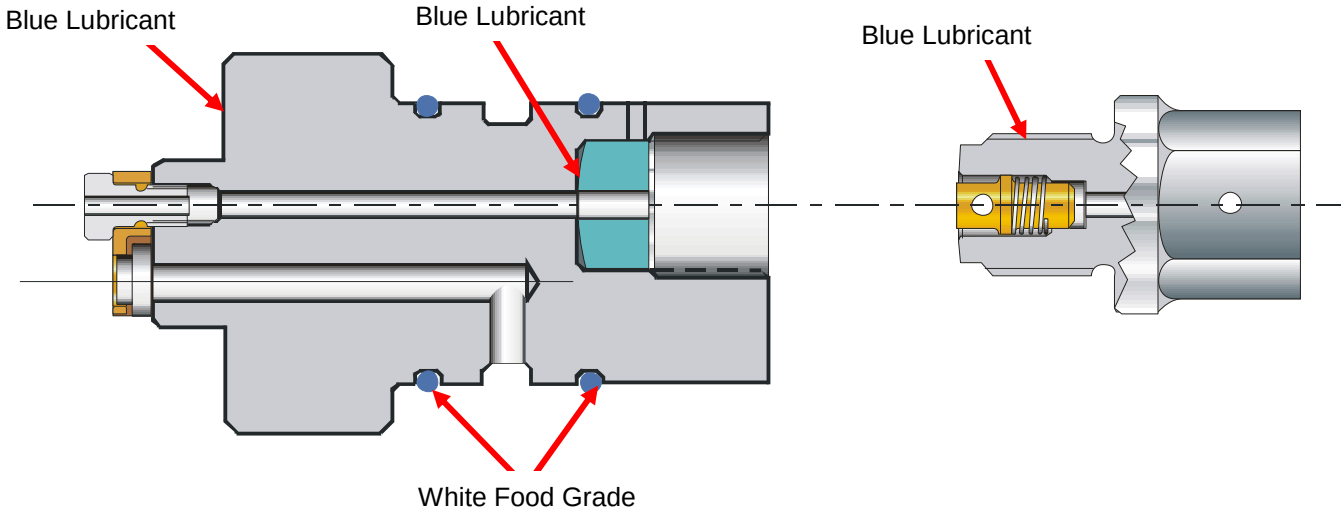



Lapped surface


Lapped surface
Lapped surface
Lapped surface

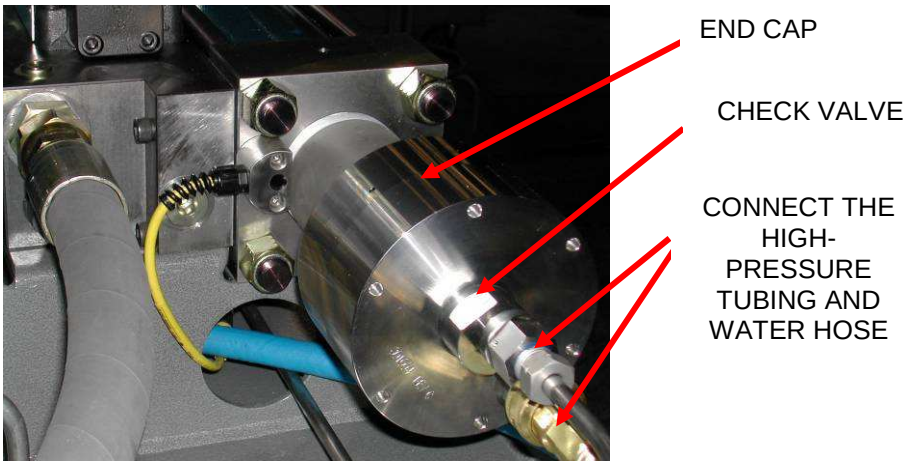

0.607"
(15.418 mm)
MINIMUM
0.350"
(8.890 mm)
MINIMUM
0.010"
(0.254 mm)
MAXIMUM DEPTH
INCREASE

5. **Torque the end cap by tapping on the spanner wrench with a mallet or 8 oz. hammer. The torque value must not exceed 55 ft-lbs (75 N-m). Apply Blue Lubricant to the radiused face of the insert and place the insert, beveled face first, into the check valve main body.Apply Blue Lubricant to the check valve outlet body face and threads. Insert the spring and poppet, small diameter of poppet first, into the outlet body as shown above. Clean all newly-lapped parts with clean solvent such as isopropyl alcohol or liquid Freon. Dry and reinspect the parts. Assemble the inlet poppet support assembly and retainer screw and apply Loctite to the screw threads. Position the support assembly on the main body with the inlet poppet in place and the roll pin in its locating hole. Tighten the retainer screw to a torque valve of 40 in-lbs (4.52 N-m). Clean and inspect all surfaces within the end cap. Check for thread damage which may prevent the end cap from becoming fully seated on the cylinder. Lubricate the o-rings on the check valve main body with White Food Grade O-Ring Lube. Apply Blue Lubricant to the check valve main body faces that contact the end cap and the high-pressure cylinder. Also, apply Blue Lubricant to the threads of the end cap.**



6. **Use your thumbs to press the check valve into the end cap.**

7. **Thread the end cap onto the cylinder. Ensure that the check valve assembly rotates with the end cap as an assembly. Orient the check valve as originally installed. If the check valve must be rotated, back off the end cap one-half turn to avoid damaging the o-rings and high-pressure seals. Using two wrenches, connect the high-pressure tubing, and connect the supply water hose at the quick disconnect fitting**



8. **Remove tools, parts, and rags from around the pump. Close the pump covers. Turn the pressure control knob fully counterclockwise to the lowest pressure setting and start the pump. Operate the pump at idle oil pressure for five minutes while checking for leaks and listening for any unusual sounds.**

Hydraulic Oil Cylinder, Piston, Plunger, and End Bell Service

The double-acting hydraulic oil cylinder is located between two high-pressure water cyliders. When the cylinder and end bells require service, one or more of the following conditions are usually present.

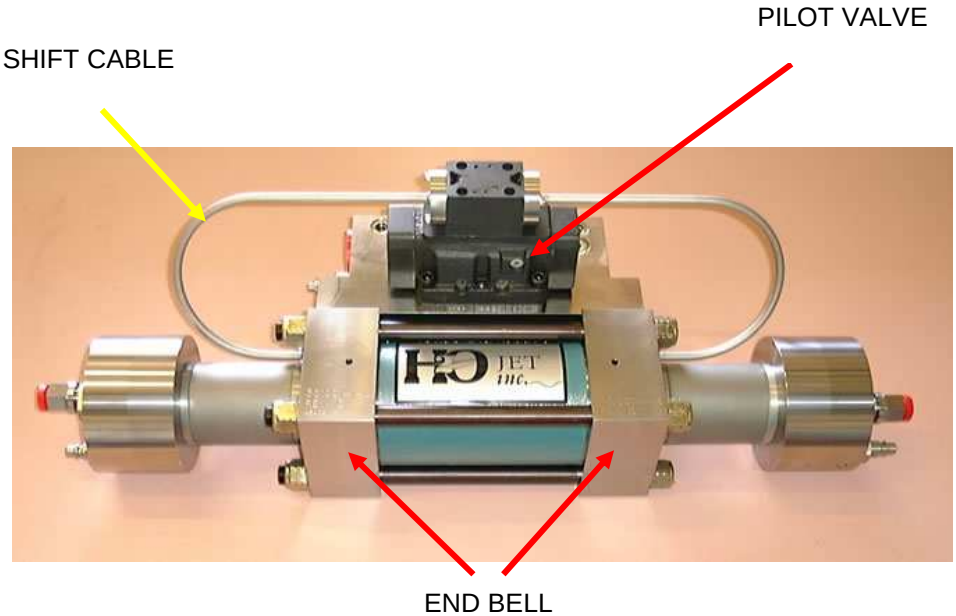
a) Excessive oil leakage from the end bell weep hole or the joint between the cylinder and end bell
b) Worm piston seals aloowing excessive oil flow athe piston
c) A jammed piston or detached plunger

1. Disassembly of the hydraulic oil cylinder, end bells and piston involves the removal of the intensifier from the cabinet and a complete teardown of the intensifier on a workbench. An experienced technician using the proper tools and procedures can usually complete the procedure in 2 ½ to 3 hours. The intensifier assembly fixture is required for this procedure. The fixture is used during the reassembly process to align the end bells square with each other and to minimize the possibility of O-ring damage as the cylinder is being drawn onto the end bells. When the end bells are assembled out of alignment, oil manifold leakage and breakage may occur.

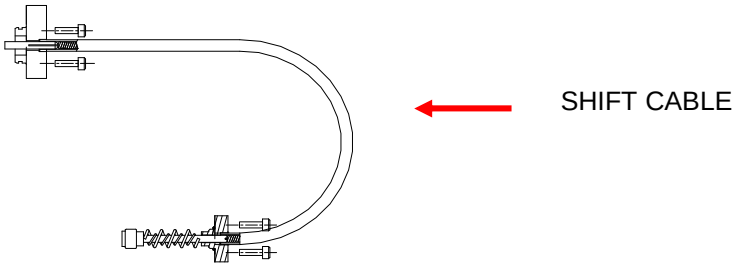


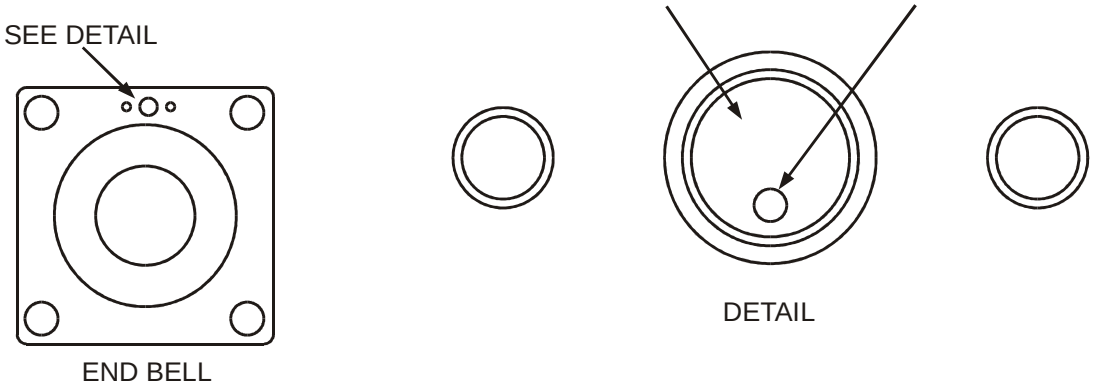
2. Shut down the system. Disconnect the supply water hose at the quick-disconnect fittings. Using two wrenches, disconnect the high-pressure water line from the check valve outlet body. Loosen the other end of the tubing and move it out of the way

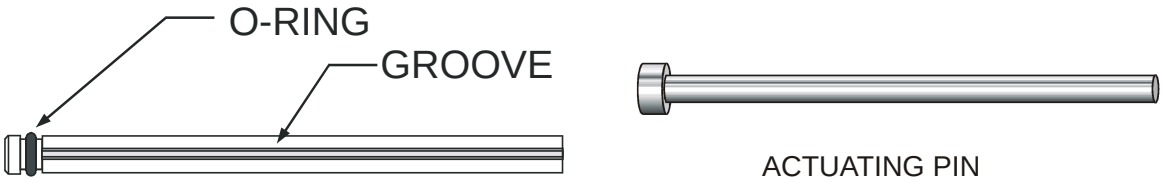
3. Remove the four allen screws securing the shift cable housing to the pilot valve and end bell.



4. **Remove the shift cables and actuating pins from the intensifier and set aside. Use the Shift Pin Tool to remove the shift pins. The shift cables can easily damaged and are removed for protection only.**





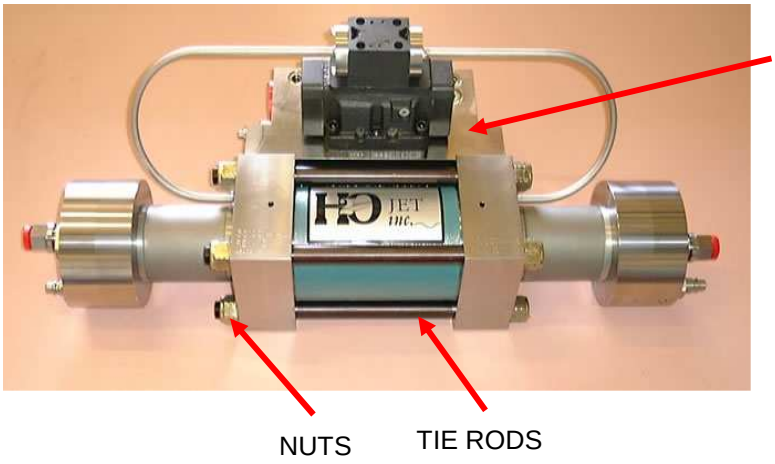


5. **Remove the oil supply and drain lines at the manifold elbows, and plug to minimize oil loss. Approximately one quart of oil may be lost when the above parts are loosened and removed.**

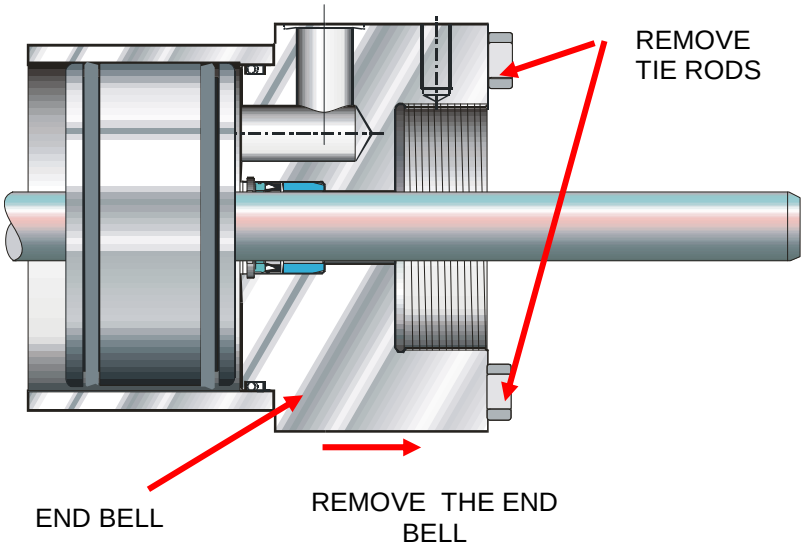
6. **Remove the mounting cap screws securing the intensifier to the manifold. The intesifier must be free to be removed from the cabinet.**

7. **Loosen the bolts and remove the intesifier from the manifold. Drain oil from the intesifier ports.**

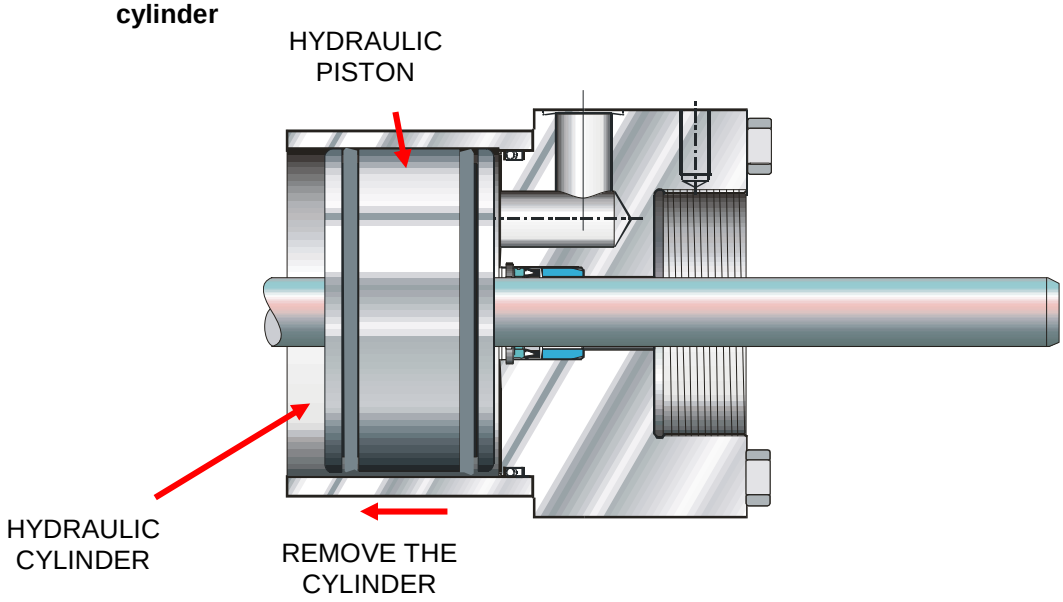
8. **Mount the intesifier assembly fixture to the workbench and place the intesifier on the assembly fixture with the manifold surface facing down. Screw the intesifier to the fixture using the provided screws. Remove the torque from the cylinder tie rods evenly using two wrenches (do not use a torque wrench) in a figure-eight pattern. Remove the nuts and tie rods.**



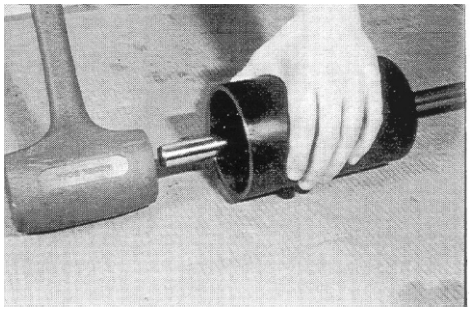
9. Remove the cap screws that hold one end bell to the intesifier assembly fixture and, using a mallet, tap the end bell free of the cylinder. Remove the end bell and set it aside.



10. Remove the cylinder from the other end bell. The piston and plungers may remain with the cylinder



11. **Drive the piston and plungers from cylinder with a soft-faced mallet and set the parts aside.**

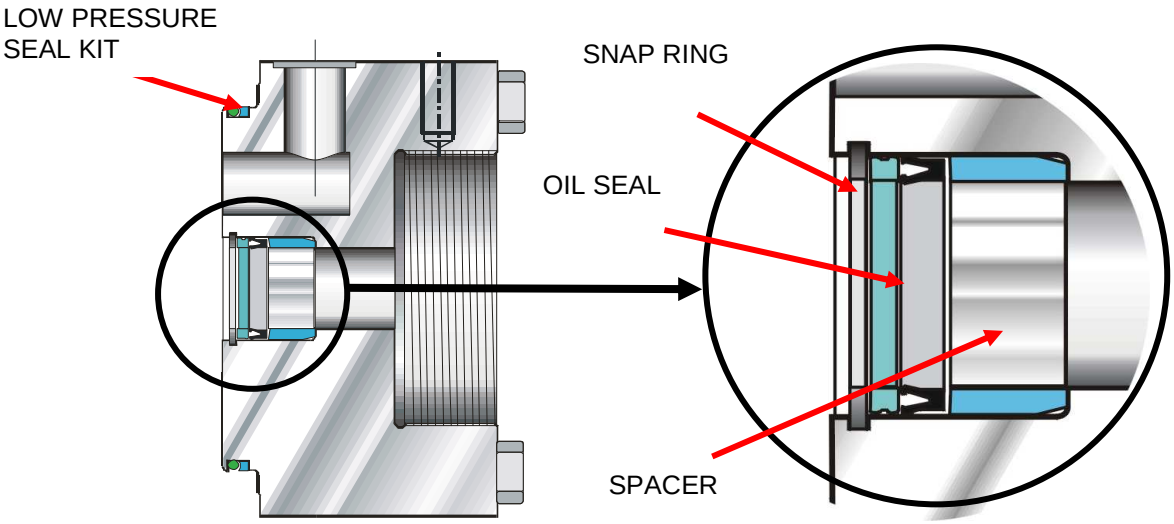


12. **Begin the inspection of the hydraulic cylinder, end bells and related parts at this point. All service must be completed in a clean, well-lit area.**

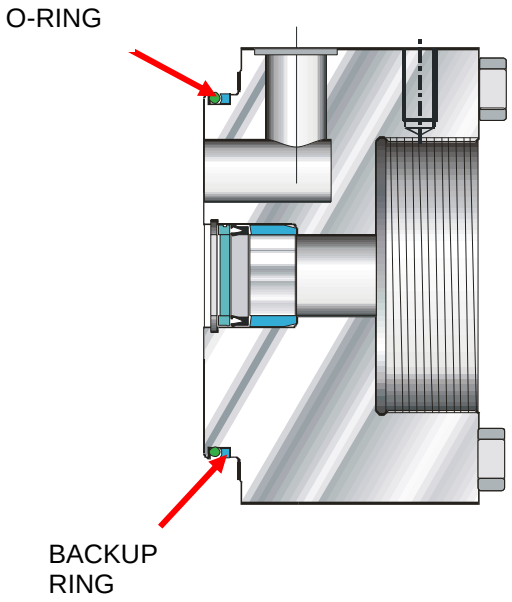
13. **Remove the snap ring, spacer and seal from the end bell, but do not use a sharp instrument as it will damage the parts.**

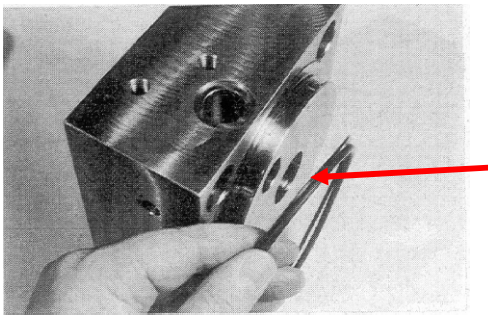
14. **Clean the end bells, seal spacers and snap rings and inspect each part.**
a) Check the end bells for damaged threads, a damaged or out-of-round shift actuating pin hole, and damage to the sealing surfaces and grooves.
b) Make sure the seal spacers are not ouf of round, warped, or nicked.
c) Make sure the snap rings are not bent and still have the approproate tension.

15. **Apply clean hydraulic oil or Parker Super O-Ring Lube to a new oil seal and install it in an end bell with the flared (o-ring) side visible. Place the spacer on top of the seal and install the snap ring.**



16. **Apply clean hydraulic oil or Parker Super O-Ring Lube to the new backup ring and note which face of the backup ring has light circular molding marks. Lay the end bell flat and install the backup ring on the end bell with the molding marks facing the end bell. Lubricate the new o-ring and place it in the groove on top of the backup ring. The o-ring must be to the outside of the backup ring, towards the low-pressure cylinder side.**

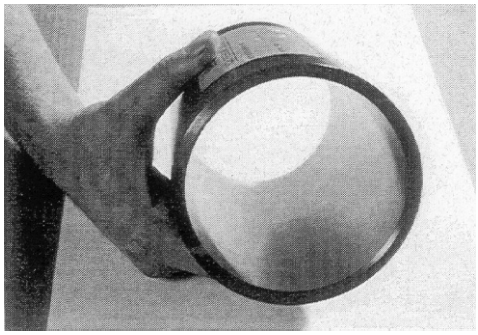




17. **Repeat Steps 15 and 16 for the other end bell, then set the end bells aside and cover with a lint-free cloth.**

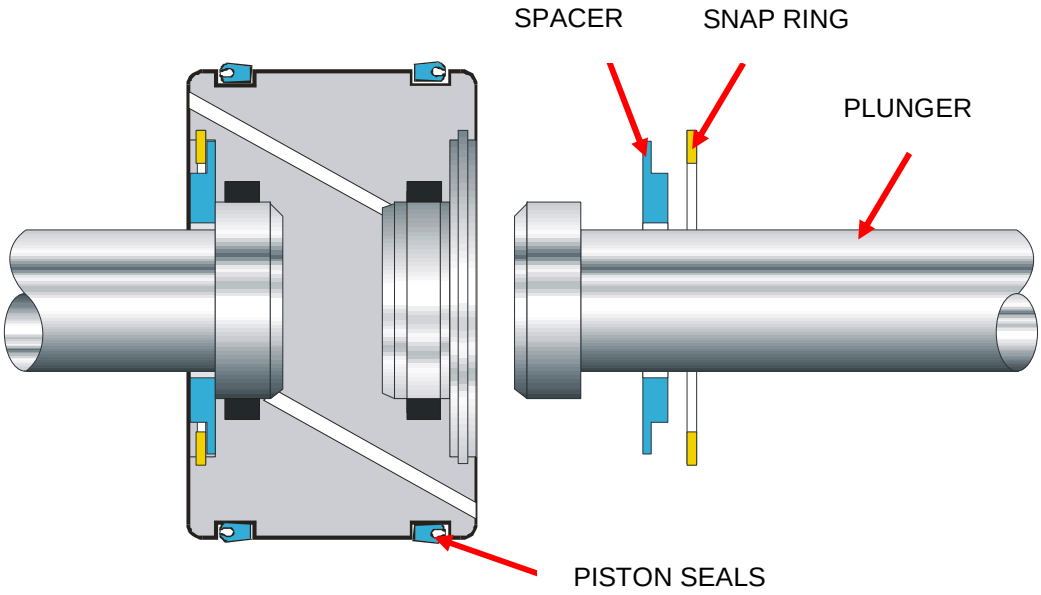
18. **Clean and inspect the hydraulic oil cylinder. Inspect all surfaces of the cylinder for any signs of damage or fatigue. Inspect the inner diameter by looking through the cylinder angled toward a brightly lit white surface. Any damage to this surface must be polished out. If this is not possible, the cylinder must be discared.**





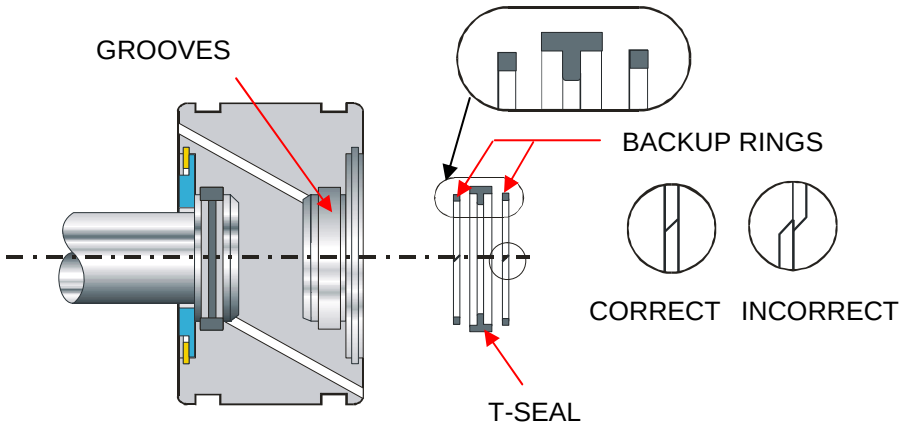
19. **Remove the snap ring and retainer. The retainer can be loosened by blowing air down between the plunger and the retainer bore.**

20. **Remove the plunger by hand.**



21. **Repeat Steps 19 and 20 for the other plunger. Set the plungers asside.**

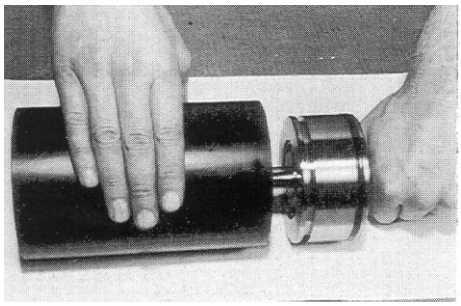
22. **Apply clean hydraulic oil or Parker Super O-Ring Lube to the head of the plunger. Install plungers into the counter bores in the piston biscuit with a turning motion. Ensure that T-seals or O-rings do not get nicked or rolled.**

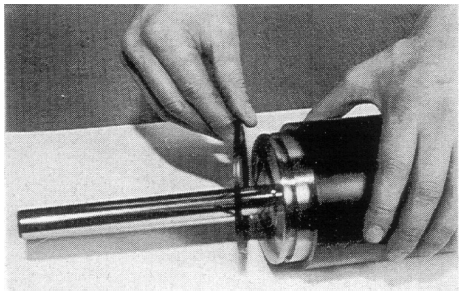


23. **Slide the piston and plunger assembly containing one seal into the cylinder oriented so that the empty groove enters the cylinder first.**

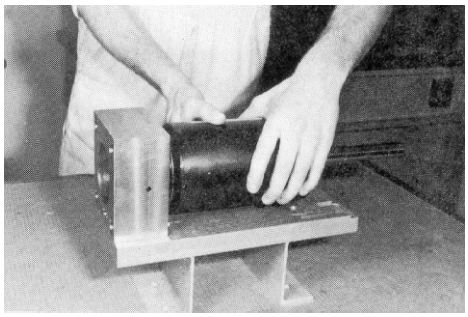
24. **Slide the piston into the cylinder until the empty seal groove on the piston appears out of the other end of the cylinder.**

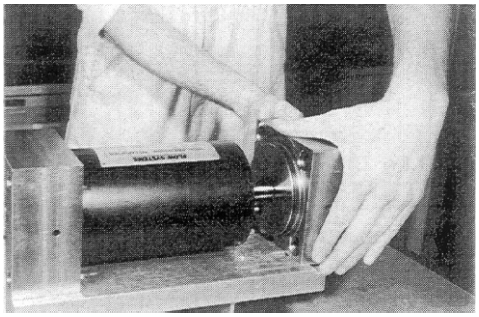
25. Apply clean hydraulic oil or Parker Super O-Ring Lube to the seal and install it in the groove oriented with the flared o-ring face towards the outside of the piston



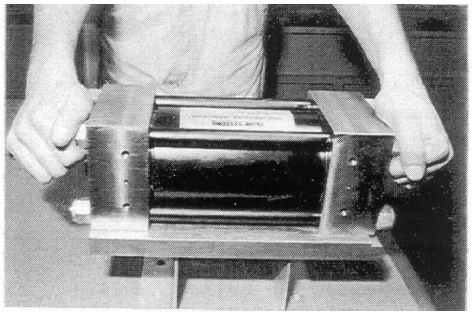


26. Mount the remaining end bell to the plunger and secure to the assembly fixture with the cap screws finger tight

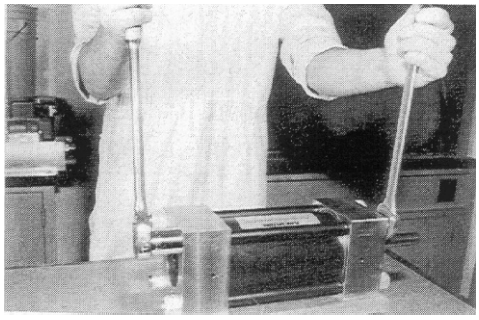




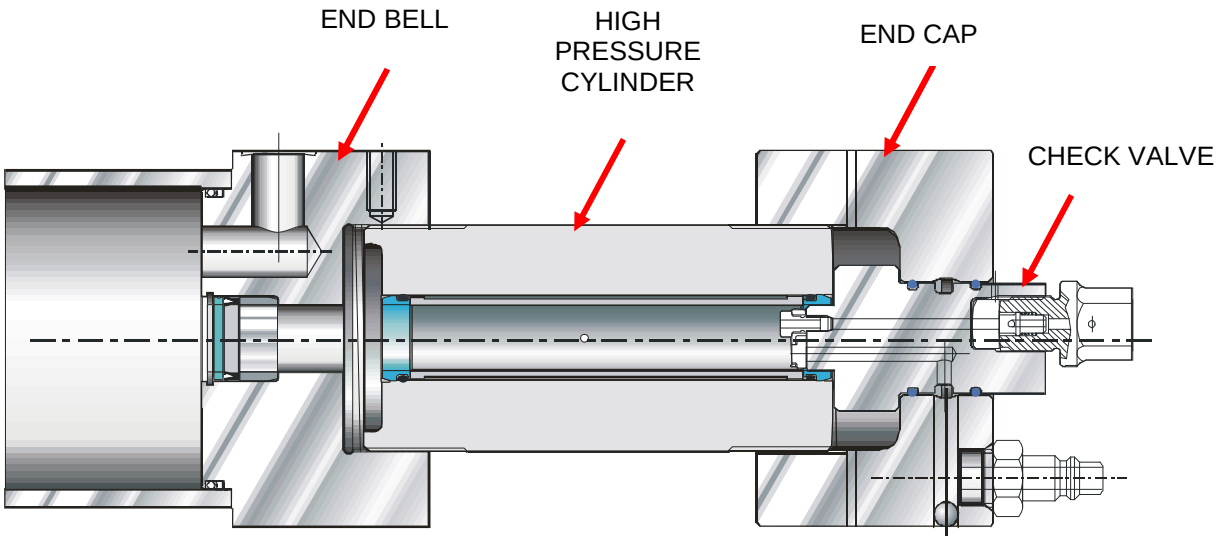
27. Install tie rods and thread on the elastic stop nuts

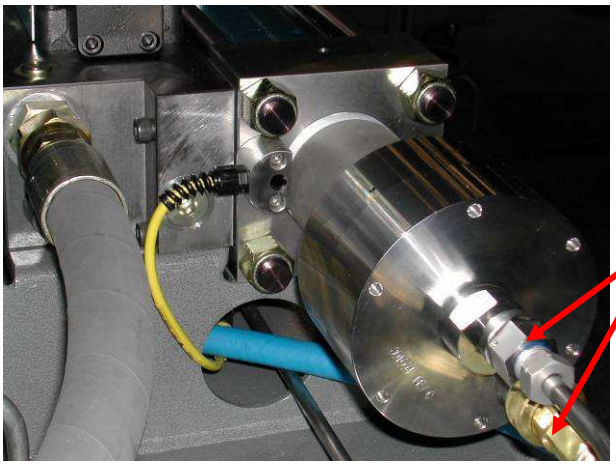


28. Draw the cylinder carefully onto the end bells by tightening the nuts evenly in a figure-eight pattern while moving end to end. Seal or backup ring debris must not appear at the cylinder/end bell interface. Check for any sign of a problem before continuing. Torque all nuts evenly to 20 ft-lb. (27 N-m) so that all parts are seated.

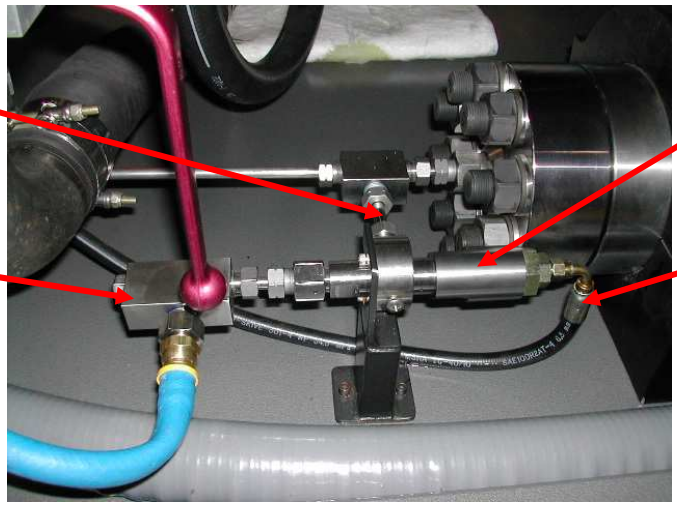


29. Install the high-pressure water cylinders and end caps.

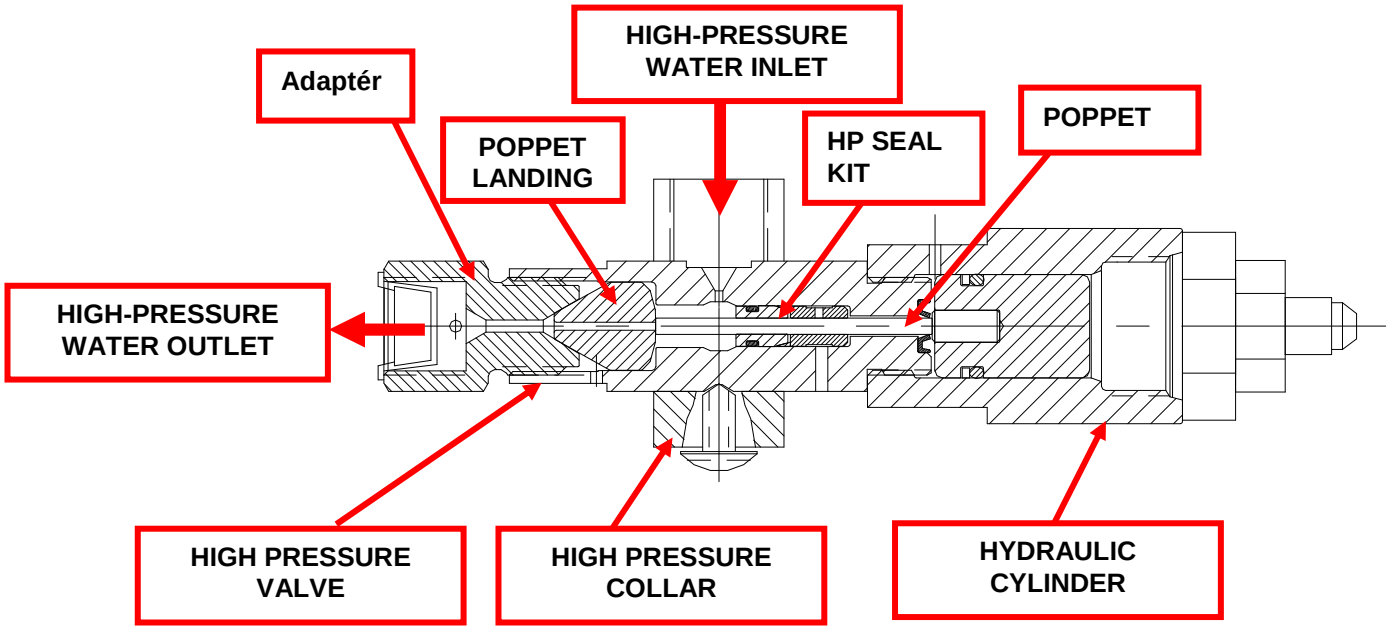




Bleed-Down Valve Service







1. **Shut down the system. Place the main electrical disconnect in the OFF position. Bleed down all high-pressure lines.**

2. **Remove the hydraulic hose from the oil port the bleed-down valve.**

3. **Remove the gland nut connecting the water drain tubing to the bleed-down valve.**

4. **Remove the gland nut from the collar at the high-pressure port of the bleed-down valve.**

5. **Place the bleed-down valve on a clean workbench.**

6. **Seperate the actuator body from the high-pressure valve assembly.**

7. **Remove the high-pressure collar from the high-pressure valve assembly.**

8. **Remove the adapter from the high-pressure body.**

9. **Remove and discard the high-pressure seat from the high-pressure body.**

10. **Remove and discard the high-pressure seal assembly (stem, seal, seal hoop, O-ring, and bushing) by inserting the extraction end of the high-pressure valve assembly tool through the oil seal and pushing on the high-pressure stem. DO NOT remove the oil seal!!!**

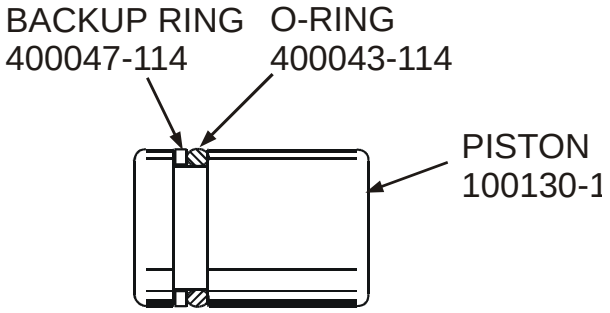
11. **Push the piston insert and piston out of the actuator body through the oil port by using a soft-faced ¼" dowel (or similar item).**

12. **Remove and discard the O-ring and backup ring from the piston.**

13. **Clean the remaining parts with an ultrasonic cleaner. Check the parts for nicks and burrs. Replace as necessary.**

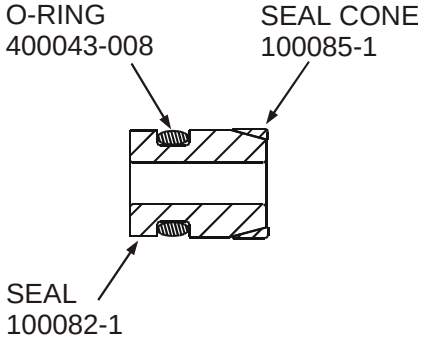
14. **Lubricate a new O-ring, backup ring, and the inner surface of the actuator body with Parker Super O Lube or equivalent.**

15. **Install the new O-ring and backup ring on the piston as shown. Install the piston insert and the piston assembly into the actuator body until seated. Make sure the piston is aligned with the actuator body to prevent damage to the O-ring.**

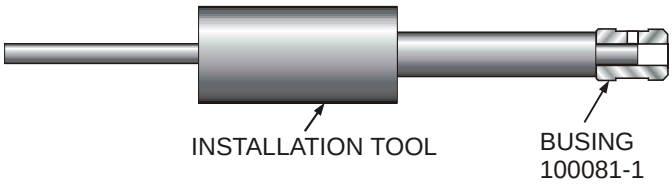


16. **Lubricate the stem, seal, seal hoop, O-ring, and bushing with White Food Grade Grease.**

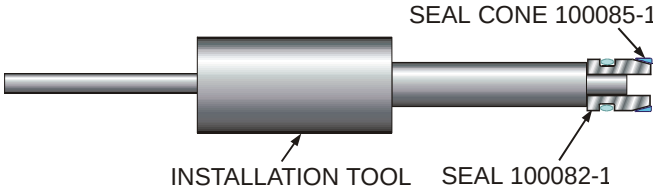
17. **Install the O-ring and seal hoop onto the high-pressure seal as shown.**



18. Install the new bushing onto the installation tool as shown. Then insert the bushing into the high-pressure body using the installation tool.



19. Place the high-pressure seal assembly onto the installation tool as shown. Then insert the high-pressure seal assembly into the high-pressure body using the installation tool. Make sure the seal assembly is installed past the high-pressure port hole.



20. Lubricate the high-pressure stem with White Food Grade Grease.
21. Install the high-pressure stem tip into the seal assembly, then use a plastic or wood dowel to push the stem through the seal assembly until it bottoms out.
22. Install the high-pressure seat into the high-pressure body.
23. Install the adapter. Torque to 35 ft-lbs (48 N-m).
24. Install the high-pressure collar.
25. Connect the high-pressure valve assembly to the actuator body.
26. Reinstall the bleed-down valve in the machine by attaching the gland nut to the high-pressure collar at the high-pressure port of the valve. Torque the gland to 35 ft-lbs (48 N-m).
27. Install the gland nut connecting the water drain tubing to the bleed-down valve.
28. Reattach the hydraulic hose to the oil port of the bleed-down valve.
29. Place intesifier pump in operation and check for water or hydraulic oil leaks. Correct leaks as required.

5. SAFETY

This section contains a comprehensive list of safety precautions that must be followed to ensure safe operation of the equipment. These precautions must be read and understood by everyone operating and maintaining the equipment – before they start working with the equipment. Applicable plant general safety precations must be followed.

The user shall practice and promote safety at all times to avoid potential injuries and unnecessary production shutdowns. The high-pressure waterjet systém and waterjet handtools are powerful cutting tools and must always be treated with respect.

SAFETY PRECAUTIONS

Throughout the manual, safety precautions and warning for specific operations are highlighted. Safety precautions are also posted on the equipment. The operator and service personnel shall pay particulary attention to these precautions at all times.

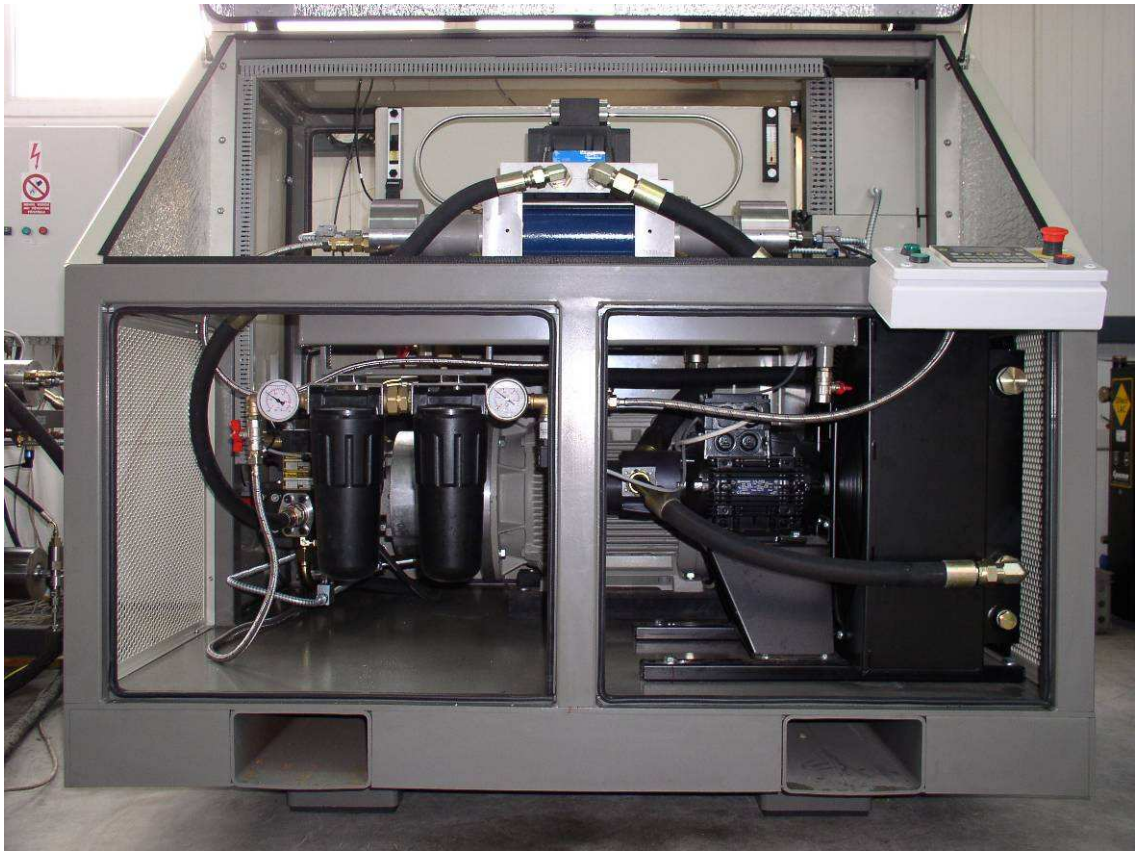
Operators of a high-pressure waterjet cutting system must treat the system as they would treat any high-speed cutting tool. Although the waterjet may appear harmless, it is a high-energy cutting tool capable of cutting many non-metallic materials such as composites, plastics, and wood products. Misuse of this equipment or carelessness in its application can be extremely hazardous to operating personnel. Always treat waterjet cutting systém with respect.



6. ANNEXES

6.1 Picture part

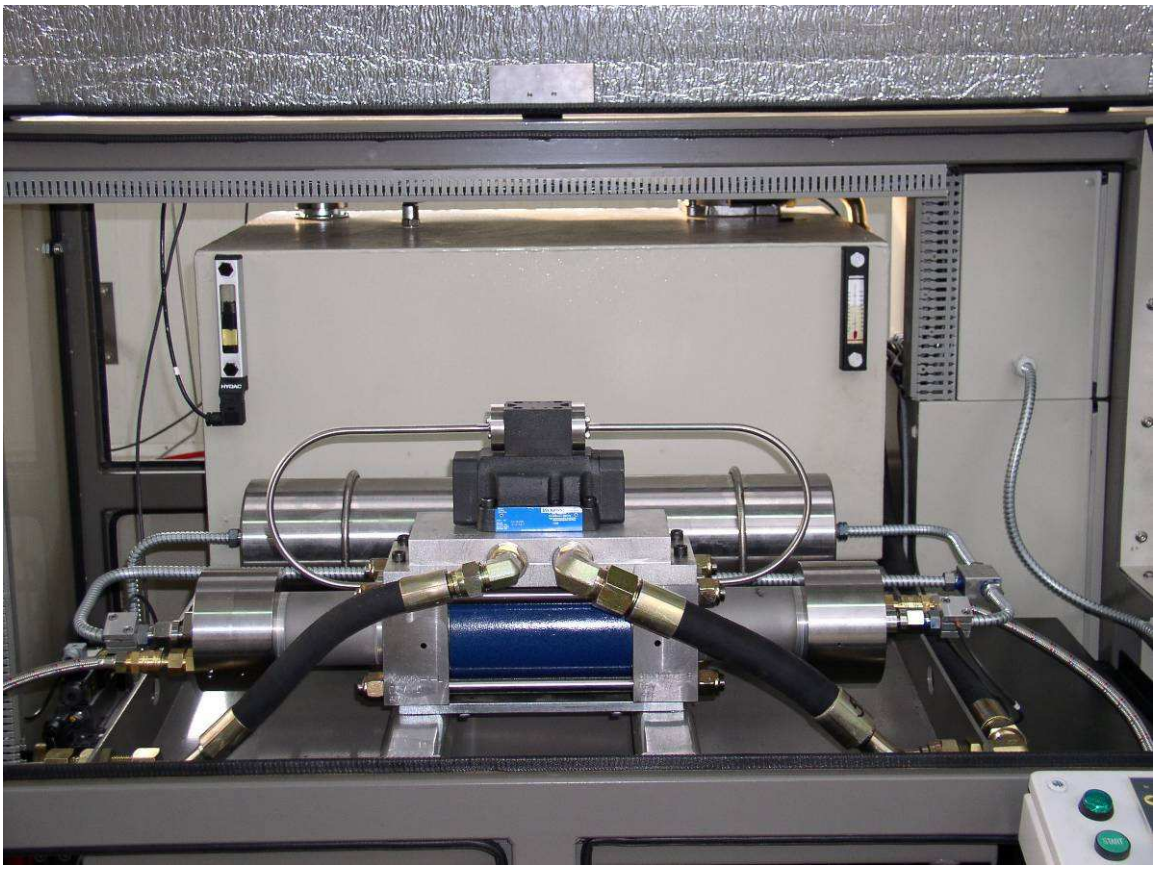
View of the machine



Frontal view of the uncovered machine



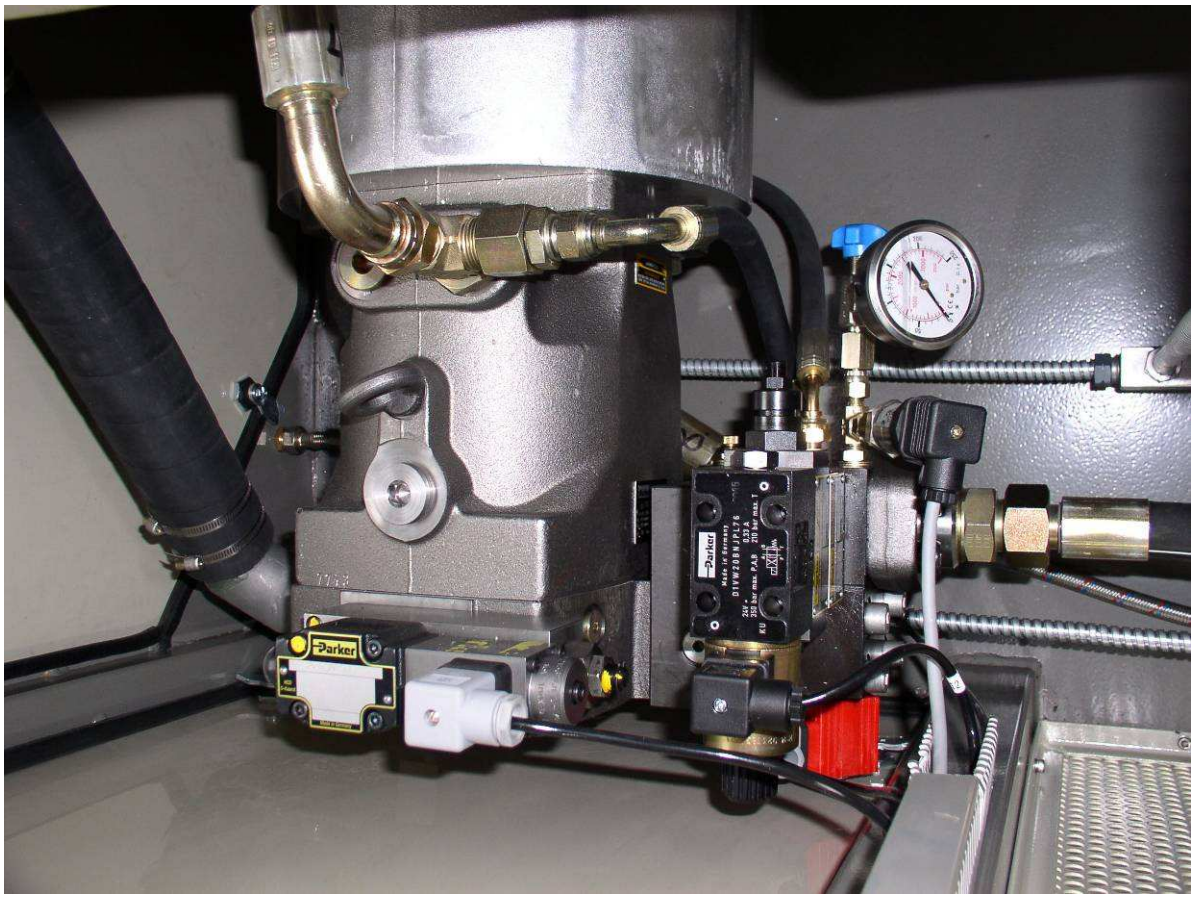
Filtering unit (FJ)



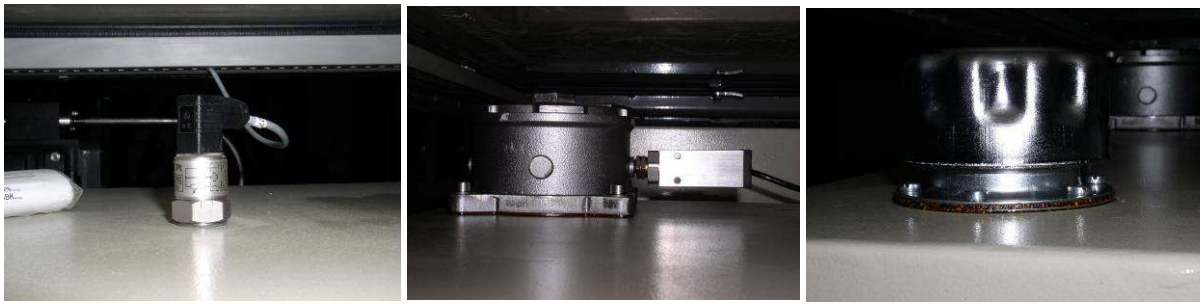
Multiplier (VTM), equalizing cylinder (VTV) and tank of the hydraulic oil (N)



Cooler of the hydraulic oil (CH)



Hydraulic pump (P) including the flush valve (PPV) and hydraulic cubes (HK1 and 2) mounted with active elements of the hydraulics on the delivery (PV, EV, DEV, DZV, DSV, P1, SV a MP2)



Temperature sensor (T1), reverse filter (ZF) and filler (NZ)



Temperature sensor (T3, T4), bleed down valve with temperature sensor (T5) and level indicator with level switch (L1)

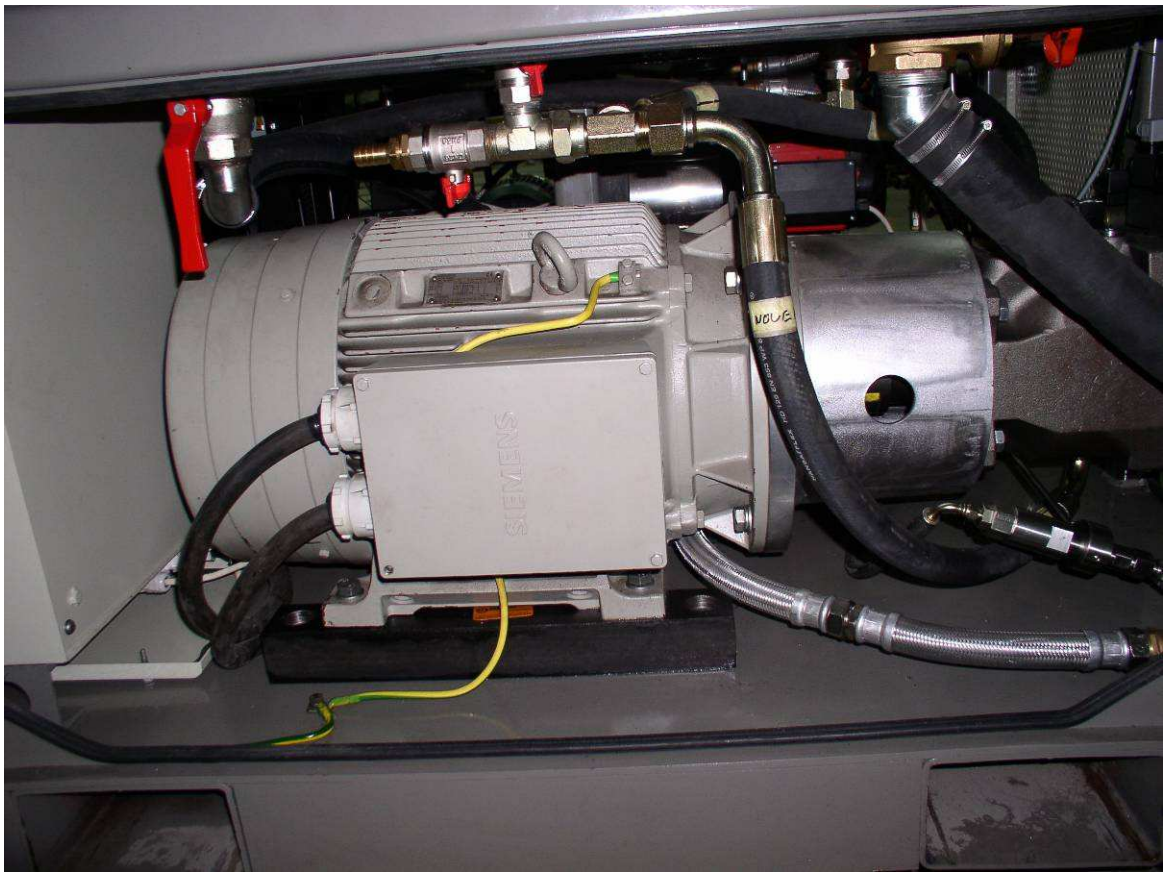




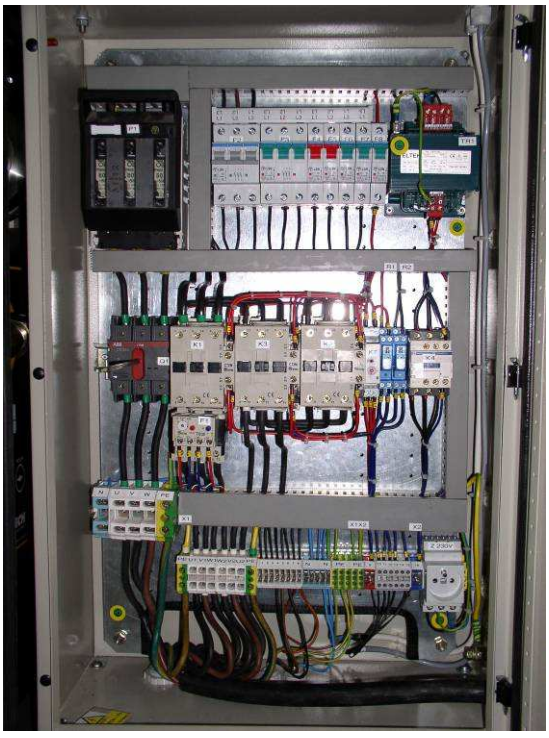
Sensor of input water pressure (P2) on the filtering unit, hydraulic oil tank (N) with openings (PN) and active elements of the hydraulics at the delivery of the hydraulic pump.



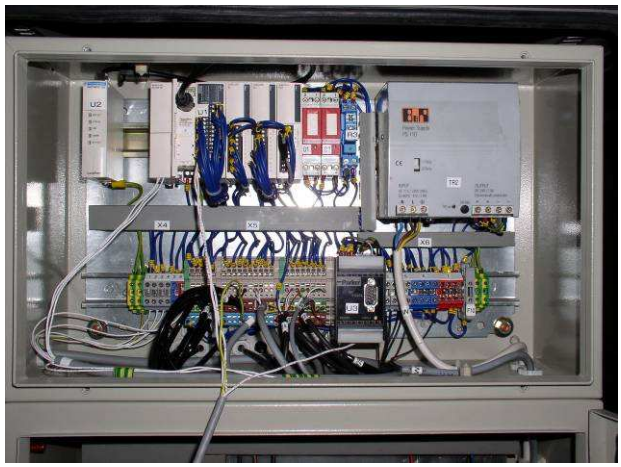
Control panel of HP pump



Electromotor (M1), hydraulic pump (P) and connection of the hydraulic pump (N)

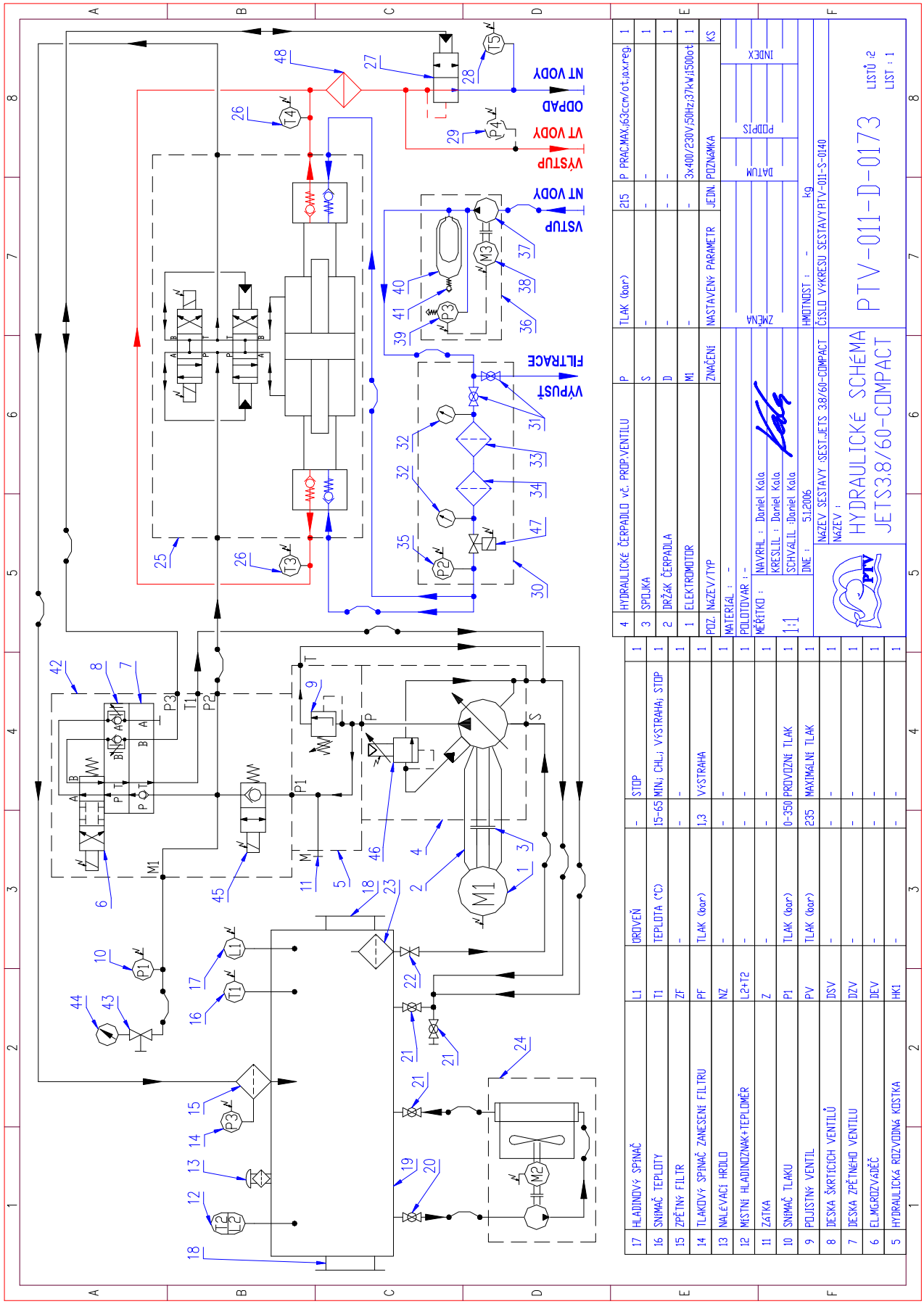


Main distributor



Control distributor

6.2 HYDRAULIC SCHEME PTV-011-D-0173 (2 pages)



POZ.	NÁZEV/TYP	ZNAČENÍ	NASTAVENÝ PARAMETR	JEDN.	POZNÁMKA	KS
48	VT VYROVNÁVACÍ VÁLEC	VTV	-	-	-	1
47	ELEKTROMAGNET. VENTIL	EVV	-	-	-	1
46	PROPORCIONÁLNÍ VENTIL	PPV	-	-	-	1
45	ELEKTROMAGNET. VENTIL	EV	-	-	-	1
44	MANOMETR	MP2	TLAK (bar)	-	0-250bar	1
43	ŠKRTÍCÍ VENTIL	SV	-	-	-	1
42	HYDRAULICKÁ ROZVODNÁ KOSTKA	HK2	-	-	-	1
41	TLAKOVACÍ VENTILEK	TV	TLAK (bar)	3,3	-	1
40	TLAKOVÁ NÁDOBA S VAKEM	TN	-	-	-	1
39	TLAKOVÝ SPÍNAČ	P3	TLAK (bar)	3,5	-	1
38	ELEKTROMOTOR	M3	-	-	-	1
37	ČERPADLO VODNÍ	PV	-	-	-	1
36	ZVYŠOVÁNÍ TLAKU VSTUPNÍ VODY	TVV	TLAK (bar)	3	P MIN.; VOLITELNÉ PŘÍSLUŠ.	1
35	TLAKOVÝ SNÍMAČ	P2	TLAK (bar)	3-6	MIN.TLAK - STOP	1
34	FILTR VSTUPNÍ VODY	FV0,5	-	-	S VLOŽKOU 0,5mikronu	1
33	FILTR VSTUPNÍ VODY	FV1	-	-	S VLOŽKOU 1mikron	1
32	MANOMETR	MP1	-	-	0-16bar	2
31	KULOVÝ VENTIL	KV3	-	-	-	2
30	FILTRAČNÍ JEDNOTKA	FJ	-	-	-	1
29	SNÍMAČ VYSOKÉHO TLAKU	P4	TLAK (bar)	-	VOLITELNÉ PŘÍSLUŠENSTVÍ	1
28	SNÍMAČ TEPLOTY	T5	TEPLOTA (°C)	35	VÝSTRAHA BD VENTIL	1
27	BLEEDOWN VENTIL	BD	-	-	-	1
26	SNÍMAČ TEPLOTY	T3+T4	TEPLOTA (°C)	35	VÝSTRAHA CHECK VALVES	2
25	MULTIPLIKÁTOR	VTM	-	-	TYP 60K (40K)	1
24	CHLADIČ HYDRAULICKÉHO OLEJE	CH	-	-	-	1
23	SACÍ FILTR	SF	-	-	-	1
22	ŠOUPÁTKO	SO1	-	-	-	1
21	KULOVÝ VENTIL	KV2	-	-	-	3
20	KULOVÝ VENTIL	KV1	-	-	-	1
19	NÁDRŽ HYDRAULICKÉHO OLEJE	N	-	-	-	1
18	PRŮLEZ	PN	-	-	-	2

MATERIÁL : -

POLOTOVAR : -

MĚŘÍTKO : 1:1

NAVRHL : Daniel Kala

KRESLIL : Daniel Kala

SCHVÁLIL : Daniel Kala

DNE : 5.1.2006

ZMĚNA | DATUM | PODPIS | INDEX

HMOTNOST : - kg

NÁZEV SESTAVY : SEST.JETS 3.8/60-COMPACT

ČÍSLO VÝKRESU SESTAVY: PTV-011-S-0140

NÁZEV :

HYDRAULICKÉ SCHÉMA JETS3.8/60-COMPACT

PTV-011-D-0173

LISTŮ : 2

LIST : 2

PTV, spol.s r.o.
Čsl. armády 23
253 01 Hostivice
Czech Republic

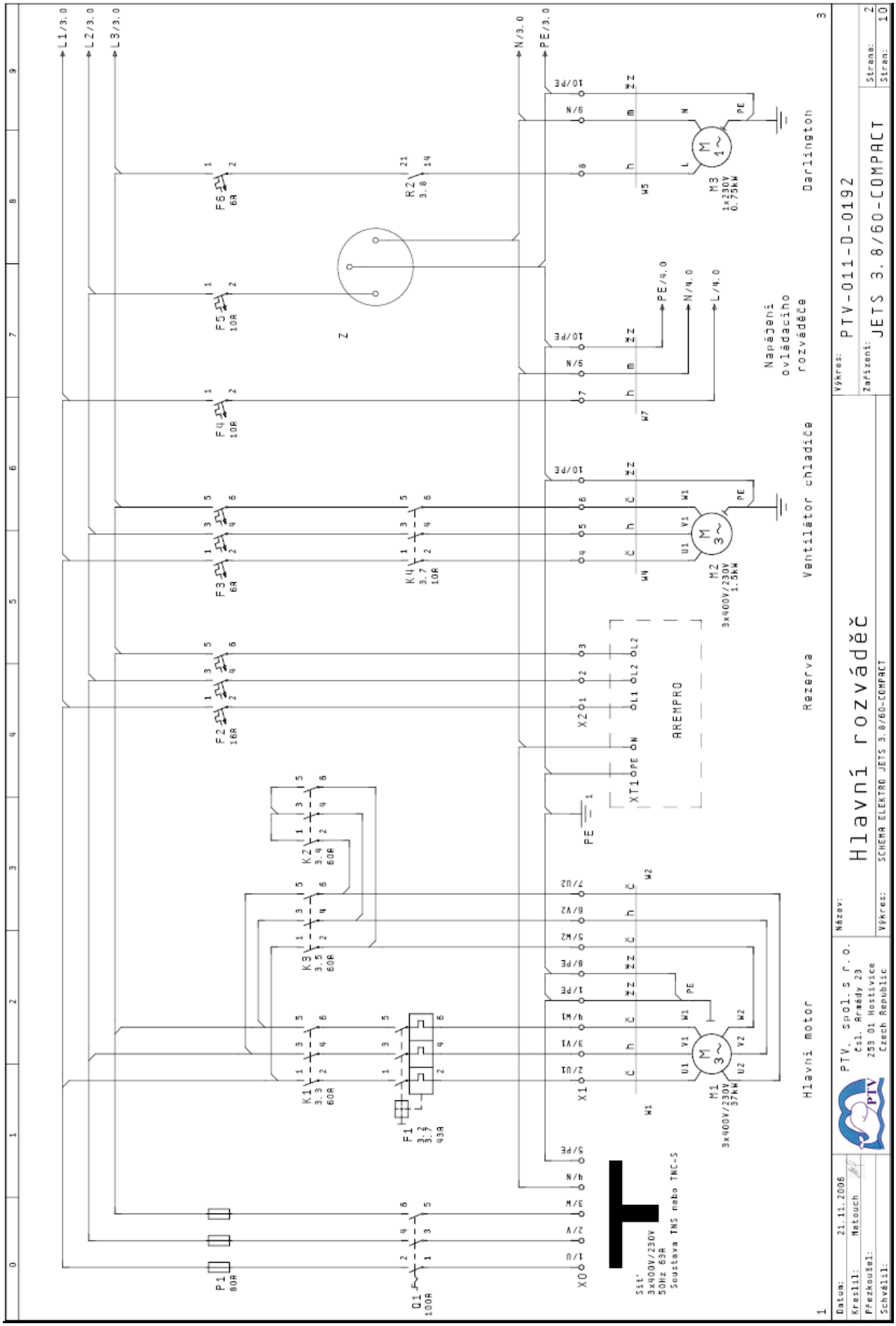
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Název sestavy	: SESTAVA ČERPADLA JETS 3.8/60-COMPACT
Číslo sestavy	: PTV-011-S-0140

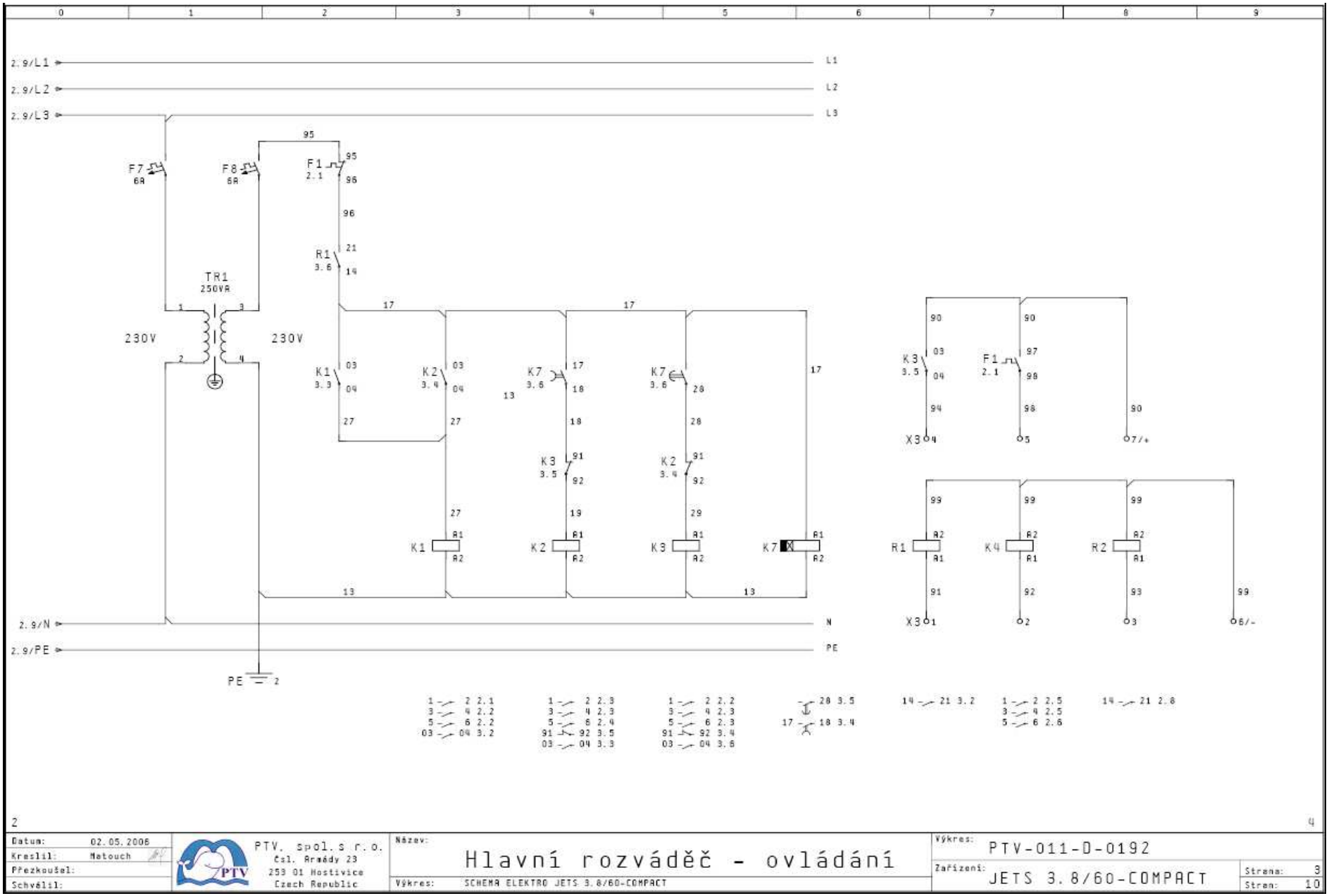
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Zodpovídá	:
Dílčí zvláštnosti	:

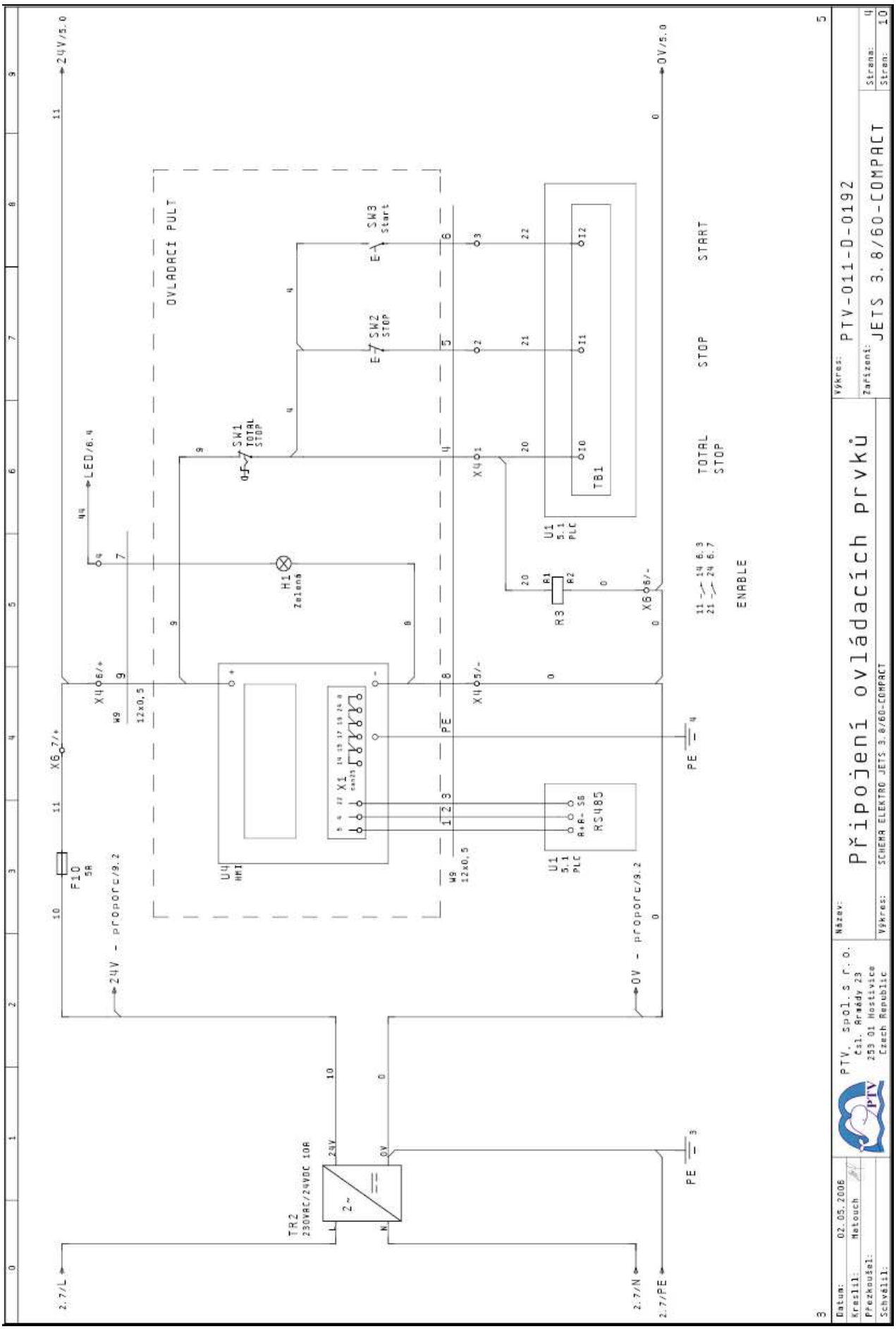
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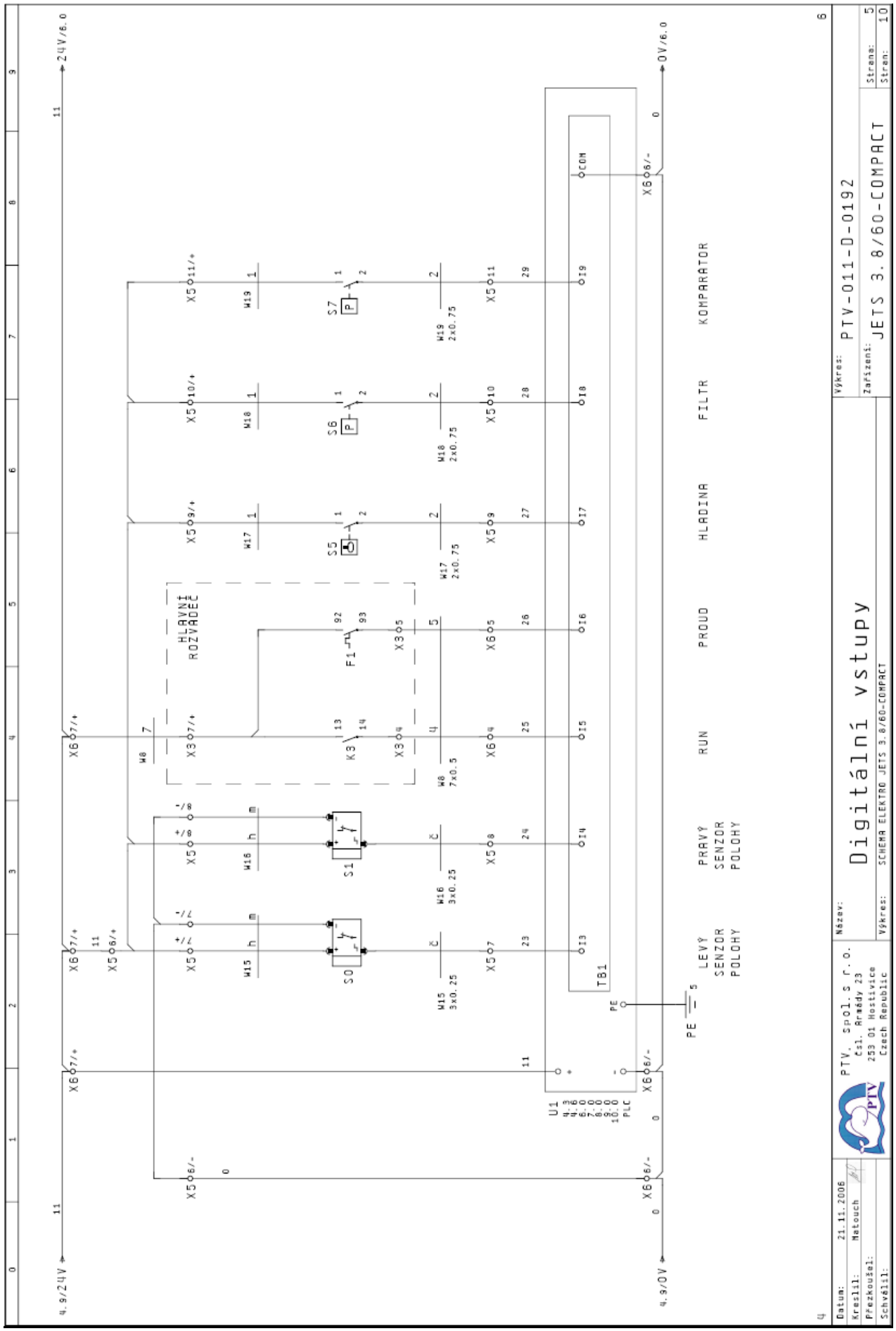
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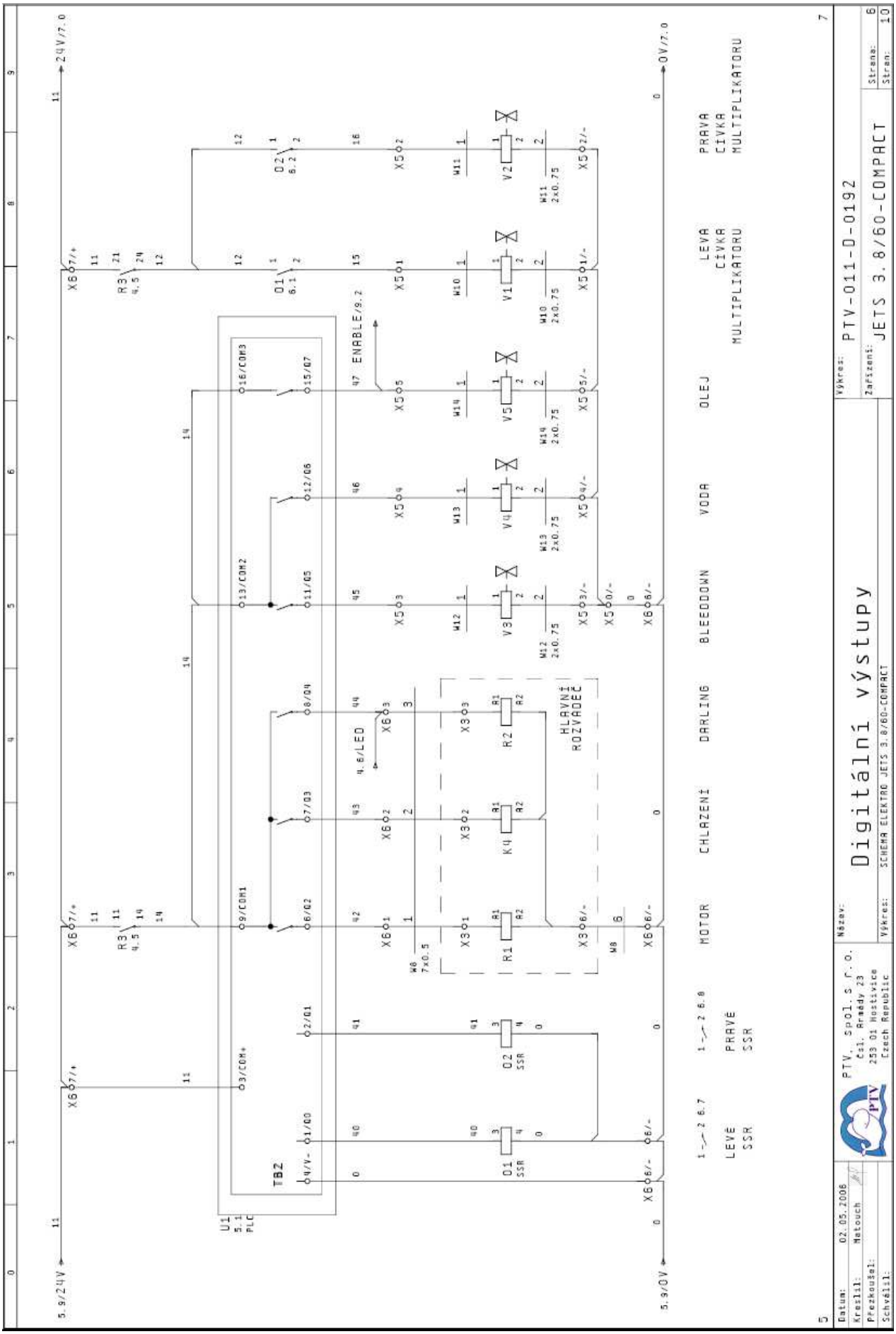
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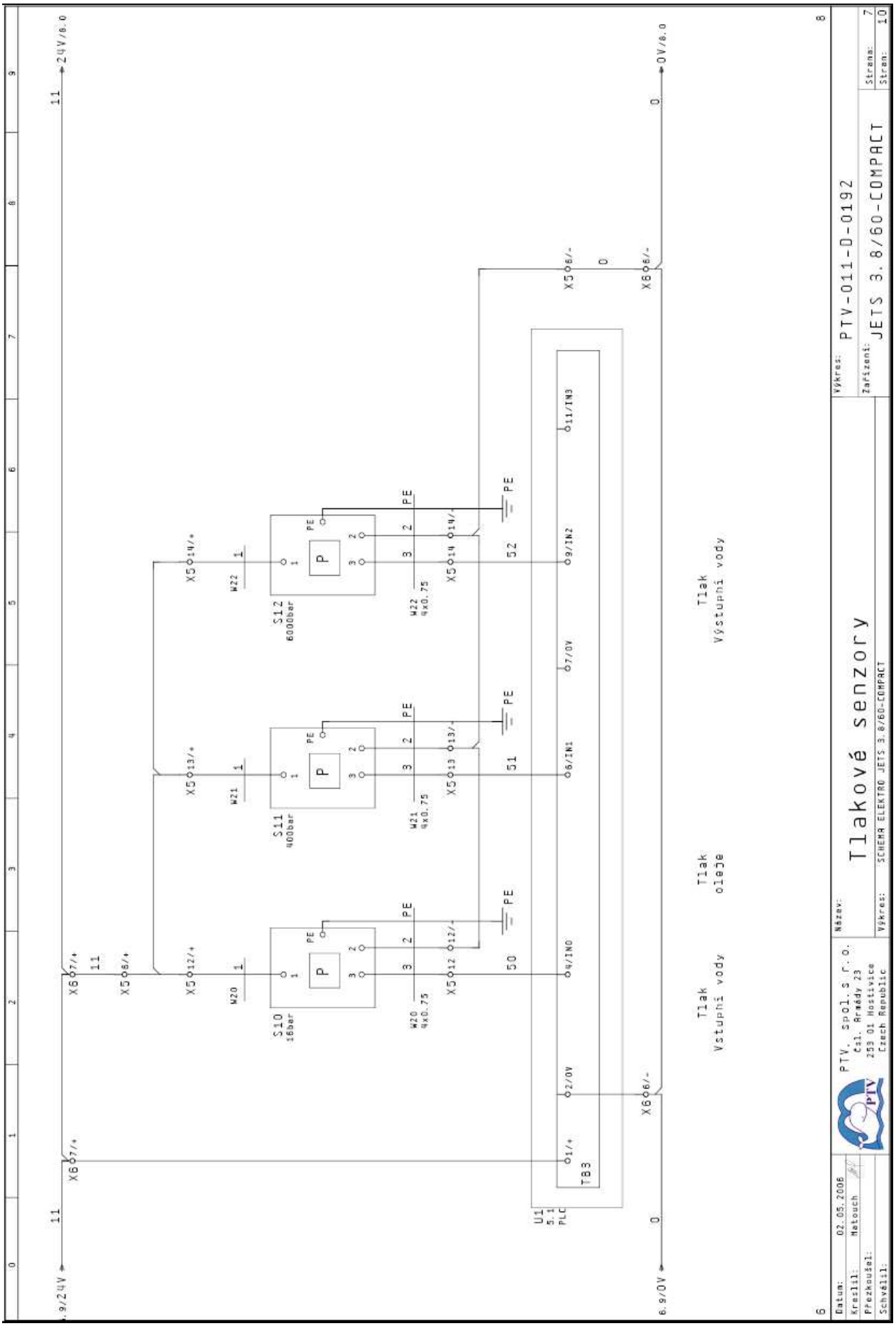

Hlavní rozváděč
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JETS 3.8/60-COMPACT
PTV, spol. s r.o.
Čsl. Armády 23
253 01 Hostivice
Czech Republic
SCHEMA ELEKTRO JETS 3.8/60-COMPACT
Hlavní motor
Rezerva
Ventilátor chladiče
Napájení ovládacího rozváděče
Darlington
AREMPRO
Síť 3x400V/230V 50Hz 63A Soustava TNS nebo TN-S
P1 80A
Q1 100A
F1 3.2 43A
K1 3.3 60A
K2 3.4 60A
K3 3.5 60A
F2 16A
F3 6A
F4 10A
F5 10A
F6 6A
K4 3.7 10A
R2 3.8
Z
M1 3x400V/230V 3kW
M2 3x400V/230V 1.5kW
M3 1x230V 0.75kW

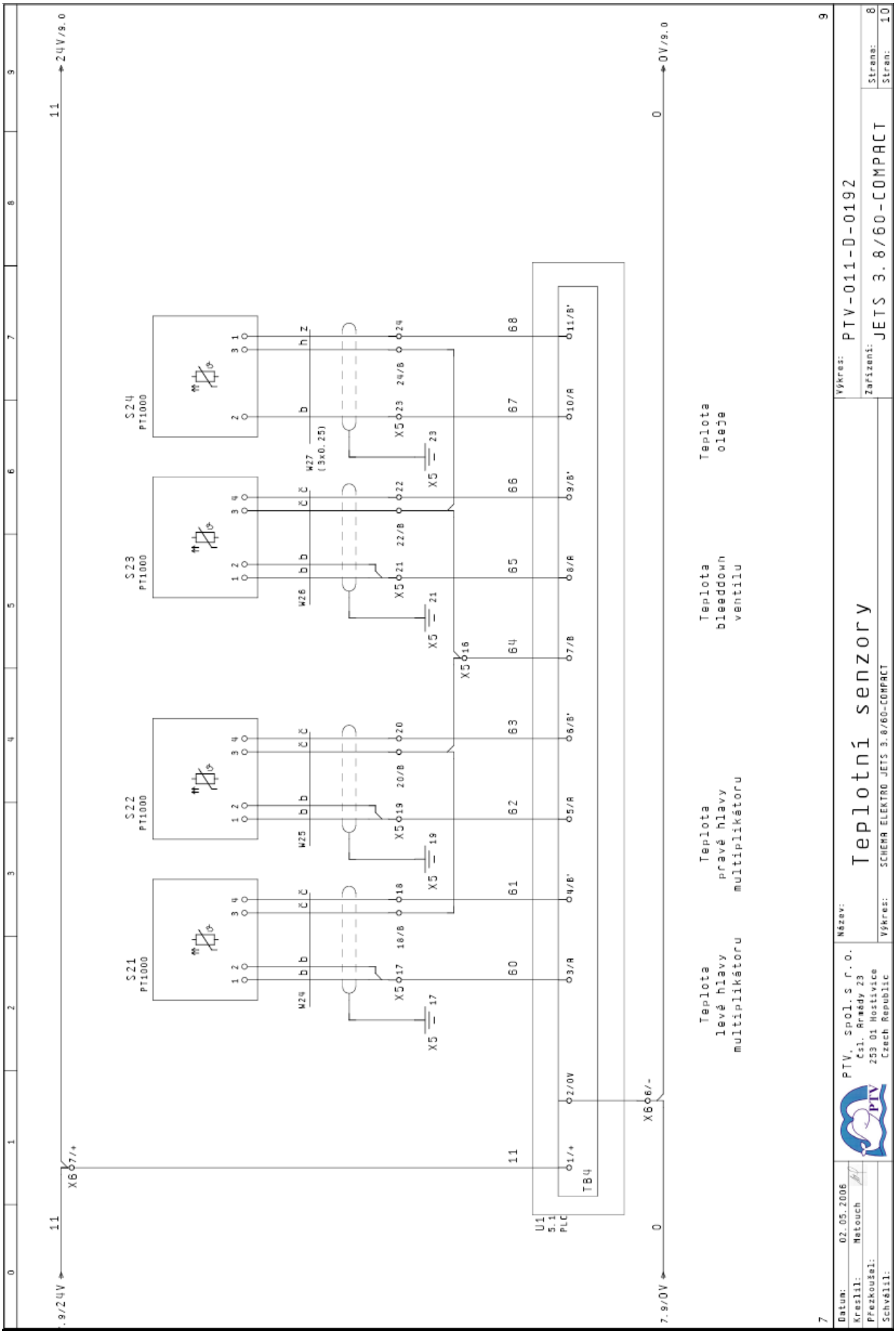

0 1 2 3 4 5 6 7 8 9
2.9/L1
2.9/L2
2.9/L3
L1
L2
L3
F7 6A
F8 6A
F1 2.1
95
96
R1 3.6
21
14
TR1 250VA
230V
230V
17
K1 3.3
K2 3.4
K7 3.6
K3 3.5
K2 3.4
K7 3.6
13
27
18
28
29
91
92
99
A1
A2
K1
K2
K3
K7
R1
K4
R2
2.9/N
2.9/PE
N
PE
X3 4
X3 5
X3 7/+
X3 1
X3 2
X3 3
X3 6/-
1 2 2.1
3 4 2.2
5 6 2.2
03 04 3.2
1 2 2.3
3 4 2.3
5 6 2.4
91 92 3.5
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253 01 Hostivice
Czech Republic
Název:
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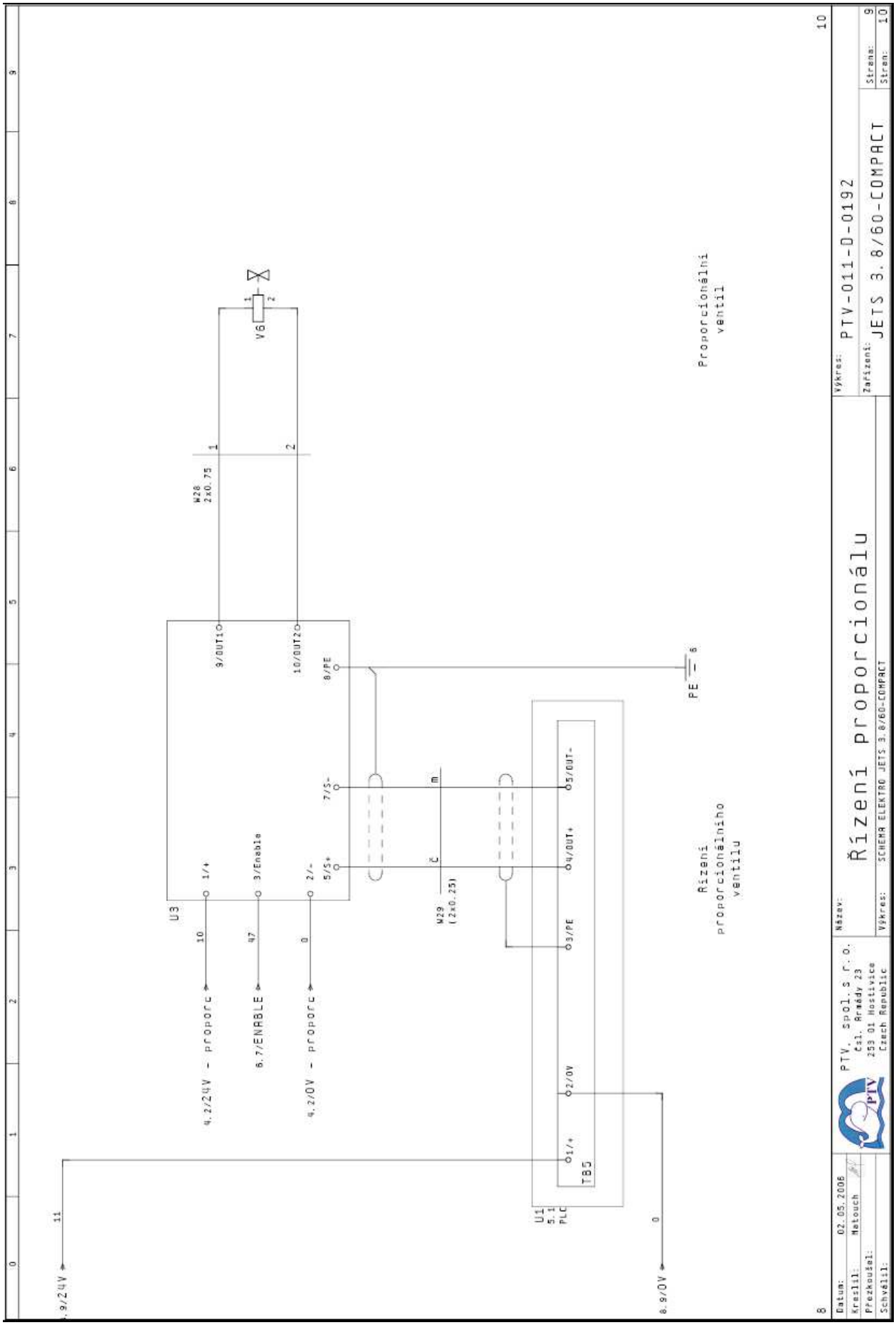

OVLADACÍ PULT
SW1 TOTAL STOP
SW2 STOP
SW3 start
H1 Zelená
R1
R2
R3
TB1
U1 5.1 PLC
U4 HMI
RS485
TR2 230VAC/24VDC 10A
F10 5A
X6 7/+
X6 8/-
X4 6/+
X4 5/-
X4 1
LED/6.4
24V/5.0
0V/5.0
24V - proporc/9.2
0V - proporc/9.2
2.7/L
2.7/N
2.7/PE
PE
TOTAL STOP
START
ENABLE
Název:
Připojení ovládacích prvků
Výkres:
PTV-011-0-0192
Zařízení:
JETS 3.8/60-COMPACT
Výkres:
SCHEMA ELEKTRO JETS 3.8/60-COMPACT
PTV, spol. s r. o.
tř. 1. Máje 23
293 01 Mladá Boleslav
Czech Republic
Datum: 02.05.2006
Kreslil: Matouch
Přezkoušel:
Schválil:
Strana: 4
Strana: 10

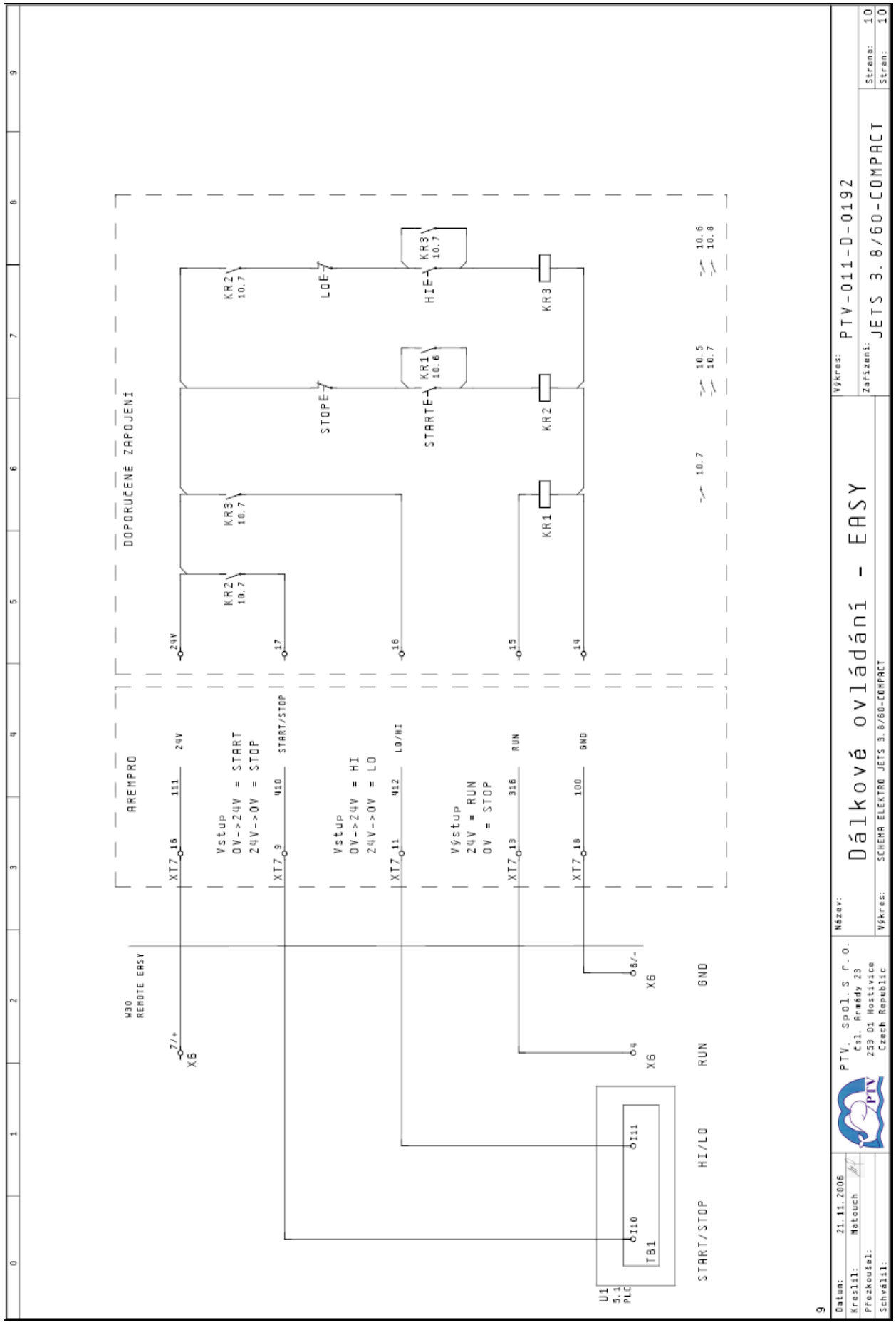

Digitální vstupy
PTV-011-D-0192
JETS 3.8/60-COMPACT
PTV, Spol. s r.o.
Čsl. Armády 23
259 01 Hostivice
Czech Republic
Datum: 21.11.2006
Kreslil: Matouch
Přezkoušel:
Schválil:
Název:
Výkres: SCHEMA ELEKTRO JETS 3.8/60-COMPACT
Zařízení:
Výkres: PTV-011-D-0192
Strana: 5
Stran: 10
HLAVNÍ ROZVADĚČ
LEVÝ SENZOR POLOHY
PRAVÝ SENZOR POLOHY
RUN
PROUD
HLADINA
FILTR
KOMPARATOR
PE
PLC
TB1
COM
24V/6.0
0V/6.0
4.9/24V
4.9/0V


Digitální výstupy
PTV-011-0-0192
JETS 3.8/60-COMPACT
PTV, spol. s r. o.
Čsl. Armády 23
253 01 Hostivice
Czech Republic
SCHEMA ELEKTRO JETS 3.8/60-COMPACT
MOTOR
CHLAZENÍ
DARLING
BLEEDDOWN
VODA
OLEJ
LEVÁ CÍVKA MULTIPLIKÁTORU
PRAVÁ CÍVKA MULTIPLIKÁTORU
LEVÉ SSR
PRAVÉ SSR
TBZ
HLAVNÍ ROZVADĚČ
ENABLE/9.2
4.6/LED
24V/7.0
0V/7.0
5.9/24V
5.9/0V

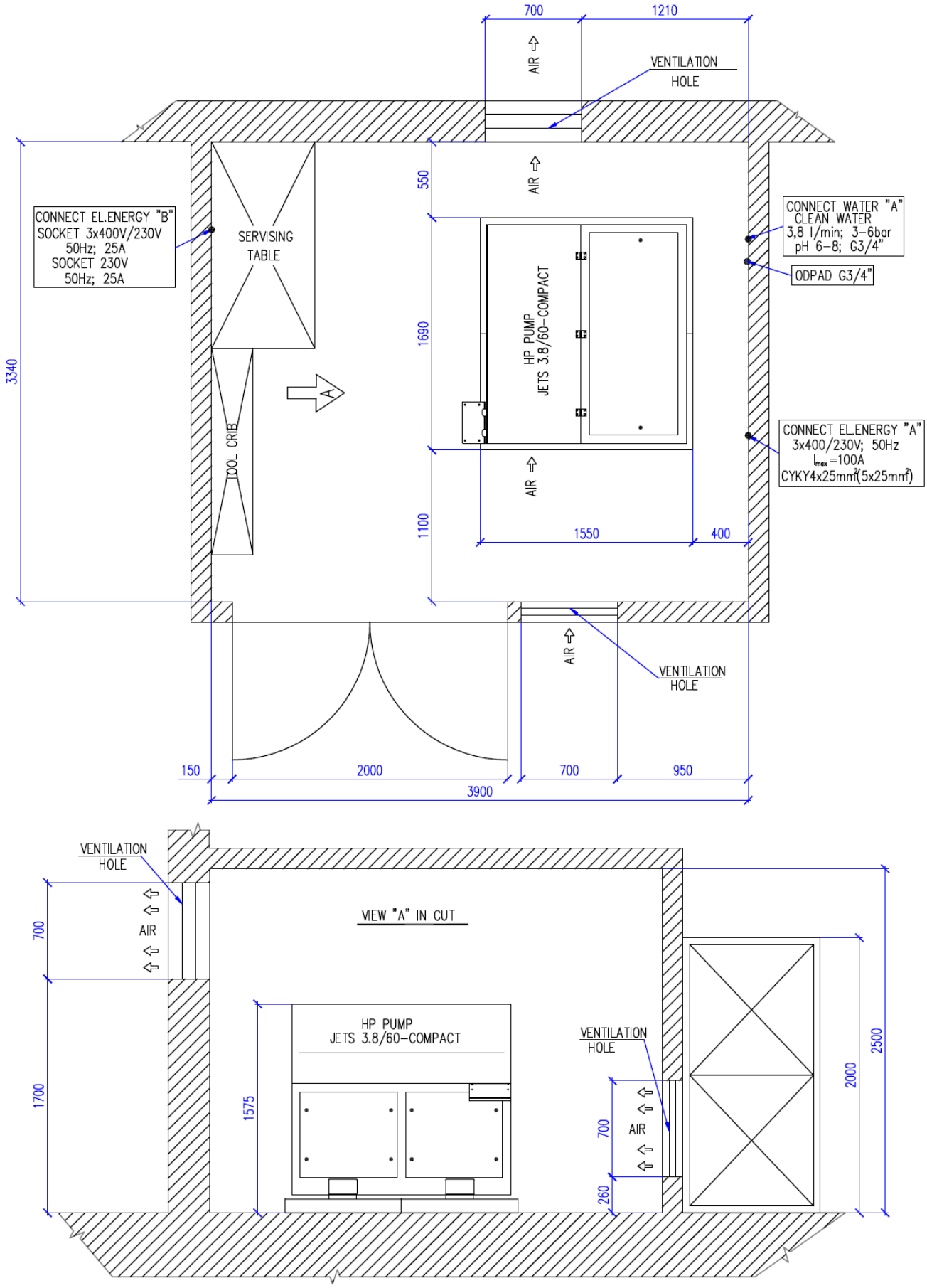

Tlakové senzory
PTV-011-D-0192
JETS 3.8/60-COMPACT
PTV, spol. s r. o.
Čsl. Armády 23
293 01 Hostivice
Czech Republic
SCHEMA ELEKTRO JETS 3.8/60-COMPACT
Tlak
Vstupní vody
Tlak
oleje
Tlak
Výstupní vody
S10
16bar
S11
400bar
S12
6000bar
TB3
U1
5.1
PLC


Teplotní senzory
PTV-011-D-0192
JETS 3.8/60-COMPACT
PTV, spol.s r.o.
Čsl. Armády 23
259 01 Hostivice
Czech Republic
SCHEMA ELEKTRO JETS 3.8/60-COMPACT
Teplota levé hlavy multiplikátoru
Teplota pravé hlavy multiplikátoru
Teplota bleaddown ventilu
Teplota oleje
S21 PT1000
S22 PT1000
S23 PT1000
S24 PT1000
TB4
U1 5.1 PLC
24V/9.0
0V/9.0


Řízení proporcionálu
PTV-011-D-0192
JETS 3.8/60-COMPACT
PTV, spol. s r. o.
Čsl. Armády 23
293 01 Hostivice
Czech Republic
SCHEMA ELEKTRO JETS 3.8/60-COMPACT
Proporcionální ventil
Řízení proporcionálního ventilu
U3
V6
W28 2x0.75
W29 (2x0.25)
PE
TB5
4.2/24V - proporc
8.7/ENABLE
4.2/0V - proporc
8.9/24V
8.9/0V

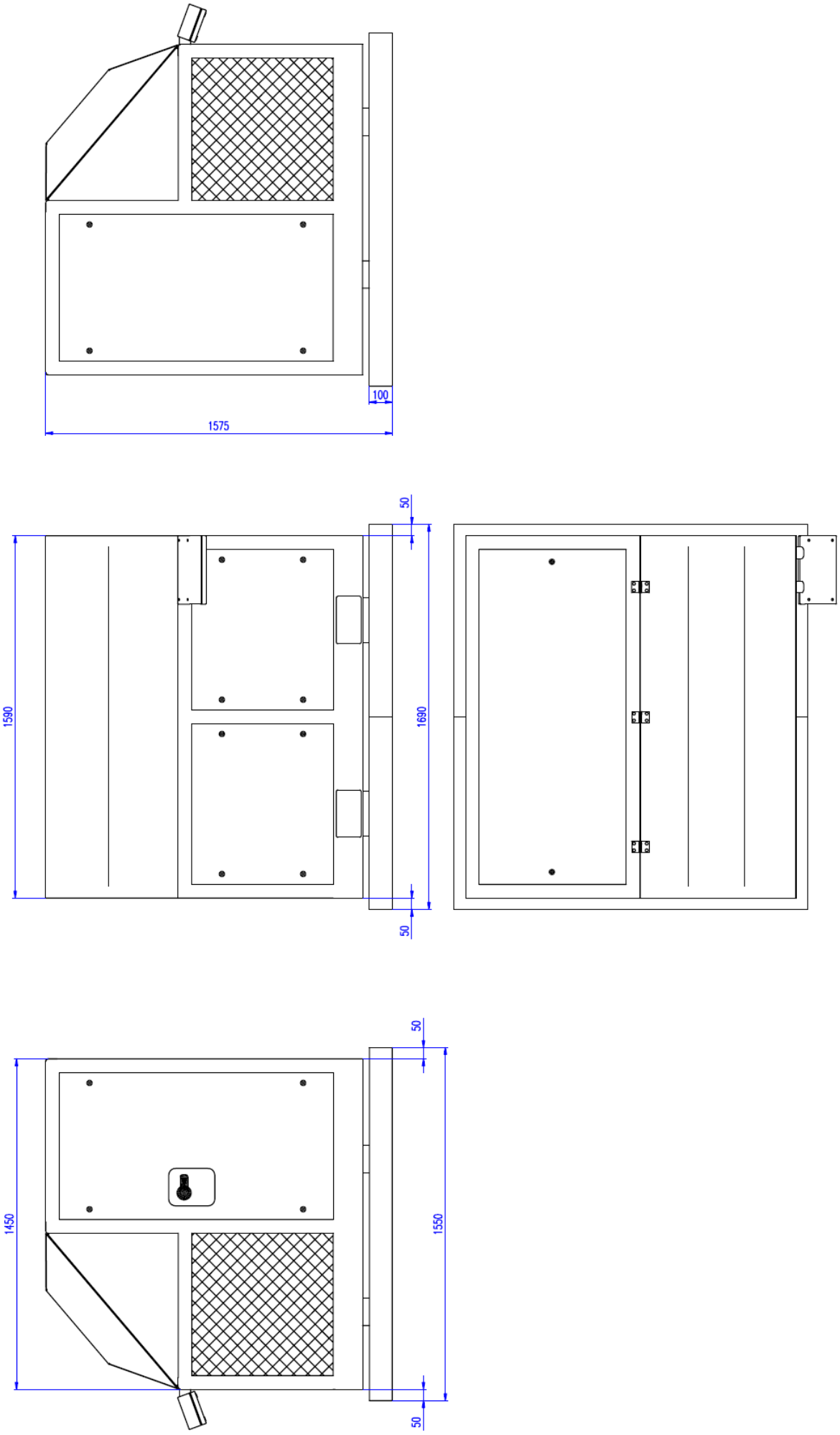

DOPORUČENÉ ZAPOJENÍ
AREMPRO
W30
REMOTE EASY
Vstup
0V->24V = START
24V->0V = STOP
Vstup
0V->24V = HI
24V->0V = LO
Výstup
24V = RUN
0V = STOP
START/STOP
HI/LO
RUN
GND
KR1
KR2
KR3
STOPE
STARTE
HIE
LOE
TB1
PTV, spol. s r. o.
Čsl. armády 23
253 01 Hostivice
Czech Republic
Dálkové ovládání - EASY
PTV-011-0-0192
JETS 3.8/60-COMPACT
SCHEMA ELEKTRO JETS 3.8/60-COMPACT
21.11.2006
Matouch

6.4 LAYOUT OF MACHINE ROOM – SAMPLE PTV-011-V-0334



TYPICAL DISPOSITION FOR HP PUMP JETS 3.8/60–COMPACT 07 15.8.2007 PTV–011–V–0334

6.5 DIMENSIONS OF PUMP JETS-3.8/60-COMPACT - PTV-011-V-0333



6.6 TABLE OF RELATIONS BETWEEN AMOUNT OF WATER AND PRESSURE AND SIZE OF USED JET NOZZLE

Závislost množství vody na průměru vodní trysky a tlaku

Tryska	Tlak	Množství
3 x 0,001 inch	20 000 psi	0,10 l/min
	25 000 psi	0,11 l/min
	30 000 psi	0,12 l/min
	35 000 psi	0,13 l/min
	40 000 psi	0,14 l/min
	45 000 psi	0,15 l/min
	50 000 psi	0,16 l/min
	55 000 psi	0,16 l/min
	60 000 psi	0,17 l/min

Tryska	Tlak	Množství
4 x 0,001 inch	20 000 psi	0,18 l/min
	25 000 psi	0,20 l/min
	30 000 psi	0,22 l/min
	35 000 psi	0,23 l/min
	40 000 psi	0,25 l/min
	45 000 psi	0,26 l/min
	50 000 psi	0,28 l/min
	55 000 psi	0,29 l/min
	60 000 psi	0,31 l/min

Tryska	Tlak	Množství
5 x 0,001 inch	20 000 psi	0,28 l/min
	25 000 psi	0,31 l/min
	30 000 psi	0,34 l/min
	35 000 psi	0,36 l/min
	40 000 psi	0,39 l/min
	45 000 psi	0,41 l/min
	50 000 psi	0,44 l/min
	55 000 psi	0,46 l/min
	60 000 psi	0,48 l/min

Tryska	Tlak	Množství
6 x 0,001 inch	20 000 psi	0,40 l/min
	25 000 psi	0,44 l/min
	30 000 psi	0,49 l/min
	35 000 psi	0,53 l/min
	40 000 psi	0,56 l/min
	45 000 psi	0,60 l/min
	50 000 psi	0,63 l/min
	55 000 psi	0,66 l/min
	60 000 psi	0,69 l/min

Tryska	Tlak	Množství
7 x 0,001 inch	20 000 psi	0,54 l/min
	25 000 psi	0,60 l/min
	30 000 psi	0,66 l/min
	35 000 psi	0,72 l/min
	40 000 psi	0,76 l/min
	45 000 psi	0,81 l/min
	50 000 psi	0,85 l/min
	55 000 psi	0,90 l/min
	60 000 psi	0,94 l/min

Tryska	Tlak	Množství
8 x 0,001 inch	20 000 psi	0,71 l/min
	25 000 psi	0,79 l/min
	30 000 psi	0,86 l/min
	35 000 psi	0,93 l/min
	40 000 psi	1,00 l/min
	45 000 psi	1,06 l/min
	50 000 psi	1,12 l/min
	55 000 psi	1,17 l/min
	60 000 psi	1,22 l/min

Tryska	Tlak	Množství
9 x 0,001 inch	20 000 psi	0,89 l/min
	25 000 psi	1,00 l/min
	30 000 psi	1,09 l/min
	35 000 psi	1,18 l/min
	40 000 psi	1,26 l/min
	45 000 psi	1,34 l/min
	50 000 psi	1,41 l/min
	55 000 psi	1,48 l/min
	60 000 psi	1,55 l/min

Tryska	Tlak	Množství
10 x 0,001 inch	20 000 psi	1,10 l/min
	25 000 psi	1,23 l/min
	30 000 psi	1,35 l/min
	35 000 psi	1,46 l/min
	40 000 psi	1,56 l/min
	45 000 psi	1,65 l/min
	50 000 psi	1,74 l/min
	55 000 psi	1,83 l/min
	60 000 psi	1,91 l/min

Tryska	Tlak	Množství
11 x 0,001 inch	20 000 psi	1,33 l/min
	25 000 psi	1,49 l/min
	30 000 psi	1,63 l/min
	35 000 psi	1,77 l/min
	40 000 psi	1,89 l/min
	45 000 psi	2,00 l/min
	50 000 psi	2,11 l/min
	55 000 psi	2,21 l/min
	60 000 psi	2,31 l/min

Tryska	Tlak	Množství
12 x 0,001 inch	20 000 psi	1,59 l/min
	25 000 psi	1,78 l/min
	30 000 psi	1,95 l/min
	35 000 psi	2,10 l/min
	40 000 psi	2,25 l/min
	45 000 psi	2,38 l/min
	50 000 psi	2,51 l/min
	55 000 psi	2,63 l/min
	60 000 psi	2,75 l/min

Tryska	Tlak	Množství
13 x 0,001 inch	20 000 psi	1,86 l/min
	25 000 psi	2,08 l/min
	30 000 psi	2,28 l/min
	35 000 psi	2,47 l/min
	40 000 psi	2,64 l/min
	45 000 psi	2,80 l/min
	50 000 psi	2,95 l/min
	55 000 psi	3,09 l/min
	60 000 psi	3,23 l/min

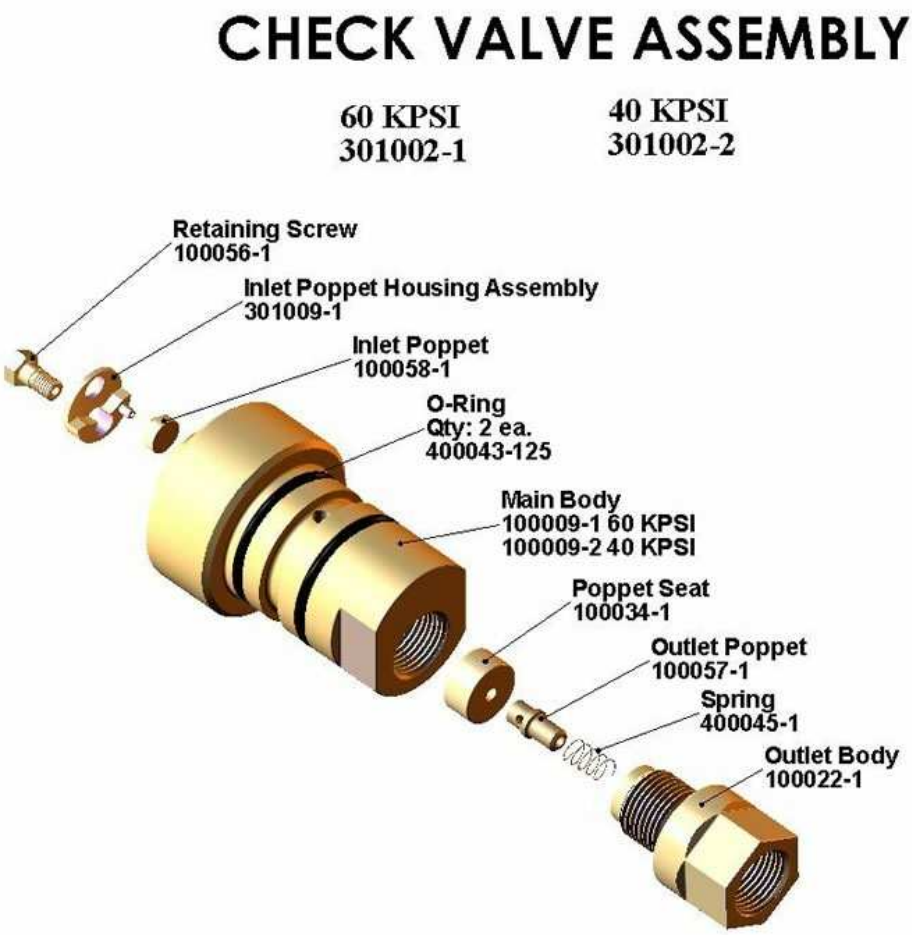
Tryska	Tlak	Množství
14 x 0,001 inch	20 000 psi	2,16 l/min
	25 000 psi	2,42 l/min
	30 000 psi	2,65 l/min
	35 000 psi	2,86 l/min
	40 000 psi	3,06 l/min
	45 000 psi	3,24 l/min
	50 000 psi	3,42 l/min
	55 000 psi	3,59 l/min
	60 000 psi	3,74 l/min

Tryska	Tlak	Množství
15 x 0,001 inch	20 000 psi	2,48 l/min
	25 000 psi	2,77 l/min
	30 000 psi	3,04 l/min
	35 000 psi	3,28 l/min
	40 000 psi	3,51 l/min
	45 000 psi	3,72 l/min
	50 000 psi	3,92 l/min
	55 000 psi	4,12 l/min
	60 000 psi	4,30 l/min

Tryska	Tlak	Množství
16 x 0,001 inch	20 000 psi	2,82 l/min
	25 000 psi	3,16 l/min
	30 000 psi	3,46 l/min
	35 000 psi	3,74 l/min
	40 000 psi	3,99 l/min
	45 000 psi	4,24 l/min
	50 000 psi	4,46 l/min
	55 000 psi	4,68 l/min
	60 000 psi	4,89 l/min

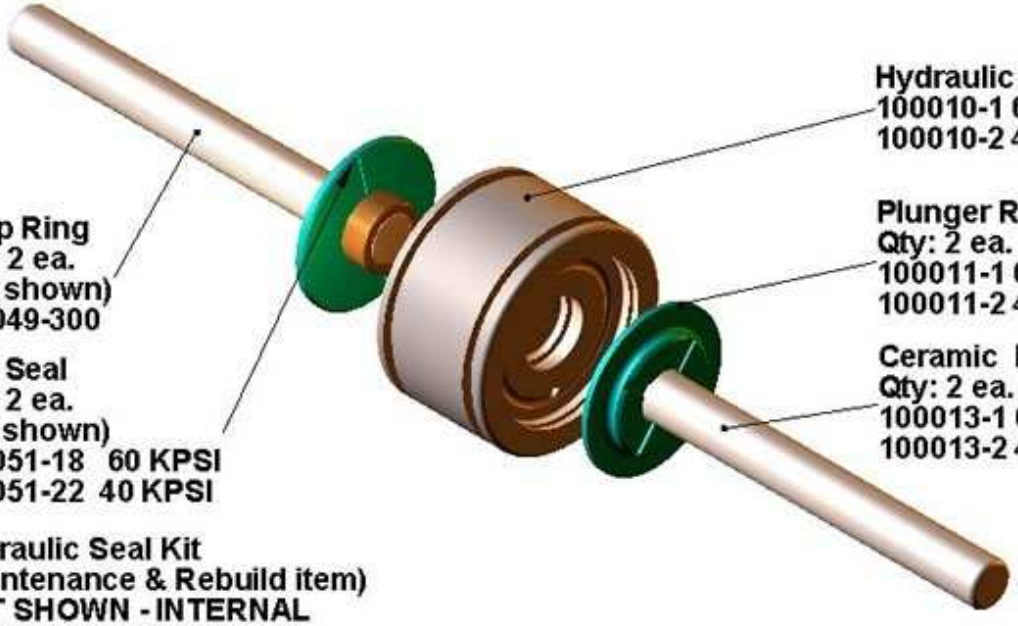
Tryska	Tlak	Množství
17 x 0,001 inch	20 000 psi	3,19 l/min
	25 000 psi	3,56 l/min
	30 000 psi	3,90 l/min
	35 000 psi	4,22 l/min
	40 000 psi	4,51 l/min
	45 000 psi	4,78 l/min
	50 000 psi	5,04 l/min
	55 000 psi	5,29 l/min
	60 000 psi	5,52 l/min

6.9 CATALOGUE OF SPARE PARTS FOR HP PART



PISTON ASSEMBLY

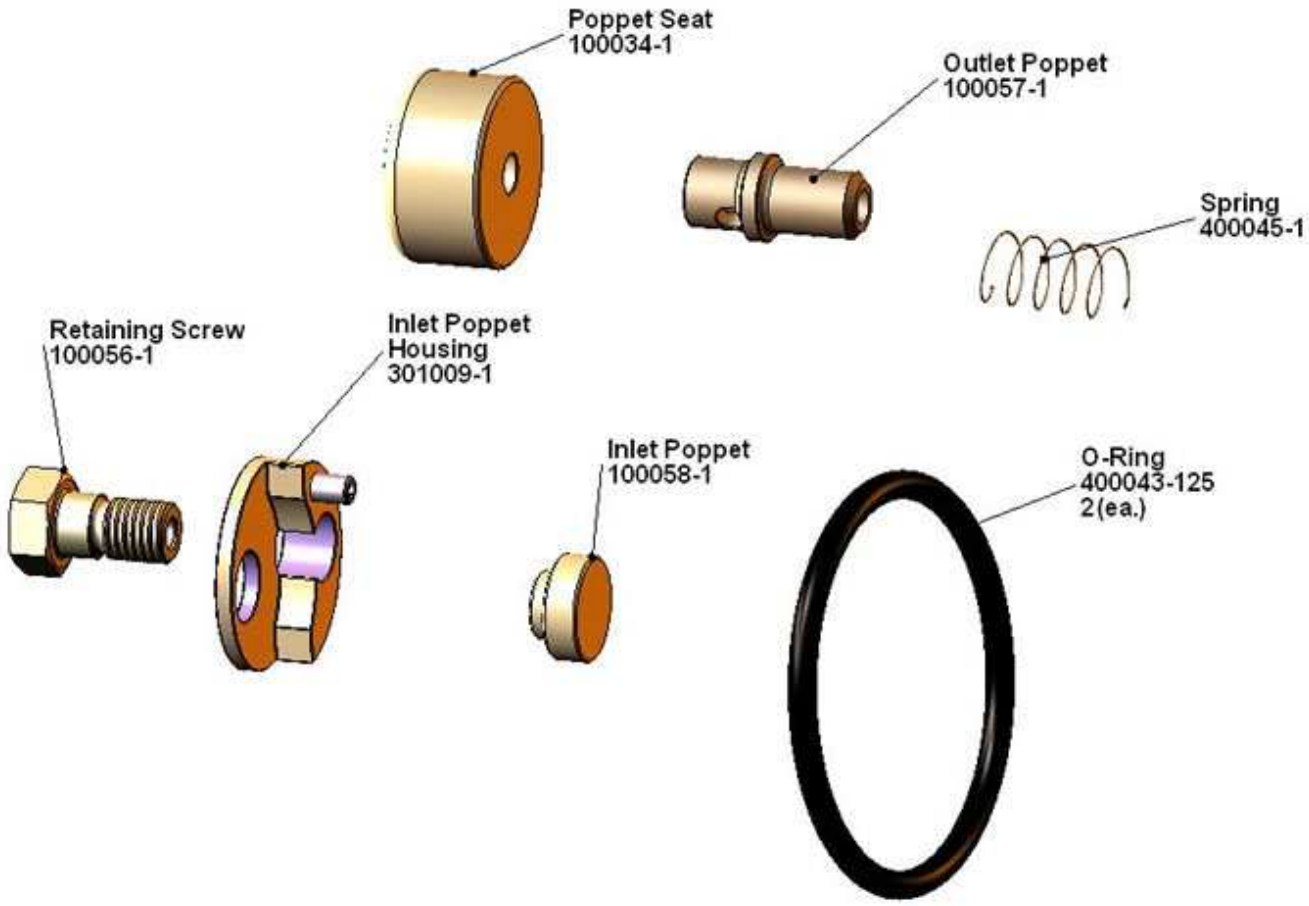
60 KPSI	40 KPSI
301005-1	301005-2



Hydraulic Seal Kit
(Maintenance & Rebuild item)
NOT SHOWN - INTERNAL
302007-1 (60 KPSI)
302007-2 (40 KPSI)

CHECK VALVE REPAIR KIT

302003-1



LOW PRESSURE SEAL KIT

55 KPSI
302007-1

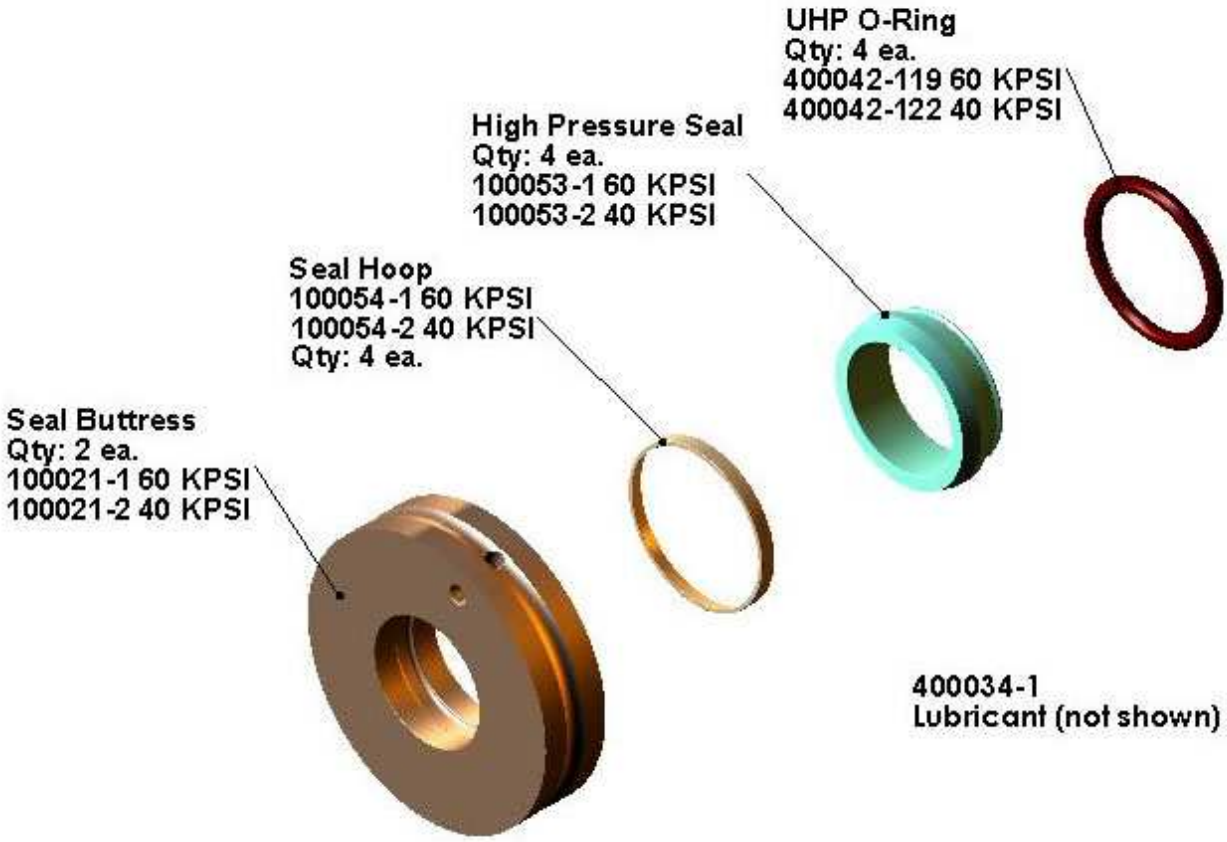
40 KPSI
302007-2



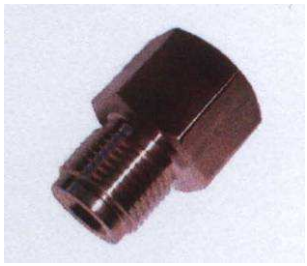
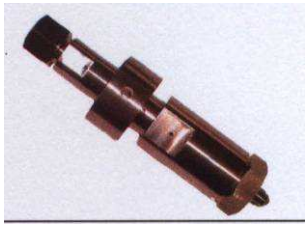
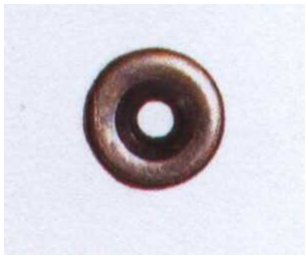
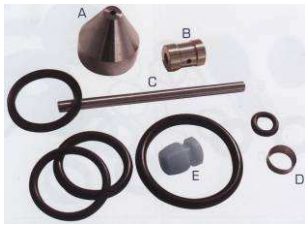
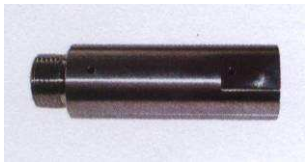
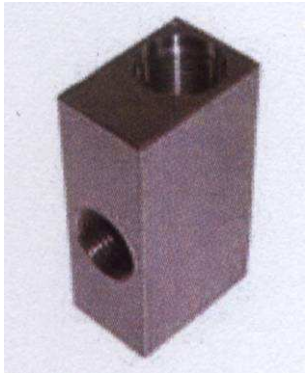
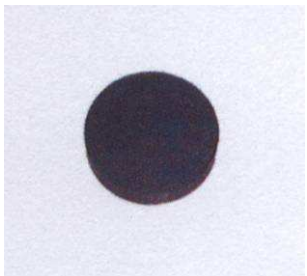
HIGH PRESSURE SEAL KIT

60 KPSI
302004-1

40 KPSI
302004-2

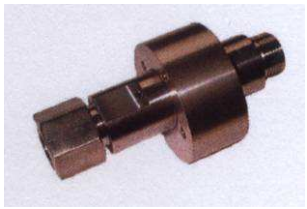


Bleed Down

Picture	Product	PLU	Description
	100134-1 A	B-5975-1	Adapter, Bleed Down Valve
	301013-1 A	C-5841-1	Bleed Down Valve, Assy
	400033-1 A	A-4241	Oil Seal, 125 Shaft Dia
	302008-1 A	004694-1	Bleed Down Valve Repair Kit, Hydraulic Auto
	100133-1 A	001822-1	Body High Pressure Valve
	100087-1 A	002878-1	Body Catcher, Bleed Down Valve
	15015 TT	A-3512	Disc Impingement, Bleed Down Valve



100084-1 A	B-5703-1	Hydraulic Auto Bleed Down Valve Poppet Landing

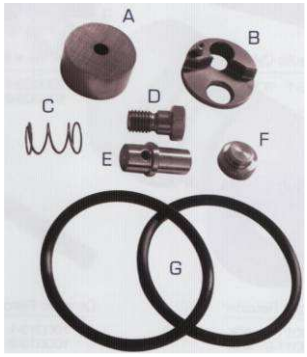
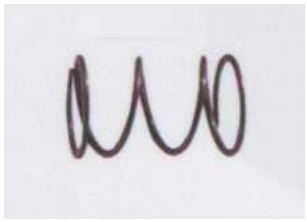
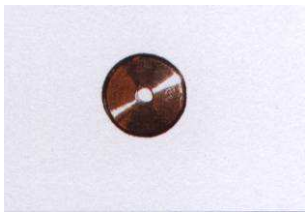
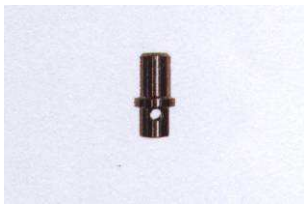
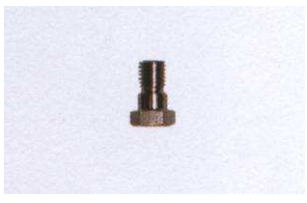


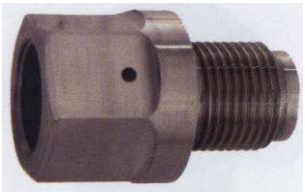
301013-2 A	C-2740-1	Body High Pressure Valve



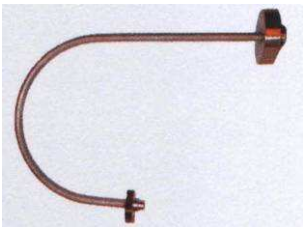
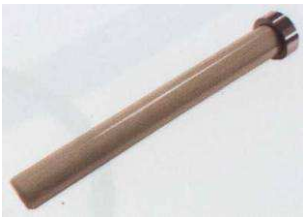
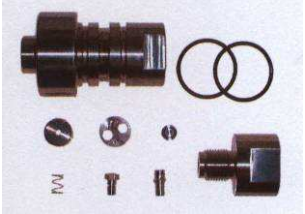
A-0641-6	A-0641-6	Plug, Hollow Hex,

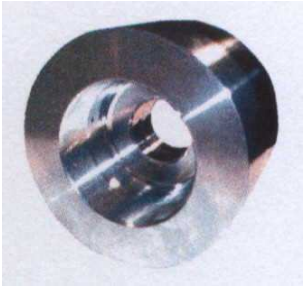
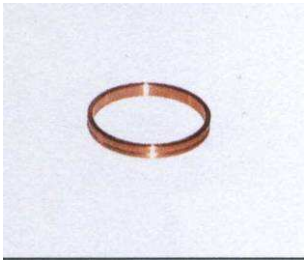
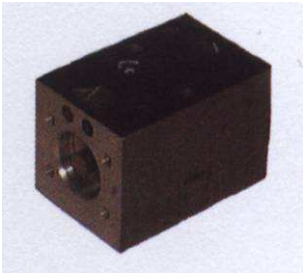
Check Valve

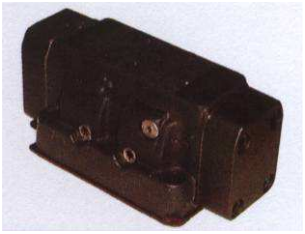
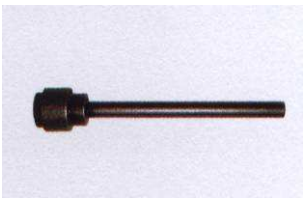
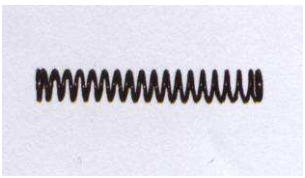
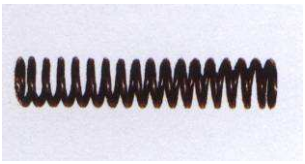
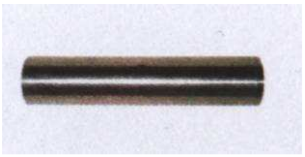
Picture	Product	PLU	Description
	301009-1 A	010564-1	Inlet Poppet Housing
	302003-1 A	010642-1	Check Valve Repair Kit, 60K, ESL
	400045-1 A	A-1606	Spring, check valve
	100009-1 A	004383-3	Check valve Main Body, 60K
	100009-2 A	004383-2	Check valve Main Body, 40K
	100034-1 A	004382-1	Poppet Seat, Check Valve
	100057-1 A	005917-1	Check Valve Outlet Poppet
	100056-1 A	004380-1	Check Valve Retaining Screw

	100058-1 A	010011-1	Poppet Inlet, Check Valve
	100022-1 A	C-1313-1	Outlet Body, Check Valve

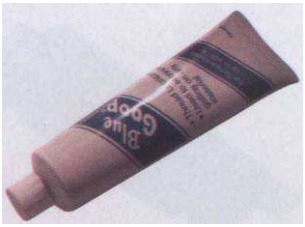
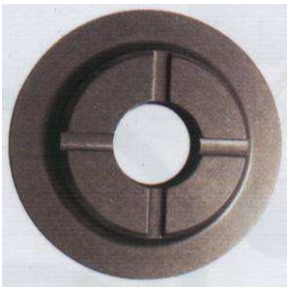
Intensifier

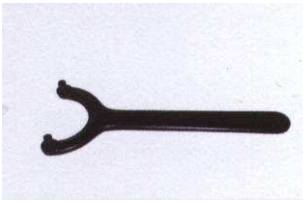
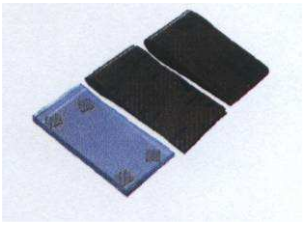
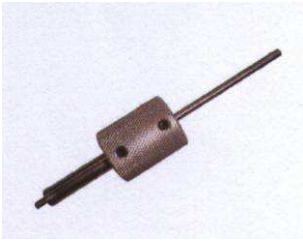
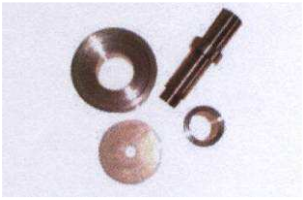
Picture	Product	PLU	Description
	100017-1 A	B-1000-2	Tie Rod
	400004-1 A	A-1000	Hex Nut, 3/4"
	A-0264-112	A-0264-112	Retaining Ring
	301011-1-LA	C-5462-1	Cable Guide Assembly, Right Hand
	301011-1-RA	C-5461-1	Cable Guide Assembly, Left Hand
	100007-1 A	B-1611-1	Shifting Pin
	100013-1 A	010253-1	Plunger, Piston, Ceramic, 60K
	301002-1 A	004384-3	Check Valve Assembly, 60K

	301004-1 A	321276-1	End Cap Assembly
	301001-1-L A	007303-2	End Bell, 60K, Left
	301001-1-R A	007303-1	End Bell, 60K, Right
	301001-2-L	007304-2	End Bell, 40K, Left
	301001-2-R	007304-1	End Bell,40K, Right
	100014-1 A	010276-1	Spacer, Oil Seal 60K
	007303/2 CZ	B-1040-2	Bushing Intensifier 60K
	007304/2 CZ	007304/2	Bushing Intensifier 40K
	301020-1A	301020-1	Intensifier 60 K -Assy
	301020-2 A	301020-2	Intensifier 40 K -Assy
	301020-4 A	301020-4	Intensifier 60 K -Assy
	400054-2 A	B-2498-1	Pilot Valve
	301018-2 A	301018-2	Electrical Pilot Shift Valve Assembly
	400054-1 A	A-10544-1	Pilot Valve; Electrical Shift

	301018-1 A	A-9183	Mechanical Pilot Shift; Valve Assembly
	301012-1 A	C-2743-1	Valve, Michanical Schift
	100051-1 A	002226-1	Firing Pin, Michanical
	100051-2 A	002226-2	Firing Pin, Electronic
	100019-1 A	B-1007-1	Actuating Plunger
	A-1713 CZ	A-1713	Compression Spring
	A-1012 CZ	A-1012	Spring
	100018-1 A	B-1002-2	Sleeve backup, 60 K
	301011-3 A	301011-3	Shift Sensor Assy, Electronic
	400057-1A	A-2124	Cartridge; Valve; Relief

Others

Picture	Product	PLU	Description
	100001-1 A	007038-3	Cylinder, HP, 60K
	100001-2 A	007038-2	Cylinder, HP, 40K
	C-1000-1	C-1000-1	Cylinder Hydraulic (ali 60K & 40K Intensifiers)
	400001-1 A	A-2185	Blue Anti-Galling Lubricant, 2 oz. Tube
	400034-2 A	A-4689	O-Ring Lubricant, Food Grade
	SE.1326	SE75361410	Filter, Tank
	FTCE2A10Q	FTCE2A10Q	Filter Element
	PP 000,5 S	PP 000,5 S	Water Filter, 0,5 um
	PP 001 S	PP 001 S	Water Filter, 1um
	PP 005 S	PP 005 S	Water Filter, 5 um
	PP 010 S	PP 010 S	Water Filter, 10 um
	100010-1 A	007026-1	Piston, Hydraulic, 60K
	100010-2 A	007201-1	Piston, Hydraulic, 40K
	100011-1 A	007029-1	Retainer, Plunger, 60K
	100011-2 A	006881-1	Retainer, Plunger, 40K

	400025-1 A	A-1628	Spanner Wrench
	301027-1 A	A-6333	Girth Wrench
	15275 TT	B-1813-1	Check Valve Lapping kit
	B-5716-1A	B-5716-1A	HP Valve Tool
	002227-1 CZ	002227-1 CZ	Shift Pin Tool
	B-2291-1 CZ	B-2291-1 CZ	HP Tool Kit 60K

Machine - type	High-pressure units FLOW type 5X, 7X, 20X. In pressure ranges 40.000 PSI, 55.000 PSI, 60.000 PSI and CNC table with a high-pressure water jet (WJ) technology
Noisiness:	Units and pump are the source of noise. By an objective measurement of noise pollution carried out by a person accredited for authorized noised measurement through measuring techniques complying with requirements defined for precision class 1, the following was proved: An equivalent level of acoustic noise A at values 88.7 dB is generated over a long-term operation in a closed machine-room bay of the JETS 3.8/60 type of pump. The machine-room, and/or ***pump are operated unattended.*** At full output and for 8-hour shift, a full-shift exposure of an attendant indicates a value of an equivalent level of acoustic pressure A – 80.1 dB, *which is in compliance even if usual measurement uncertainties E = 1.6 dB are taken into consideration, with a sufficient allowance for requirements determined to hygienic limits set out in the Governmental Regulation No. 148/2006 Coll. on protection of health against adverse noise and vibration effects.* (Remark: subject to the recommendation of the World Health Organization, the same hygienic limits de facto apply in most states).
Dust nuisance:	No dust is produced when the unit is in operation.
General safety:	The unit and CNC tables are designed as per ČSN EN ISO 12100-1 (83 3001) Machine and Equipment Safety - Basic Terms, General Design Principles and ČSN EN ISO 12100-2 Machine and Equipment Safety – Basic Terms, General Design Principles – Part 2: Technical Principles. Maximum available level of safety is therefore provided when the device is used in compliance with instructions. Residual risks are acceptable.
Electrical safety:	Electrical equipment of units and CNC tables are designed as per ČSN EN 60204-1 (33 2200) – March 2000. Safety of Machine and Equipment – Electrical Equipment of Machines - Part 1: General Requirements. This standard is a Czech version of the European standard EN 60204:1997 including amendments from September 1998. European standard EN 60204:1997 has a status of a Czech technical standard. The unit has undergone initial revision of electrical installation as per ČSN 33 1500 Electrical Engineering Directives. Revision of Electrical Equipment. Environment as regards electrical equipment "basic, normal" is in accordance with ČSN 33 2000-3 - AA4, AB4, AB5, XX1, for external effects AC to AR (except of AQ), BA1, BC1, BC2, BD1, BE1, CA1, CB1; as well as BA4, BA5
Humidity :	Not relevant for this type of equipment
Fire safety:	No special fire safety required

AKMEST Ing. Stuchlík, CSc. č.a. 23/1993 AUTORIZOVANÁ ÚŘEDNÍ MĚŘENÍ HLUKU

Ing. Václav Stuchlík, CSc.

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